When Does Sentiment Predict Volatility?

A Negative Result and a Caution on Regime-Conditional Forecast Evaluation

Frances (Su) Liu[♯] Vitali Alexeev[♯,†*] Katja Ignatieva[‡]

♯ School of Accounting and Finance, UTS Business School, University of Technology Sydney, NSW, Australia

† Department of Economics and Finance, University of Guelph, Ontario, Canada

‡ School of Risk and Actuarial Studies, UNSW Australia, NSW, Australia

November 6, 2026

Abstract

Does media sentiment improve realised volatility forecasts, and if so, when? We construct 27 media-based predictors in four families (direction, salience, instability and cross-channel disagreement) from firm-level news and social-media sentiment and attention indices, compress them by principal components, and add them to HAR models of daily realised volatility for DJIA constituents over nineteen years. In sample, the predictors raise conditional-quantile fit in the upper tail, most for the least persistent stocks. Out of sample, no augmented model robustly beats a daily re-estimated HAR benchmark in any state identifiable at the forecast origin. A regime-conditional evaluation that sorts forecast errors by the realised volatility of the target day shows gains where high volatility meets low persistence; we show that selecting on the outcome produces this pattern by rewarding any forecast shifted towards the realised surprise. A pre-specified forecast combination recovers a small gain in one real-time state.

*Corresponding author, Email: *vitali.alexeev@uts.edu.au*; Phone: +61(2) 9514 7778; Postal address: UTS Business School, 14-28 Ultimo Rd., Ultimo, NSW, 2007, Australia.

JEL classification: C22, C52, C53, G14, G12.

Keywords: Media sentiment; Volatility forecasting; Realised volatility; Volatility persistence; State-dependent predictability; HAR model; Forecast evaluation.

1 Introduction

Can narrative information improve volatility forecasts when price-based dynamics fail? A large and growing literature augments standard volatility models with news and sentiment variables, yet the gains are typically modest and inconsistent (Audrino et al., 2020; Bodilsen & Lunde, 2025). This has generated scepticism about the incremental value of media-based signals for volatility forecasting. A natural response to this scepticism is that it reflects a misspecified question: rather than asking whether sentiment improves volatility forecasts *on average*, one should ask *when* it does. This paper takes that response seriously, tests it as carefully as we are able, and reports that it fails: for firm-level media sentiment and daily realised volatility of DJIA constituents, no sentiment-augmented model shows a robust gain over the benchmark in any state identifiable at the forecast origin; the one real-time gain we find, about two per cent under QLIKE in one state for a pre-specified combination of the augmented and benchmark forecasts, is small and confined to one loss function; and the state-dependent gains that a conventional regime-conditional evaluation displays are an artefact of the evaluation itself. The paper is therefore about two things, the forecasting value of media sentiment, which turns out to be negligible once the state has to be known in advance, and the evaluation of state-dependent predictors, where the way the state is defined decides the answer.

Our central hypothesis is that media sentiment becomes valuable in the states in which the predictive content of past volatility is low. The heterogeneous autoregressive (HAR) model of Corsi (2009), by far the dominant framework in realised volatility forecasting, rests on a powerful but restrictive assumption: that historical price variation alone is the best predictor of future volatility. The model performs well because realised volatility exhibits strong and persistent long-memory dynamics (Andersen et al., 2003; Barndorff-Nielsen & Shephard, 2002). But persistence is not constant, across firms or over time. When it is low, less of next-day volatility is accounted for by its own past, and there is correspondingly more scope for other signals to matter. In such states, forward-looking narrative signals that aggregate the collective expectations, beliefs, and mood of market participants may carry information about evolving uncertainty that is not yet reflected in realised price dynamics. This conditional hypothesis, that the incremental value of media signals rises as the persistence of volatility falls, is the hypothesis of the paper; it is also the hypothesis that the evidence rejects out of sample.

We test this hypothesis using nearly two decades of daily realised volatility for DJIA constituents and firm-level media sentiment and attention measures from the Thomson Reuters MarketPsych Indices (TRMI). We begin by establishing the baseline: consistent with the prior literature, augmenting HAR-type models with sentiment delivers only modest improvements in average predictive performance.[1] An average, however, cannot show whether the contribution varies across market states, and that is the question the scepticism leaves open.

We therefore examine the fit of the media predictors across the conditional distribution of realised volatility and across firms sorted by the persistence of their volatility. In sample,

[1] Section 1.1 reviews the evidence. For U.S. equities the closest results are Audrino et al. (2020), Behrendt and Schmidt (2018), Bodilsen and Lunde (2025) and Rahimikia and Poon (2026); evidence of the same modest average contribution in other markets includes crude oil (Jiang et al., 2025), crypto-assets (Brauneis & Şahiner, 2026), rare-earth equities (Dutta et al., 2025) and Chinese equities (Lei & Song, 2023).

the incremental fit of the media predictors is negligible around the median of the volatility distribution and rises sharply in its upper quantiles (Koenker & Bassett, 1978), and these upper-tail gains are concentrated where persistence is low. Across firms, the upper-tail gain in fit is largest for the stocks whose volatility is least persistent and negligible for the most persistent stocks at every quantile.

The in-sample evidence is where the paper starts, not where it ends. Conditional-quantile fit is an in-sample description of the conditional distribution, and the cross-sectional relationship with persistence rests on 33 firms. Whether either translates into forecast gains available at the forecast origin is a separate question, and it is the one that matters for the literature's scepticism.

Out of sample the answer is no, and the paper devotes most of its length to establishing how firmly. Unconditionally, every sentiment-augmented specification forecasts worse than the daily re-estimated HAR benchmark, by one to four per cent of the benchmark's loss; model confidence sets retain the benchmark for every one of the 33 stocks under both loss functions; a test for superior predictive ability finds nothing to correct for, because no specification has a positive mean gain; and a fluctuation test finds no window of the evaluation period in which any specification did better. Conditioning on persistence alone, measured in real time by a rolling fractional-integration estimate ranked across firms, makes matters worse: the augmented specifications lose most in the high-persistence states, as the hypothesis predicts, but they also lose in the low-persistence states, which it does not. A conventional regime-conditional evaluation, which sorts forecast errors into cells defined by the realised volatility of the target day and by persistence, does display the pattern the hypothesis predicts: gains of about one per cent in the cell that combines high realised volatility with low persistence, for a majority of stocks. We show that this pattern is an artefact of conditioning on the outcome. Selecting the days on which realised volatility was extreme selects the days on which every model's error was large and of one sign, and on those days any forecast shifted in the direction of the surprise wins whatever its information content: the augmented specifications are shifted upward by about 0.4% on low-persistence days, a placebo forecast equal to the benchmark plus that shift reproduces the cell-by-cell pattern, a one-line proposition predicts the placebo's relative loss to the third decimal, and the pattern vanishes when the volatility state is defined on the volatility observed at the forecast origin, at every threshold and cut-off we try. Pooled tests that respect the cross-sectional dependence of the loss differentials find nothing in real time; under the outcome-based classification they do still reject equal accuracy under squared-error loss for the three factor specifications, though not under QLIKE, and we report this rather than let it stand for the hypothesis, because a gain that exists only once the outcome is known is not one a forecaster could have used. A Monte Carlo study of the evaluation design confirms both halves: real-time classification produces no spurious gains, and outcome-based classification rewards whatever component of a forecast difference is correlated with the realised surprise.

This paper contributes to the literature in three ways, and we are explicit about which rest on in-sample and which on out-of-sample evidence. First, it develops a framework for constructing media-based predictors that is, to our knowledge, more extensive than any in the volatility-forecasting literature. From the two raw series that commercial sentiment vendors supply per channel, tone and attention, we construct 27 predictors in four informational families: the *direction* of coverage over several horizons, its *salience* (attention, and attention-weighted and

signed tone), its *instability* (surges and collapses of attention and of attention-weighted tone), and cross-channel *disagreement* between news and social media. The framework is vendor-agnostic and documented to the last formula (Section 3 and Appendix A), and its in-sample payoff is clear: the predictors raise conditional-quantile fit in the upper tail of the volatility distribution by up to a quarter for the least persistent stocks, their family-level components rotate in importance over time, and compressing them into a handful of factors is what makes them usable in a rolling window. Second, it provides what we believe is the most thorough negative out-of-sample result to date on firm-level media sentiment as a predictor of daily realised volatility: the modest average contribution found by Audrino et al. (2020), Behrendt and Schmidt (2018) and Bodilsen and Lunde (2025) is confirmed on a different dataset and vendor, and the absence of robust gains is shown to hold in every real-time state defined by volatility, persistence or both, in every subperiod, and against the multiple-comparison and dependence-robust tests that the forecasting literature now expects. One qualification is reported in full: the benchmark does not encompass the augmented forecasts, and a pre-specified rolling combination recovers a gain of about two per cent in one real-time state under one loss function, consistent with information that is present but too noisy to use at full weight. Third, it identifies and quantifies a failure mode of regime-conditional forecast evaluation. Sorting forecast errors by the realised value of the target variable, a device that appears in evaluations of volatility forecasts "on high-volatility days", manufactures state-dependent gains for any forecast that differs from the benchmark in level within the selected state; we give the closed-form expression, a placebo that isolates the channel on real data, a Monte Carlo design that reproduces it, and the real-time alternative that removes it.

1.1 Related evidence

Three strands of evidence bear on our question, and they point in different directions. The first establishes that public information and the attention it attracts are tightly linked to volatility, contemporaneously and at high frequency. Antweiler and Frank (2004) find that an increase in internet message-board postings predicts higher trading volume and volatility on the following day, while the messages have no forecasting power for returns; Groß-Klußmann and Hautsch (2011) show, with a 20-second vector autoregression on London Stock Exchange stocks, that items an automated analytics engine classifies as highly relevant produce robust responses in volatility and volume; Boudoukh et al. (2019) show that firm-specific news identified by textual analysis accounts for about half of overnight idiosyncratic volatility and an eighth of trading-hours idiosyncratic volatility; Engle et al. (2021) formalise a public information processing hypothesis in which the arrival of public news raises volatility immediately and through subsequent private processing, and find that indicators of public information arrival explain an additional five to twenty per cent of the variation in changes of firm-specific volatility; and Ballinari et al. (2022) find that retail attention around news releases raises post-announcement volatility. At the market level, Da et al. (2015) show that a search-based index of household economic fears predicts short-term return reversals and temporary increases in volatility, and Sun et al. (2016) that intraday sentiment from a commercial text-analytics provider predicts intraday index returns. None of this is in doubt, and our in-sample results (Sections 5.2 and 5.3) are consistent with it.

The second strand asks the forecasting question that this paper asks: whether media-based measures improve volatility forecasts *relative to a model that already uses the history of*

volatility. Here the gains are small. Audrino et al. (2020) combine sentiment and attention measures from social media, news, information-consumption and search data with a large set of financial controls, select predictors by penalised regression, and find that the measures improve realised-volatility forecasts significantly but by amounts that are small from an economic point of view, with attention measures (Google searches and StockTwits message volume) rather than sentiment carrying most of the gain. Behrendt and Schmidt (2018), on the DJIA constituents we also study, find a significant correlation between stock-related Twitter sentiment and activity and intraday volatility, economically negligible effects, and no significant improvement in out-of-sample forecasts. Bodilsen and Lunde (2025) find that firm-specific news sentiment has only modest predictive power for the realised volatility of individual stocks, that the overnight count of firm-specific news items improves one-step-ahead forecasts, and that the sentiment of macroeconomic news improves forecasts substantially at long horizons. Rahimikia and Poon (2026) find that news adds information beyond a strong HAR-family benchmark, that news volume is the most robust signal and sentiment the less consistent one, and that the gains are concentrated on high-volatility days. The common pattern is that attention and arrival matter more than tone, that the forecast gains at daily frequency are of the order of a per cent, and that the benchmark is hard to beat; Chassot and Audrino (2026) show, across 1,445 stocks, that a HAR model re-estimated daily is not beaten even by machine-learning methods with extensive tuning.

The third strand finds that textual measures forecast volatility at longer horizons or at the aggregate level. Glasserman and Mamaysky (2019) show that unusual negative news about large financial firms forecasts both company-specific and aggregate volatility several months ahead, and Calomiris and Mamaysky (2019) that a compact set of news measures predicts country-level returns, volatility and drawdowns, with larger effects at the year-ahead than at the monthly horizon. Our evidence concerns daily forecasts of firm-level volatility and does not speak to these horizons; we return to them in the conclusion. Finally, the "when" question has typically been addressed by conditioning on sub-samples: recessions, for the predictive content of news sentiment for returns (García, 2013), and high-volatility days, for the forecast gains from news (Rahimikia & Poon, 2026). Section 6.2 shows that the answer depends on whether the sub-sample is identified before or after the outcome is observed, and that the difference is not small. More broadly, the results are a reminder that a daily re-estimated HAR model is a demanding benchmark (Chassot & Audrino, 2026), that the states in which it is least reliable are also the states in which any competitor's forecast is least reliable, and that an evaluation which identifies those states after the fact will flatter whichever competitor happened to lean the right way.

The remainder of the paper is organised as follows. Section 2 describes the data. Section 3 presents the construction of the media-based predictors and their compression. Section 4 presents the forecasting framework. Section 5 reports the in-sample evidence and Section 6 the out-of-sample evidence, including the real-time evaluation, the placebo, the model confidence sets and the encompassing and combination results. Section 7 concludes. An online appendix provides complete definitions and distributional diagnostics for all media-based predictors (Appendix A), technical details of the one-stage and two-stage PCA procedures with an illustrative analysis for AAPL (Appendix B), parameter stability tests (Appendix C), additional in-sample and out-of-sample results (Appendices D and E), and the Monte Carlo study of the evaluation design (Appendix F).

2 Data

We analyse 19 years of daily observations for constituents of the DJIA, covering January 1998–December 2016. Media sentiment and attention are sourced from the Thomson Reuters MarketPsych Indices (TRMI), and price data come from Refinitiv DataScope.[2] The data were obtained through a proprietary vendor archive. The sample ends in 2016 because the vendor archive from which the firm-level indices were obtained covers this period. The analysis therefore does not speak to the post-2016 media environment, and we do not claim that the magnitudes reported below carry over to it. The period does, however, contain the variation that our question requires. It spans the Dot-com episode, the pre-crisis expansion, the Global Financial Crisis, and the post-crisis recovery, so that persistence and volatility states are observed in several distinct configurations rather than in a single regime.

Our focus on DJIA constituents is deliberate. These highly visible, large-capitalisation firms attract intensive and continuous media coverage across both traditional news and social media, yielding dense and reliable sentiment signals throughout the sample. Restricting attention to this set minimises the sentiment sparsity and microstructure noise that afflict smaller stocks, and allows us to isolate firm-specific narrative effects in a controlled, information-rich setting. The aim is to establish whether, and when, media information adds to a volatility forecast for well-covered firms, before asking whether the answer generalises to the broader cross-section. The panel comprises the firms that were DJIA constituents during the sample period and for which both intraday price data and firm-level TRMI coverage are available over 1998–2016; Table 1 lists the constituents and reports the resulting firm-day count.

All variables are aggregated to a daily horizon to align with the HAR volatility framework and with tactical risk management cycles. Our central objective is to assess whether sentiment- and attention-based predictors observed on day t improve out-of-sample forecasts of realised volatility on day $t+1$.

2.1 Realised Volatility

Daily realised volatility (RV) is computed from 5-minute intraday price data, a sampling frequency that balances estimation accuracy against microstructure noise (Liu et al., 2015). To mitigate the influence of data errors and outliers, 5-minute returns are winsorised at the 0.005% level prior to aggregation. This a deliberately conservative threshold designed to remove only observations most plausibly attributable to data recording errors rather than genuine price movements.[3] For each stock on day t, RV is defined as the sum of squared

[2] In August 2023, Refinitiv was rebranded as LSEG following its 2021 acquisition by the London Stock Exchange Group (LSEG). We retain the Refinitiv label to reflect the data vendor's name during our sample period.

[3] Winsorisation is applied stock by stock. For a typical constituent with roughly 4,500 trading days of 78 five-minute intervals (about 349,000 five-minute returns), the 0.005% threshold replaces approximately 17 observations in each tail, about 34 per stock and about 1,100 across the 147,610 firm-days in the panel (Table 1). A threshold of 0.5%, as often applied in the literature (e.g., Liu et al., 2015; Zhang et al., 2023), and even 1.0% (Van Kervel & Menkveld, 2019), proved too aggressive in our data, removing observations that appear to reflect genuine intraday price variation rather than recording errors.

5-minute log-returns:

$$RV_t = \sum_{j=1}^{M} r_{t,j}^2, \tag{1}$$

where M denotes the number of 5-minute intervals in a trading day, $r_{t,j} = \log P_{t,j} - \log P_{t,j-1}$ is the log-return over interval j, and $P_{t,j}$ the previous-tick price at the close of that interval.

2.2 Sentiment and Attention Data

We extract two primary metrics from TRMI at the firm level: *Sentiment* and *Buzz*, both derived from automated natural language processing (NLP) of firm-level news and social media text by Refinitiv's proprietary text analytics engine.[4] Sentiment is the score of positive references to the firm net of negative references, divided by Buzz, and lies in $[-1, 1]$. Buzz measures the volume of scored references to the firm over the preceding 24 hours and is non-negative.

Both metrics are provided separately for the *news* and *social media* channels, enabling us to capture potential cross-channel discrepancies in the information environment. This distinction is economically relevant because the two media environments may convey different information. Traditional news tends to be more curated and institutionally filtered, whereas social media often reacts faster, is more fragmented, and may better capture shifts in investor attention, disagreement, or emerging narratives (Bollen et al., 2011). Full details of the underlying data sources, textual feeds, and proprietary scoring methodology are provided in Peterson (2016). Because the scoring engine is proprietary, we set out what is and is not known about the construction of the two raw series. What is documented is the following: each index is computed at the entity (firm) level from the text items in which the firm is identified; *Buzz* is the sum of the absolute scores of all such references over the preceding 24 hours, so that it measures the volume of scored content rather than a raw count of articles, and *Sentiment* is a net measure of positive relative to negative references, normalised by the volume of references so that it is bounded in $[-1, 1]$; the two channels draw on separate source sets, with the news channel built from Reuters and other professional news feeds (Reuters only in 1998–2002, Reuters and selected third-party wires from 2003, and internet news sites from 2005) and the social channel from public internet sources (with an aggregated social media feed added from 2009); and the same lexicon and aggregation rules are applied to every firm at a given time, while the lexicon is revised every two to three years and the source set expands over the sample. What is not disclosed is the lexicon itself, the grammatical rules that score each reference, and the list of scored categories that enter Sentiment; the

[4] We rely on vendor-constructed sentiment scores rather than constructing our own measures from raw text. This choice is deliberate: commercial vendors provide a standardised, consistently applied scoring methodology across all firms and time periods, eliminating the researcher degrees of freedom inherent in bespoke NLP pipelines (including choices over sentiment analysers, weighting schemes, and aggregation methods) that could otherwise be tuned, even inadvertently, to favour particular empirical results. A measure fixed before the analysis began cannot have been tuned to it, which reduces the risk that the results reflect choices made in constructing the sentiment score. The sensitivity of results to the choice of sentiment construction method is well established: Bollen et al. (2011) show that only specific mood dimensions from Twitter are predictive of DJIA movements, while others are not, an example of the discretion that a pre-committed vendor score removes. A widely used commercial product also makes the results relevant to practitioners who rely on the same inputs.

aggregation itself is documented: each category enters with a sign of $+1$ or -1. We do not attempt to reverse-engineer these. All 27 predictors used in the paper are transparent, fully documented transformations of the four raw series (Appendix A.2), so the only opaque step is the one performed by the vendor, and it is common to every specification we compare. Replicating the analysis with an open-source sentiment engine, so that every construction detail is known, is a natural extension that we leave for future work.

The TRMI variables are mapped to DJIA constituents using firm-level ticker identifiers. While the raw TRMI feed is available at intraday frequency, we aggregate the variables to the daily level to align them with the realised volatility forecasting framework and to reduce the influence of very short-lived textual noise. This aggregation also improves comparability across firms and ensures that the media-based signals correspond to the same daily horizon as the realised volatility measures used in the forecasting models.

Table 1 reports stock-level descriptive statistics for realised volatility, sentiment, and buzz across all DJIA constituents. From these four raw series ($sent^{\mathrm{N}}$, $sent^{\mathrm{S}}$, $buzz^{\mathrm{N}}$ and $buzz^{\mathrm{S}}$) we construct the 27 media-based predictors of Section 3.

3 Media Signal Construction and Extraction

The first contribution of the paper is the framework developed in this section. Studies that add media information to volatility models typically enter one or two series per source, a sentiment score and a count of items, in levels. We start from the same two raw series per channel and ask what distinct informational dimensions they can be made to reveal: not only whether coverage is favourable, but whether it is loud, whether it is changing, and whether the two media ecosystems agree. The answer is a set of 27 predictors in four families, summarised in Table 2, listed one by one in Table A.2 and defined formula by formula in Appendix A.2, together with a two-stage compression that keeps the families' economic interpretation while making the set usable in a rolling window. The framework uses nothing specific to the vendor: any provider that reports tone and volume by channel supports it. Its in-sample value is documented in Sections 5.2 and 5.3; Sections 6.2 and 6.3 show that this value does not translate into one-day-ahead forecast gains, a finding that the framework makes more, not less, informative, since it rules out the objection that the sentiment information was entered too crudely to be found.

The raw series are highly persistent, strongly correlated and contaminated by short-lived fluctuations in news flow (Gan et al., 2020); Section 3.1 constructs the predictors and Section 3.2 compresses them.

3.1 Constructing sentiment signals

We begin with daily sentiment and buzz measures from both traditional news and social media sources available from TRMI. These series capture two distinct dimensions of media information: the *direction* of narrative tone and the *intensity* of information flow. On their own, however, raw sentiment levels and attention counts are blunt instruments. A neutral average sentiment score may conceal a firm whose coverage has swung from uniformly positive to uniformly negative within the same week. High buzz on one day says nothing about whether

TABLE 1: DESCRIPTIVE STATISTICS FOR DJIA CONSTITUENTS. Daily summary statistics over January 1998–December 2016 for realised volatility and raw TRMI sentiment and attention scores. Specifically, RV is daily realised volatility (%), $sent^{N}$ and $sent^{S}$ are the raw daily sentiment scores from the Thomson Reuters MarketPsych Indices (TRMI) news and social media channels, respectively; both lie in $[-1, 1]$. $buzz^{N}$ and $buzz^{S}$ denote the corresponding raw daily article and post counts (thousands). The panel comprises 147,610 firm-day observations across 33 DJIA constituents. No transformations are applied; the full set of engineered predictors derived from these raw scores is defined in Table A.3. Missing observations are negligible ($< 0.25\%$) for all firms except `UNH` ($\approx 5.1\%$). The maximum RV of 17.12% occurs on 16 September 2008 for `AIG`, reflecting the extreme volatility surrounding the U.S. government bailout.

	RV (%)				$sent^{N}$				$sent^{S}$				$buzz^{N}$				$buzz^{S}$			
RIC	mean	std	min	max	mean	std	min	max	mean	std	min	max	mean	std	min	max	mean	std	min	max
AAPL.OQ	0.06	0.09	0.00	1.49	0.01	0.16	-1.00	1.00	-0.01	0.05	-0.22	0.21	8.07	10.37	0.00	141.94	7.94	8.45	0.10	119.88
AIG.N	0.08	0.41	0.00	17.12	-0.01	0.30	-1.00	1.00	-0.01	0.23	-1.00	1.00	0.65	2.77	0.00	97.84	0.42	1.74	0.00	78.34
AXP.N	0.04	0.07	0.00	1.27	0.05	0.30	-1.00	1.00	0.02	0.23	-1.00	1.00	0.25	0.53	0.00	11.93	0.11	0.17	0.00	3.70
BA.N	0.03	0.03	0.00	0.49	-0.04	0.15	-0.60	0.67	-0.04	0.15	-0.88	1.00	1.83	2.89	0.00	54.91	0.56	0.71	0.00	12.82
BAC.N	0.05	0.13	0.00	2.14	-0.06	0.21	-1.00	1.00	-0.06	0.10	-0.58	0.71	1.13	2.00	0.00	40.02	1.81	3.13	0.00	35.21
C.N	0.06	0.17	0.00	4.56	-0.03	0.14	-1.00	1.00	-0.04	0.12	-1.00	0.93	2.15	4.21	0.00	111.48	1.46	2.64	0.00	48.82
CAT.N	0.03	0.04	0.00	0.72	-0.01	0.32	-1.00	1.00	-0.00	0.24	-1.00	1.00	0.33	0.99	0.00	15.83	0.17	0.37	0.00	9.28
CSCO.OQ	0.05	0.09	0.00	0.92	0.12	0.20	-1.00	1.00	-0.08	0.08	-0.52	0.48	1.26	2.21	0.00	34.49	2.80	2.77	0.03	32.39
CVX.N	0.02	0.04	0.00	0.91	-0.06	0.25	-1.00	1.00	-0.02	0.16	-1.00	0.75	0.58	0.84	0.00	10.43	0.19	0.22	0.00	3.61
DD.N	0.03	0.03	0.00	0.64	-0.02	0.41	-1.00	1.00	-0.06	0.21	-1.00	1.00	0.03	0.06	0.00	1.22	0.17	0.19	0.00	2.22
DIS.N	0.03	0.04	0.00	0.63	0.01	0.18	-1.00	1.00	-0.00	0.14	-0.75	0.71	0.93	1.44	0.00	36.62	0.38	0.54	0.00	11.19
GE.N	0.03	0.05	0.00	1.01	0.07	0.17	-0.67	1.00	-0.05	0.09	-0.50	0.51	1.16	1.91	0.00	44.52	1.72	1.75	0.01	28.05
GS.N	0.04	0.08	0.00	2.14	-0.04	0.15	-0.57	0.75	-0.04	0.11	-0.75	0.59	1.95	3.63	0.00	88.03	1.42	2.25	0.00	44.96
HD.N	0.03	0.04	0.00	0.80	0.09	0.31	-1.00	1.00	0.02	0.16	-0.75	1.00	0.40	1.20	0.00	21.18	0.29	0.41	0.00	13.14
HON.N	0.03	0.04	0.00	0.83	0.10	0.32	-1.00	1.00	0.06	0.31	-1.00	1.00	0.19	0.39	0.00	5.61	0.05	0.09	0.00	1.57
HPQ.N	0.03	0.04	0.00	0.57	0.04	0.18	-1.00	1.00	-0.05	0.08	-0.32	0.68	1.73	3.16	0.00	49.61	1.07	1.11	0.01	23.16
IBM.N	0.02	0.03	0.00	0.52	0.09	0.19	-1.00	1.00	0.01	0.10	-0.46	0.77	1.33	1.96	0.00	23.49	0.82	0.88	0.01	19.24
INTC.OQ	0.05	0.08	0.00	1.14	0.06	0.18	-1.00	1.00	-0.00	0.08	-0.41	0.62	1.82	2.76	0.00	36.09	1.82	1.87	0.07	30.32
IP.N	0.04	0.06	0.00	0.80	0.02	0.38	-1.00	1.00	0.04	0.32	-1.00	1.00	0.06	0.15	0.00	3.65	0.04	0.06	0.00	0.59
JNJ.N	0.01	0.02	0.00	0.35	0.00	0.26	-1.00	1.00	0.03	0.15	-1.00	1.00	0.42	0.76	0.00	10.47	0.22	0.23	0.00	2.95
JPM.N	0.04	0.09	0.00	1.75	-0.06	0.15	-1.00	1.00	-0.07	0.10	-0.50	0.93	1.90	4.11	0.00	108.27	1.09	1.96	0.00	77.02
KO.N	0.02	0.02	0.00	0.44	-0.01	0.20	-1.00	1.00	-0.00	0.12	-0.47	1.00	0.68	1.10	0.00	27.62	0.30	0.26	0.01	4.10
MCD.N	0.02	0.03	0.00	0.54	-0.03	0.22	-1.00	1.00	-0.02	0.15	-1.00	1.00	0.72	1.16	0.00	16.80	0.25	0.33	0.00	7.20
MMM.N	0.02	0.03	0.00	0.63	0.05	0.25	-1.00	1.00	0.03	0.24	-1.00	1.00	0.33	0.53	0.00	7.04	0.08	0.12	0.00	1.96
MRK.N	0.02	0.03	0.00	0.72	-0.01	0.24	-1.00	1.00	-0.00	0.13	-0.98	1.00	0.67	1.43	0.00	28.10	0.32	0.49	0.00	12.87
MSFT.OQ	0.04	0.06	0.00	1.03	0.05	0.12	-0.63	1.00	-0.02	0.07	-0.34	0.33	5.15	4.31	0.01	43.31	2.33	1.45	0.37	18.98
NKE.N	0.03	0.03	0.00	0.50	0.05	0.30	-1.00	1.00	0.04	0.28	-1.00	1.00	0.23	0.59	0.00	10.38	0.10	0.27	0.00	6.28
PFE.N	0.02	0.03	0.00	0.60	-0.03	0.21	-1.00	1.00	-0.02	0.10	-0.51	0.58	0.71	1.58	0.00	40.49	0.77	0.67	0.00	8.67
PG.N	0.02	0.02	0.00	0.45	0.06	0.26	-1.00	1.00	0.03	0.19	-1.00	1.00	0.36	0.74	0.00	10.53	0.18	0.27	0.00	6.46
UNH.N	0.03	0.05	0.00	0.85	0.08	0.34	-1.00	1.00	0.03	0.31	-1.00	1.00	0.18	0.51	0.00	16.50	0.09	0.25	0.00	5.56
VZ.N	0.02	0.03	0.00	0.72	0.02	0.18	-1.00	1.00	0.01	0.10	-0.39	0.50	0.94	1.31	0.00	25.04	0.57	0.51	0.01	5.37
WMT.N	0.02	0.03	0.00	0.49	-0.01	0.20	-1.00	1.00	-0.01	0.10	-1.00	0.65	1.54	2.42	0.00	31.45	0.77	0.94	0.00	25.05
XOM.N	0.02	0.03	0.00	0.90	-0.09	0.21	-1.00	1.00	-0.06	0.10	-0.47	0.58	0.86	1.49	0.00	54.97	0.52	0.44	0.01	9.23

that attention is rising, falling, or split across channels sending opposite signals. And a moderately negative tone carries very different information when it is broadcast widely than when it appears in a handful of obscure posts.

To capture these distinctions, we construct a set of transformed predictors designed to isolate four informational dimensions of the media environment. *Direction* captures whether the prevailing narrative tone is favourable or unfavourable, aggregating sentiment signals across horizons to smooth noise and identify dominant coverage trends. *Salience* measures whether sentiment is accompanied by unusually high media attention, reflecting the intuition that tone is economically relevant primarily when it is widely disseminated (Da et al., 2011). *Instability* captures whether tone or attention is changing rapidly or inconsistently over time, with sudden surges or collapses in coverage signalling informational uncertainty that may presage elevated volatility. *Disagreement* measures whether the news and social media channels convey conflicting signals (whether in tone, intensity, or attention allocation) with cross-channel divergence indicating an ambiguous informational environment that amplifies uncertainty (Gan et al., 2020; García, 2013; Jiao et al., 2020).

Table 2 summarises the four families and names the predictors in each; Table A.2 in Appendix A.2 lists the 27 predictors individually by family and channel. Full definitions, construction formulae, and economic interpretations for all 27 predictors are provided in Table A.3; descriptive statistics are reported in Table A.4.

Table 2: The Four Families of Media-Based Predictors. All 27 predictors are transformations of the four raw daily TRMI series, sentiment and buzz for the news (N) and social (S) channels; $c \in \{\mathrm{N}, \mathrm{S}\}$ denotes a predictor constructed for each channel. MCS is the media coverage sentiment score (a non-linear balance of positive against negative sentiment days over a weekly or monthly window), RB the buzz-weighted three-day mean of sentiment, sgn the signed log buzz, and D an absolute cross-channel difference, with $\star$ its polarity-magnitude version. The predictors are listed one by one in Table A.2 and defined, with imputation rules and economic rationales, in Table A.3; descriptive statistics are in Appendix A.3.

Family	What it measures	Predictors	N
Direction	Whether the prevailing tone of coverage is favourable or unfavourable, over one day and over longer windows	$sent^c$; MCS^c_w, MCS^c_m	6
Salience	How loud the coverage is, and tone weighted by how widely it is disseminated	$buzz^c$, $\ln buzz^c$; RB^c, $\ln RB^c$; sgn^c	10
Instability	Whether attention, or attention-weighted tone, is surging or collapsing	$\Delta \ln buzz^c$, $\Delta^2 \ln buzz^c$; $\Delta \ln RB^c$	6
Disagreement	Whether the news and social channels send conflicting signals, in tone, in attention, and in their combination	D^{Sent}, $D^{\mathrm{Sent}\star}$; D^{Buzz}; D^{RB}, $D^{\mathrm{RB}\star}$	5

3.2 Signal Extraction via Dimension Reduction

Constructing 27 transformed predictors from four raw series solves one problem, representing several dimensions of the media environment at once, while creating another. The resulting feature set is not large in absolute terms (27 predictors in a 252-day estimation window),

but it is internally correlated and partly redundant by design: all 27 predictors are transformations of four underlying series, so a high degree of collinearity is built into the design rather than being an incidental feature of the data. Directly incorporating all predictors into the forecasting equation overstates the true dimensionality of the media environment, inflates estimation noise, and, as the stability diagnostics in Appendix C confirm, induces severe parameter instability that renders coefficient estimates unreliable. The challenge is to compress the predictor space without discarding the economically meaningful variation it contains.

We focus on principal component analysis (PCA) rather than sparsity-based alternatives such as LASSO for a reason grounded in the structure of the signal itself. Variable selection assumes that predictive content is concentrated in a small subset of individually informative variables. But the feature set is designed to capture the *collective* informational state of the media environment: tone, attention, instability, and disagreement are all partial reflections of the same underlying narrative dynamics and are expected to co-move. In such a setting, discarding variables selectively, as LASSO does, risks eliminating signal that is only recoverable in aggregate (Audrino et al., 2020; Stock & Watson, 2002). This is a hypothesis about the particular set of constructed predictors, whose collinearity is largely mechanical, and not a claim about media information in general; the comparison between PCA-based and LASSO-based specifications in Section 5 should be read in that light. We retain a fixed number K of leading components, chosen once after inspecting the variance contributions so that the retained components explain about 80% of predictor variance on average across estimation windows ($K = 4$ in the one-stage specification), a conventional cumulative-variance rule (Jolliffe, 2002) applied to the average rather than window by window, so that the dimension of the media component does not change from one window to the next. Information criteria for the number of factors (Bai & Ng, 2002) offer a data-driven alternative; we prefer the fixed-share rule because it is transparent and behaves predictably when re-applied in every rolling estimation window.

Although one-stage PCA is statistically efficient, it treats the full predictor set as a single undifferentiated block. This risks blending economically distinct informational channels into a single set of components (e.g., averaging across Salience and Disagreement signals that operate through different mechanisms). To preserve more economic structure, we also implement a two-stage hierarchical PCA that first compresses variation *within* groups of related predictors (defined below) and then enters the retained within-family components directly into the forecasting equation. There is no further compression across families: the second stage is the forecasting regression itself, which weights the family-level components by their estimated coefficients. This is what is estimated in every two-stage specification (Appendix B gives the formal description). This layered strategy mirrors the factor-based approach of Fan et al. (2020), which demonstrates that isolating distinct information channels prior to aggregation prevents subtle signals from being diluted by globally dominant factors.

Within the two-stage framework, we explore two partitioning methods. The first uses data-driven clustering: feature loadings from the one-stage PCA are grouped via hierarchical agglomerative clustering (HAC) with cosine distance (Oti & Olusola, 2024), so that the clusters reflect systematic co-movement in the predictor space rather than our prior economic assumptions. The second imposes an economic structure directly, partitioning features by information channel (News vs. Social) and signal type (tone vs. attention vs. cross-channel discrepancy) into six groups: news sentiment, social sentiment, news attention, social atten-

tion, attention-weighted (relative) sentiment, and cross-channel disagreement, ensuring that the economic interpretation of each group is preserved in the extracted components. The six groups are a finer partition of the four informational families of Section 3.1: Direction and Salience are each split by channel, and the relative-buzz predictors, which combine tone and attention, form a group of their own. For each partitioning, we compare retaining only the single leading component per cluster against retaining a fixed number of components per group, chosen so that the retained components explain at least about 80% of within-group variance on average (91% over the full sample), testing whether the predictive signal is concentrated in a single dominant factor or distributed across secondary components. Full technical details, including the formal PCA decomposition, implementation of HAC clustering, and the AAPL illustrative example, are provided in Appendix B.

4 Empirical Framework

This section outlines the forecasting framework used to evaluate whether media-based sentiment signals improve realised volatility forecasts and, more importantly, whether their contribution varies systematically across market states. Our empirical design is built around a standard HAR benchmark, which we augment with media-based predictors. We then examine whether the contribution of sentiment differs across the conditional distribution of volatility and across regimes characterised by different levels of volatility persistence.

4.1 Benchmark volatility forecasting model

Our benchmark specification is the heterogeneous autoregressive (HAR) model of Corsi (2009), which is widely used in realised volatility forecasting because it parsimoniously captures the multi-horizon persistence structure of volatility through a simple additive framework. The canonical HAR model is linear in levels, however, financial volatility typically exhibits positive skewness and heavy tails that can distort estimation and inflate forecast errors. We therefore adopt the logarithmic HAR specification of Corsi et al. (2012), which stabilises the variance, enforces a positivity constraint on predicted volatility, and mitigates the influence of extreme observations.[5] For each firm on day t, the benchmark model is:

$$\log RV_{t+1} = \alpha + \beta_d \log RV_t^{(d)} + \beta_w \log RV_t^{(w)} + \beta_m \log RV_t^{(m)} + \varepsilon_{t+1}, \tag{2}$$

where $RV_t^{(d)} \equiv RV_t$ denotes daily realised volatility, and the weekly and monthly components are defined as averages of daily log-volatilities over the past 5 and 22 trading days, respectively:

$$\log RV_t^{(w)} = \frac{1}{5}\sum_{j=0}^{4} \log RV_{t-j}^{(d)}, \qquad \log RV_t^{(m)} = \frac{1}{22}\sum_{j=0}^{21} \log RV_{t-j}^{(d)}. \tag{3}$$

The pure HAR model in Eq.(2) provides a natural benchmark for our setting because our central question is whether media-based predictors add value beyond the autoregressive per-

[5] Gonçalves and Meddahi (2011) provide a detailed econometric justification for the log transformation in this context. Its practical importance is evident in our sample: the AIG event of September 2008 generates a maximum daily RV of 17.12% (Table 1), which would exert considerable leverage on a levels-based estimator.

sistence already embedded in realised volatility. The benchmark is deliberately restricted to the autoregressive information set, because our hypothesis concerns the relationship between the informativeness of lagged volatility and the incremental value of media signals, and that relationship is only cleanly defined against a benchmark that contains nothing but lagged volatility. Market-level predictors such as the VIX, earnings-announcement indicators and economic-uncertainty measures are known to improve HAR forecasts out of sample (see, e.g., Chassot & Audrino, 2026); a benchmark that includes them would answer a different question, whether firm-level media signals carry information beyond both lagged volatility and market-level state variables, which we do not address, and since the attention (buzz) measures in particular are likely to co-move with earnings announcements and market-wide stress, nothing below should be read as establishing that sentiment is informative relative to such a richer benchmark.

4.2 Sentiment-augmented specifications

To evaluate whether media-based information improves volatility forecasts, we augment the HAR benchmark with an exogenous component capturing media signals. The general forecasting equation takes the form:

$$\log RV_{t+1} = \underbrace{\alpha + \beta_d \log RV_t^{(d)} + \beta_w \log RV_t^{(w)} + \beta_m \log RV_t^{(m)}}_{\text{HAR benchmark}} + \underbrace{\boldsymbol{\gamma}' \mathbf{F}_t}_{\text{media component}} + \varepsilon_{t+1}, \tag{4}$$

where $\mathbf{F}_t$ is a $K \times 1$ vector of exogenous media predictors and $\boldsymbol{\gamma}$ is the corresponding vector of coefficients. The media component is general: $\mathbf{F}_t$ may contain the raw and engineered features directly (Section 3.1), a sparse subset selected via regularisation, or latent factors extracted through one-stage or two-stage PCA (Section 3.2). This generalisation is what distinguishes the model specifications summarised in Table 3.

We estimate all specifications equation-by-equation for each firm, allowing coefficients to vary across firms while preserving a common model structure. Table 3 summarises the six specifications (in addition to the HAR baseline), which we organise into two groups. The first group, HARSent and HARLASSO, serves as a methodological baseline, contrasting the unrestricted kitchen-sink approach with sparsity-based selection, and is intended primarily to diagnose the consequences of multicollinearity when no dimension reduction is applied. The second group, HARPCA, HAR2PCA, HARmPCA, and HARmPCAOpt, implements progressively structured dimensionality reduction strategies, moving from a single global PCA to hierarchical two-stage extraction based on either data-driven or economically motivated feature clusters.

TABLE 3: HAR MODEL SPECIFICATIONS. This table outlines the seven model specifications estimated in this study. $\mathbf{F}_t$ denotes the vector of exogenous media predictors added to the HAR benchmark. N denotes the total number of engineered media predictors ($N = 27$). For one-stage PCA models, K denotes the number of retained global components. For two-stage models, G denotes the number of feature clusters and $n_g \geq 1$ the number of components retained from cluster g, where $g = 1, \ldots, G$.

Model	**Media component $\mathbf{F}_t$**	**Description**
HARPURE	—	Baseline; no media predictors.
Group 1: Baseline comparators		
HARSENT	All N features	"Kitchen sink" diagnostic; includes all raw and engineered predictors. Illustrates parameter instability under high multicollinearity.
HARLASSO	LASSO-selected subset ($K^* < N$)	Sparsity-based alternative; regularisation selects a small subset of individually predictive features.
Group 2: Dimension-reduction specifications		
HARPCA	K global PCs	One-stage PCA; retains the leading K principal components ($K = 4$, fixed; about 80% of total variance on average across windows).
HAR2PCA	G cluster PC_1s (data-driven)	Two-stage PCA; clusters features via Hierarchical Agglomerative Clustering (HAC) on first-stage PCA loadings, then retains the leading component $F_t^{(g)}$ from each of the G clusters.
HARMPCA	G cluster PC_1s (manual)	Two-stage PCA; uses $G = 6$ groups defined by channel and signal type (Section 3.2) and retains the leading component $F_t^{(g)}$ from each group.
HARMPCAOPT	$\sum_{g=1}^{G} n_g$ cluster PCs (manual)	Two-stage PCA; as HARMPCA but retains the top n_g components $F_{1,t}^{(g)}, \ldots, F_{n_g,t}^{(g)}$ from each group, with n_g fixed per group so that the retained components explain about 80% or more of within-group variance on average (91% over the full sample).

The specifications above define the conditional mean structure of our forecasting framework. However, estimating these equations via OLS captures only the average effect of sentiment on volatility, which risks masking regime-specific dynamics. García (2013) shows that the predictive content of news sentiment for stock returns is concentrated almost entirely during recessions, with negligible predictive power in expansions, suggesting that sentiment's signal is state-dependent rather than uniformly informative. To test for these non-linear tail dynamics without imposing rigid regime-switching assumptions, we extend the framework to quantile regression in Section 4.3.

4.3 Quantile Regression Specification

Quantile regression (Koenker & Bassett, 1978) provides a natural solution. Rather than estimating a single conditional mean, it maps the full conditional distribution of volatility, allowing the contribution of sentiment to vary freely across quantiles. Deveikyte et al. (2022) find that sentiment's influence on volatility varies systematically with the intensity of information flow, with the strongest effects observed during high-attention episodes, consistent with the hypothesis that the signal is concentrated in the upper tail of the volatility distribution rather than distributed uniformly across it. Quantile regression recovers this heterogeneity without requiring ex-ante regime classification or non-linear model architectures.

Conditional correlations between the media metrics and index-level volatility, reported in Appendix D.1 (Table D.1), motivate this step. The correlation between news sentiment and next-day volatility is negligible under low media attention (-0.075) and -0.308 under high

attention, with the same amplification in social media; and the sign of the attention–volatility correlation depends on tone, negative when news tone is positive and positive when social-media tone is negative. These are unconditional, index-level, in-sample correlations that do not control for lagged volatility, and they say nothing about forecast value; what they suggest is that the contribution of the media predictors should be allowed to vary across the conditional distribution of volatility rather than be averaged by least squares.

Formally, the estimator replaces the squared-loss objective of OLS with an asymmetric check function, yielding estimates of the conditional quantiles of the response distribution rather than its conditional mean. For quantile $\tau \in (0,1)$, the model takes the form:

$$Q_\tau(\log RV_{t+1} \mid \mathcal{I}_t) = \underbrace{\alpha(\tau) + \beta_d(\tau)\log RV_t^{(d)} + \beta_w(\tau)\log RV_t^{(w)} + \beta_m(\tau)\log RV_t^{(m)}}_{\text{HAR benchmark}} + \underbrace{\boldsymbol{\gamma}(\tau)'\mathbf{F}_t}_{\text{media component}} + \varepsilon_{t+1}(\tau), \tag{5}$$

where $Q_\tau(\cdot \mid \mathcal{I}_t)$ denotes the conditional τ-quantile given the information set $\mathcal{I}_t$, and all other notation follows Eq. (4). The coefficients $\alpha(\tau)$, $\beta_d(\tau)$, $\beta_w(\tau)$, $\beta_m(\tau)$, and $\boldsymbol{\gamma}(\tau)$ are estimated by minimising the standard check-function objective:

$$\min_{\alpha(\tau),\boldsymbol{\beta}(\tau),\boldsymbol{\gamma}(\tau)} \sum_t \rho_\tau\Big(\log RV_{t+1} - Q_\tau(\log RV_{t+1} \mid \mathcal{I}_t)\Big), \tag{6}$$

where $\rho_\tau(u) = u\big(\tau - \mathbb{I}(u<0)\big)$ is the asymmetric check function and $\mathbb{I}(\cdot)$ is the indicator function. We estimate Eq. (5) across quantiles $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.85, 0.90, 0.95, 0.975\}$ for each firm separately, applying the same model specifications as in Table 3. Goodness of fit in the quantile setting is measured by the $R^1(\tau)$ statistic of Koenker and Machado (1999), the quantile analogue of R^2, defined as one minus the ratio of the minimised check-function objective of the full model to that of a model containing only an intercept, whose fitted value is the unconditional τ-quantile of the dependent variable. We report it adjusted for the number of regressors p, $\bar{R}^1(\tau) = 1 - \big(1 - R^1(\tau)\big)(n-1)/(n-p-1)$, by analogy with the least-squares adjusted R^2, so that specifications with more media components are not favoured mechanically; throughout the paper, the "incremental fit" of a sentiment-augmented specification at quantile τ is the percentage change in this statistic relative to the HARPure benchmark, $100 \times \big[\bar{R}^1_m(\tau) - \bar{R}^1_{\text{HARPure}}(\tau)\big]/\bar{R}^1_{\text{HARPure}}(\tau)$ for specification m, and we refer to it as the incremental pseudo-R^2. It is not the least-squares adjusted R^2 and should not be compared with the in-sample statistics in Table 4. The grid extends to $\tau = 0.975$ so that the extreme upper tail, where the state-dependent role of sentiment is hypothesised to be strongest, is covered.

4.4 Measuring volatility persistence

The conjecture behind the design is that media signals add most when the persistence of volatility is low. The HAR model is built on the assumption of long memory, that is the past realised volatility is the most reliable predictor of its future path. We argue, however, that this picture may change when autoregressive structure weakens. When persistence is lower,

the share of next-day variation that lagged volatility can account for is smaller, and there is correspondingly more scope for an exogenous signal to matter. Whether media signals actually occupy that scope is an empirical question, and the evidence below should be read as a conditional association between persistence and the incremental value of sentiment rather than as a demonstration of a causal mechanism.

To quantify this volatility persistence, we estimate the fractional differencing parameter d for each firm using an autoregressive fractionally integrated moving average (ARFIMA) framework (Hirshleifer et al., 2024; Izzeldin et al., 2019; Varneskov & Perron, 2018). The ARFIMA specification estimates a continuous parameter that captures the intermediate long-memory dynamics characteristic of realised volatility ($0 < d < 0.5$), rather than forcing a choice between short memory ($d = 0$) and a unit root ($d = 1$).

Formally, we estimate the ARFIMA$(0, d, 0)$ model:

$$(1 - L)^d (\log RV_t - \mu) = \varepsilon_t, \tag{7}$$

where L is the lag operator, μ is the unconditional mean of the log-volatility series, and ε_t is a white noise error term. A higher estimated d indicates strong persistence, implying that volatility shocks decay slowly and that past realised volatility remains highly informative. A lower d signals faster mean reversion. We do not equate low persistence with forecast failure of the benchmark. As Varneskov and Perron (2018) document, structural instability can coexist with high estimated persistence, and a low d is compatible with a benchmark that forecasts well; d is a measure of how much of the predictable variation is carried by the past, not a measure of forecast loss. The conjecture we take to the data is the weaker one that the incremental value of media signals is larger, on average, in states in which d is low.

This persistence measure plays a central role in our empirical design, and it enters in two distinct ways that we keep separate throughout. In the in-sample analysis of Section 5.3, d is estimated once per firm over the full sample and used to compare firms in the cross-section: the question there is whether sentiment adds more for *stocks* whose volatility is less persistent. In the out-of-sample analysis of Section 6.1, $d_{i,t}$ is re-estimated for every firm within the 252-day rolling window and firms are ranked cross-sectionally on each day: the question there is whether sentiment adds more on *firm-days* on which recent persistence is low relative to the rest of the panel. Both cuts bear on the same conjecture, but they are not the same evidence and we do not treat them as interchangeable. We use the estimated d series to partition the sample into persistence regimes and evaluate whether the forecasting gains from sentiment-augmented specifications vary systematically across those regimes. If the incremental value of media signals rises when persistence falls, sentiment-augmented models should outperform the HAR benchmark by more in low-persistence states than in high-persistence states; if it does not, the ordering should be flat or reversed.

4.5 Out-of-Sample Evaluation Design

To assess whether the predictive role of sentiment survives beyond in-sample fit, we conduct a rolling out-of-sample forecasting exercise. This is a stricter test: it asks whether the patterns identified in-sample translate into genuine forecasting gains in unseen data, in a setting where overfitting is a natural concern.

Rolling estimation scheme. We employ a fixed rolling window of 252 trading days (approximately one calendar year) to generate one-step-ahead forecasts, with the out-of-sample evaluation period covering January 2004 to December 2016. For each new window, all model parameters, standardisation factors, and principal components are re-estimated using only information available at the time of forecasting, strictly avoiding look-ahead bias. Daily re-estimation follows the first and most important recommendation of Chassot and Audrino (2026). Our window is shorter than the two and a half to four years that they find optimal unconditionally, for two reasons specific to our question, the persistence classification and the decay of media-predictor importance with the lookback, that Appendix E.1 sets out; because the same window is used for every specification, the choice affects the level of forecast losses rather than their comparison. For each firm, the one-step-ahead forecast of log-volatility takes the form:

$$\hat{y}_{t+1} = \hat{\alpha} + \hat{\beta}_d \log RV_t^{(d)} + \hat{\beta}_w \log RV_t^{(w)} + \hat{\beta}_m \log RV_t^{(m)} + \hat{\boldsymbol{\gamma}}' \mathbf{F}_t, \tag{8}$$

consistent with the general specification in Eq. (4), where all quantities are estimated over the current rolling window.

Loss functions. We evaluate forecast accuracy using three loss functions (Patton, 2011). Defining the log-space forecast error as $e_{t+1} = \log RV_{t+1} - \hat{y}_{t+1}$, the mean squared error (MSE) and mean absolute error (MAE) are:

$$\text{MSE} = \frac{1}{T_{\text{oos}}} \sum_{t=1}^{T_{\text{oos}}} e_{t+1}^2, \qquad \text{MAE} = \frac{1}{T_{\text{oos}}} \sum_{t=1}^{T_{\text{oos}}} |e_{t+1}|, \tag{9}$$

where T_{oos} denotes the number of out-of-sample observations. We also report the Quasi-Likelihood (QLIKE) loss, which is robust to extreme observations and has been shown to have superior power for discriminating between volatility forecasts (Diebold & Mariano, 1995; Patton & Sheppard, 2009; West, 1996). Defining the raw-scale forecast as $F_{t+1} = \exp(\hat{y}_{t+1})$, which is the conditional median rather than the conditional mean of RV_{t+1} under a symmetric log-scale error distribution, QLIKE is:

$$\text{QLIKE} = \frac{1}{T_{\text{oos}}} \sum_{t=1}^{T_{\text{oos}}} \left(\frac{RV_{t+1}}{F_{t+1}} - \log\left(\frac{RV_{t+1}}{F_{t+1}} \right) - 1 \right). \tag{10}$$

Because the models are estimated for log volatility, $\exp(\hat{y}_{t+1})$ understates the conditional mean of RV_{t+1} by a retransformation factor (Granger & Newbold, 1976) that differs across specifications. We report QLIKE without the correction, for comparability with the HAR literature, treat agreement between the three loss functions rather than QLIKE alone as the relevant criterion, and show in Appendix E.5 that the correction narrows the regime-conditional gains further; Appendix E.1 discusses the issue.

Regime definitions. Our primary interest is not whether sentiment improves forecasts unconditionally, but whether it does so in the specific market states where the HAR benchmark is least reliable. We therefore evaluate out-of-sample performance both unconditionally and conditional on two regime classifications, defined consistently with the rolling estimation scheme. The classifications serve a diagnostic purpose: they ask in which states the forecasting gains from media signals are located. They are not a real-time switching rule, and we do

not evaluate one. The two classifications differ in how much of their construction is available in real time. The persistence classification uses only information available on day t: $d_{i,t}$ is estimated within the rolling window ending at t and compared with the same day's cross-sectional distribution. The volatility classification used in Tables E.1 and 5 is different in kind: it sorts firm-days by the *realised* volatility of the day being forecast, $\log RV_{i,t+1}$, relative to the firm's full-sample distribution. It therefore conditions on the outcome, not on a state observable at the forecast origin. Sorting forecast errors by the realised outcome is a common descriptive device, but it is not a state that a forecaster can identify at t, and because the forecast error is $\log RV_{i,t+1} - \hat{y}_{i,t+1}$, selecting on a high outcome selects on a positive error: in the high-volatility cell every model under-forecasts on average, and a model whose forecasts happen to be higher in that state is favoured whatever its information content. We therefore also report, in Section 6.2, a real-time version of the classification that uses the volatility observed at the origin, $\log RV_{i,t}$, with full-sample and with trailing thresholds, together with a placebo that isolates the outcome-selection channel. The persistence classification is real-time in both versions.

Volatility regimes. For each firm, we classify the forecast made on day t as belonging to the *high-volatility* regime if the realised target-day volatility $\log RV_{t+1}$ falls at or above the 90th percentile of that firm's log-volatility distribution over the evaluation period, and to the *low-volatility* regime if it falls at or below the 10th percentile; the real-time variant of Section 6.2 applies the same percentiles to $\log RV_t$.

Persistence regimes. On each day t, we compute the cross-sectional distribution of the ARFIMA persistence estimates d_t across all firms, re-estimated within the same 252-day rolling window used for the HAR models. A firm is classified as *high-persistence* on day t if its d_t exceeds the 90th cross-sectional percentile, and *low-persistence* if it falls below the 10th percentile. This cross-sectional approach captures relative persistence rankings on any given day rather than imposing time-invariant thresholds.

These definitions allow us to isolate the *interaction regime*, jointly characterised by high volatility and low persistence, in which we hypothesise sentiment provides the greatest incremental forecasting value. The cut-offs are conventional rather than optimised, and nothing below depends on their exact values in a way that we have tuned; Appendix E.1 discusses the choice, the Monte Carlo evidence on neighbouring cut-offs, and a threshold-free alternative that models the firm-day loss differential directly as a function of persistence and volatility, which is reported in Appendix E.5. Because regime sample sizes vary, we evaluate performance within each regime using the relative loss ratio: the average loss of the sentiment-augmented model divided by the average loss of the HARPure benchmark within the same regime. A ratio below one indicates superior performance relative to the baseline.

Statistical inference. Forecast comparisons are based on the loss differential between each sentiment-augmented specification and the HARPure benchmark, and three points matter for reading the tables; Appendix E.1 develops them. First, because the comparison is between forecasting methods estimated in a fixed-length rolling window, the Diebold–Mariano statistic retains its standard limiting distribution although the benchmark is nested in every augmented specification (Giacomini & White, 2006). Second, all tests are two-sided: the null is equal predictive accuracy, and a rejection in a cell whose relative loss exceeds one indicates that the augmented model is significantly *worse* than the benchmark, not better.

Third, the significance stars in the regime tables come from a pooled test that averages the loss differentials across the stocks in a regime on each day and applies a single Diebold–Mariano test with a Newey–West variance to the daily series, which absorbs the cross-sectional dependence of the loss differentials; Fisher's combination of stock-level p-values, the natural alternative, rejects a true null about 15% of the time at the nominal 5% level under the dependence typical of index constituents (Appendix F) and is not used. The proportions of stocks with a loss ratio below one, which the tables also report, are descriptive summaries of the cross-section, not tests.

5 In-Sample Evidence

This section and the next report the evidence, and the division between them is the organising principle of the paper. The evidence here is in-sample: it measures how much of the variation in realised volatility the media predictors account for on the estimation sample, on average (Section 5.1), across the conditional quantiles of volatility (Section 5.2) and across firms sorted by the persistence of their volatility (Section 5.3), and which predictors carry that fit. None of it measures forecast accuracy. Section 6 asks the separate question of whether one-day-ahead forecasts made with information available at the forecast origin are more accurate with the media predictors than without them, and Table 11 at the end of that section sets out, for every result in both sections, which question it answers and whether it survives the real-time classification of Section 6.2.

5.1 Average Fit

We begin with the benchmark comparison across model specifications. Table 4 reports the in-sample goodness-of-fit of the HARPure baseline and each sentiment-augmented extension, averaged across all DJIA constituents.

Table 4: In-Sample Relative Goodness-of-Fit. This table reports the average percentage improvement of each sentiment-augmented specification over the HARPure baseline, computed as $(\text{Metric}_{\text{model}} - \text{Metric}_{\text{baseline}})/|\text{Metric}_{\text{baseline}}|$ and averaged across all DJIA constituents. Positive $\Delta R^2\%$ and $\Delta\text{AdjR}^2\%$ indicate higher explanatory power; negative ΔAIC% and ΔBIC% indicate improved model efficiency. Statistical significance is assessed using Fisher's combined probability test (Becker, 1994), which aggregates firm-level p-values via the statistic $\chi^2 = -2\sum_i \ln(p_i)$, distributed as χ^2 with $2N$ degrees of freedom under the joint null of no improvement. $^{***}p < 0.001$.

Model	$\Delta R^2\%$	$\Delta\text{AdjR}^2\%$	ΔAIC%	ΔBIC%
HARSent	0.605***	0.325***	−0.004	2.613
HARLasso	0.220***	0.173***	−0.157	0.295
HARPCA	0.132***	0.093***	−0.070	0.291
HAR2PCA	0.124***	0.084***	−0.047	0.314
HARmPCA	0.126***	0.068***	−0.004	0.538
HARmPCAopt	0.387***	0.232***	−0.060	1.384

The first result is straightforward: on average, augmenting the HAR benchmark with media-based predictors produces only modest improvements in explanatory power. Across specifications, the increase in adjusted R^2 is positive and statistically significant but economically limited. The HARSent kitchen-sink model achieves the largest raw gain ($\Delta R^2 = 0.605\%$),

but this apparent advantage is offset by severe parameter instability: the model simultaneously records the largest increase in BIC (ΔBIC $= 2.613$) and a much smaller adjusted gain ($\Delta \text{AdjR}^2 = 0.325\%$), indicating that the raw feature set captures noise rather than a robust signal.[6]

Among the viable specifications, the sparsity-based HARLASSO achieves the sharpest reduction in AIC (ΔAIC $= -0.157\%$), indicating strong parsimony. However, the density-based HARmPCAopt delivers a higher adjusted explanatory gain ($\Delta \text{AdjR}^2 = 0.232\%$ versus 0.173% for LASSO). This ranking is consistent with the hypothesis advanced in Section 3.2 that the incremental information carried by the constructed predictors is spread across the set rather than concentrated in a few variables: isolating a handful of variables via LASSO discards information that PCA-based compression aggregates (Burnham & Anderson, 2002; Yang, 2005). The ranking is descriptive: the differences between specifications are small and are not tested formally in sample (the out-of-sample comparison of Section 6.3 uses a model confidence set for that purpose).

If taken in isolation, these modest average gains could be interpreted as evidence that firm-specific sentiment contains little incremental information beyond lagged realised volatility, consistent with the sceptical strand of the literature (Bodilsen & Lunde, 2025).

5.2 Quantile Variation in Incremental Fit

To examine whether the incremental in-sample fit of the media predictors varies across the conditional distribution of volatility, we estimate quantile regressions of realised volatility on the HAR benchmark and sentiment-augmented specifications, as described in Section 4.3. This allows the contribution of sentiment to differ across the conditional distribution of volatility rather than being restricted to average effects.

[6] Stability diagnostics for HARSent confirm severe multicollinearity. The full-sample mean VIF is 339.5, compared to 2.6 for the PCA-based HARmPCAopt. In rolling-window estimation, HARSent violates the VIF > 10 threshold in 47.9% of windows, against 2.5% for the PCA specification. Refer to Appendix C for more details. We therefore retain HARSent solely as a cautionary diagnostic; all substantive inference is drawn from the dimension-reduction specifications.

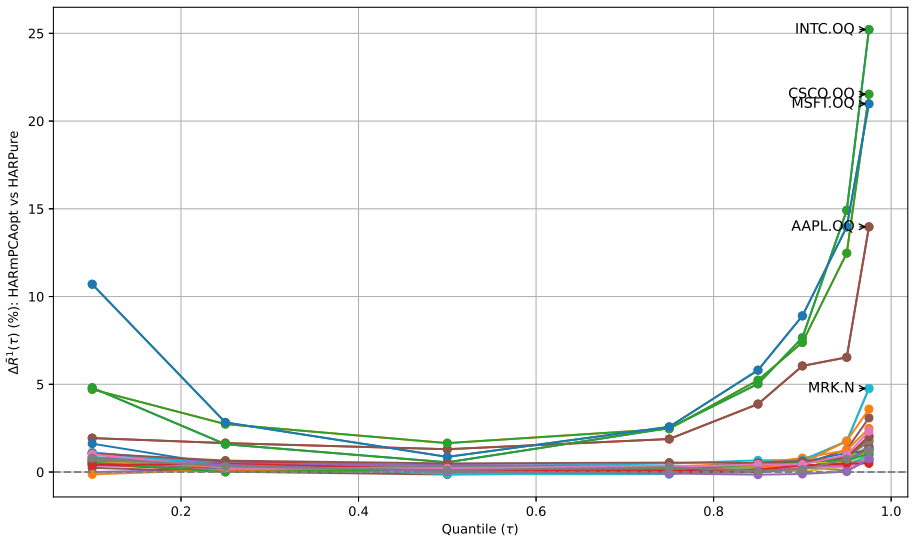


FIGURE 1: QUANTILE-DEPENDENT INCREMENTAL EXPLANATORY POWER. Each line plots the percentage improvement in the quantile pseudo-R^2 of Koenker and Machado (1999) (Section 4.3) of the HARMPCAOPT specification relative to the HARPURE benchmark for each individual DJIA constituent, evaluated across conditional volatility quantiles $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.85, 0.90, 0.95, 0.975\}$. Gains are negligible around the median but rise sharply in the upper tail ($\tau \geq 0.85$), with the strongest incremental fit concentrated in the extreme quantiles ($\tau \in [0.90, 0.975]$). The pronounced cross-sectional heterogeneity is evident: select constituents such as INTC and CSCO exceeding 20% incremental pseudo-R^2 at $\tau = 0.975$ while others remain broadly flat. This motivates the cross-sectional analysis by persistence examined in Section 5.3. The figure is drawn for HARMPCAOPT, the specification that retains the largest number of media components and therefore has the most in-sample flexibility; see Section 6.1 for the relationship between in-sample and out-of-sample rankings across specifications.

Figure 1 shows a clear pattern. The incremental explanatory power of the HARMPCAOPT specification relative to HARPURE is negligible across the lower and middle quantiles of the volatility distribution, but rises sharply and monotonically in the upper tail. Gains remain broadly below 2% around the median ($\tau = 0.5$), but surge in the extreme upper quantiles ($\tau \in [0.90, 0.975]$). This non-linearity is not uniform across constituents: while firms such as Merck (MRK) exhibit a relatively flat response even in the tails, a distinct subset, including Intel (INTC) and Cisco (CSCO), displays a sharply convex increase with incremental pseudo-R^2 exceeding 20% at $\tau = 0.975$.

This result has two implications. First, it explains why unconditional averages understate the role of sentiment: the gains are heavily concentrated in high-volatility tail episodes, which represent a small fraction of total observations. Second, it suggests that media-based predictors add most to in-sample fit when a firm's volatility is in the upper tail of its own distribution. This state should not be equated with market-wide stress: a firm's volatility can be in its upper decile because of an earnings announcement, firm-specific news, or a market-wide episode, and the quantile analysis does not distinguish between these causes. In such states, shifts in narrative tone, disagreement across information channels, and surges in media attention may convey information about evolving uncertainty that has not yet been absorbed into lagged realised volatility. This is a statement about in-sample conditional fit; whether it translates into out-of-sample forecast gains, and for which specifications, is the question of Section 6, and the answer there is that it does not.

The upper-tail evidence establishes that the in-sample fit contribution of sentiment is condi-

tional on state. It does not say anything about persistence. Elevated volatility and weakened persistence often coincide, but they are conceptually distinct. The next subsection examines whether, across firms, the upper-tail gains are larger for stocks whose volatility is less persistent.

5.3 Incremental Fit, Persistence and Predictor Importance

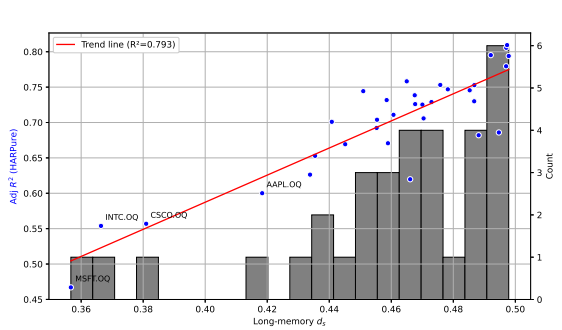
Our central conjecture, set out in Section 4.4, is that sentiment provides more incremental value when the persistence of volatility is low. When persistence weakens, the autoregressive structure becomes less informative, and there is more scope for an exogenous signal to add to it. This subsection examines the conjecture in the cross-section of firms: d is estimated once per firm over the full sample, and each panel of Figure 2 relates a firm-level statistic to the firm's d, so that every relationship reported below is a regression across the 33 constituents. It therefore asks whether sentiment adds more for less persistent *stocks*; the out-of-sample analysis in Section 6.1 asks the separate question of whether it adds more on less persistent *days*.

Figure 2 examines the conjecture by relating the stock-specific fractional integration parameter d, estimated via the ARFIMA$(0, d, 0)$ specification in Eq. (7), to model fit and coefficient estimates at a low and a high conditional quantile.

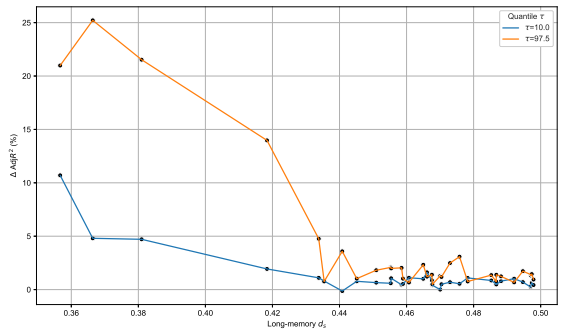
The baseline HAR model fits well for high-persistence stocks and less well for low-persistence ones. Panel A reveals a strong positive relationship ($R^2 = 0.79$) between d and the in-sample adjusted R^2 of the HARPure baseline. Stocks with higher long memory behave more predictably from their own history; for stocks with lower persistence a larger share of next-day variation is left unexplained by the past. This shows that the explanatory power of the HAR benchmark is not constant across firms. It is also, in part, arithmetic: for a fixed set of regressors, less persistence means less explained variation. The interesting question is whether the residual variation is one that media signals can explain, which Panel B addresses.

Incremental sentiment gains are concentrated in those low-persistence stocks, and only at the upper quantile. Panel B plots the incremental pseudo-R^2 of HARmPCAopt relative to HARPure separately at $\tau = 0.10$ (blue) and $\tau = 0.975$ (orange), as a function of d. For stocks with high persistence ($d > 0.44$), sentiment offers negligible incremental fit at either quantile. As persistence falls below approximately 0.42, however, the two curves diverge sharply: gains at the lower quantile remain modest, while incremental explanatory power at the upper quantile rises to over 20% for the lowest-persistence stocks. In the cross-section, then, the in-sample contribution of media signals is largest for the stocks whose volatility is least persistent, and it is largest for those stocks in the upper tail of their volatility distribution. With 33 firms the relationship rests on a small number of low-persistence stocks, several of which (INTC, CSCO) are technology firms, and we do not exclude that industry composition contributes to it.

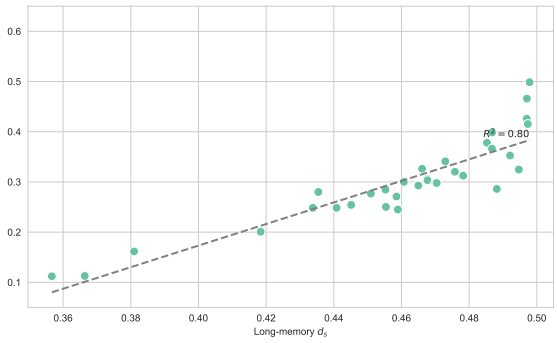
Panels C and D offer a partial account of where this divergence comes from. At the lower quantile ($\tau = 0.10$, Panel C), the short-term HAR coefficient β_d is tightly linked to d ($R^2 = 0.80$): stocks with higher persistence load more heavily on yesterday's volatility. At the upper quantile ($\tau = 0.975$, Panel D), this relationship is largely absent, with R^2 falling to 0.03. At the upper quantile the weight placed on yesterday's volatility is no longer explained by the



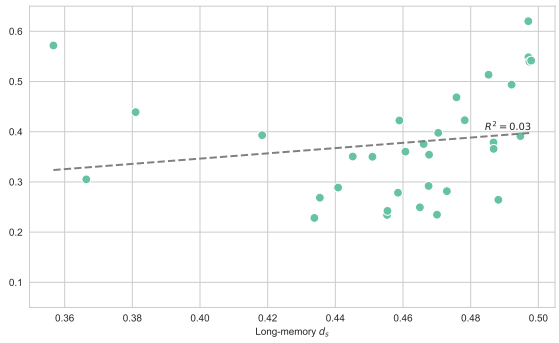
PANEL A: HARPure: Adj-R^2 against d



PANEL B: HARmPCAopt: ΔAdj-R^2% against d



PANEL C: Calm Regimes ($\hat{\beta}_d$ relative to d)



PANEL D: Turbulent Regimes ($\hat{\beta}_d$ relative to d)

FIGURE 2: VOLATILITY PERSISTENCE, MODEL FIT, AND THE BREAKDOWN OF AUTOREGRESSIVE STRUCTURE. Each panel plots a firm-level statistic against the stock-specific ARFIMA fractional integration parameter d (described in Section 4.4), estimated once per firm over the full sample; each panel is a cross-sectional relationship across the 33 constituents, and the R^2 values quoted are those of the fitted line across firms. Panel A: adjusted R^2 of the HARPURE benchmark; the strong positive relationship ($R^2 = 0.79$) shows that the autoregressive model fits best for high-persistence stocks and fits less well as persistence weakens. Panel B: incremental pseudo-R^2 of HARMPCAOPT relative to HARPURE at a low (blue, $\tau = 0.10$) and a high (orange, $\tau = 0.975$) conditional quantile; gains are negligible for high-persistence stocks but exceed 20% at the upper quantile for low-persistence stocks ($d \lesssim 0.42$). Panel C: short-run HAR coefficient β_d at $\tau = 0.10$ against d; the tight relationship ($R^2 = 0.80$) indicates that the daily coefficient tracks long-memory persistence closely at the lower quantile. Panel D: the same relationship at $\tau = 0.975$; the much weaker relationship ($R^2 = 0.03$) indicates that, at the upper quantile, the daily coefficient is no longer aligned with the firm's long-memory persistence.

firm's long-memory persistence, which is consistent with, but does not by itself establish, the view that lagged volatility is a less complete summary of the relevant information on high-volatility days. This evidence is in-sample, cross-sectional and based on 33 firms, and it is compatible with explanations other than a change in the informativeness of lagged volatility, for instance a change in the composition of the high-volatility days themselves.

This pattern is consistent with models of information ambiguity (Epstein & Schneider, 2008; García, 2013), in which the weight that investors place on narrative information rises when the informativeness of the usual signals falls. When persistence is high, historical prices summarise most of the predictable variation and leave little for narrative signals to add. When persistence is low, more of the predictable variation is left for other signals, and narrative information is one candidate. We do not claim that it is the dominant one.

Two variable-importance diagnostics, reported in Appendix D.3, complete the in-sample picture. In a static full-sample decomposition (Figure D.2) the three HAR volatility lags dominate and the media families share the remainder, which is the in-sample counterpart of the modest average gains of Table 4. In rolling windows of three, six and twelve months (Fig-

ure D.3) the ranking is not stable: bursts of sentiment and attention importance periodically overtake the autoregressive components, the importance of the media families decays quickly as the window lengthens, and it collapses in the 2008 crisis itself, when volatility momentum dominated every other signal. These are statements about which predictors carry the in-sample fit and how quickly that changes; they motivate the one-year rolling window of the out-of-sample design, in which the media component is re-estimated daily, and they are not evidence of forecast value.

6 Out-of-Sample Evidence

This section asks whether the in-sample patterns of Section 5 translate into more accurate one-day-ahead forecasts. Everything in it is computed from forecasts made with the 252-day rolling scheme of Section 4.5, re-estimated daily, and compared with the HARPure benchmark by the loss ratios and the pooled tests defined there. Section 6.1 reports the conventional regime-conditional comparison, in which the volatility state is the realised volatility of the target day; Section 6.2 replaces it by a classification available at the forecast origin and shows what produces the difference; Section 6.3 adds model confidence sets, encompassing and a pre-specified combination; Section 6.4 summarises what the evaluation design can and cannot establish.

6.1 Regime-Conditional Forecasting Performance

The regime-conditional comparison changes the unit of analysis relative to Section 5.3: persistence is now measured in real time within the rolling window, and the objects classified are firm-days rather than firms (Section 4.4). Table 5 conditions jointly on the volatility and the persistence state and displays the pattern that the hypothesis predicts. As Section 4.5 explains, the persistence state is measured at the forecast origin but the volatility state is the realised volatility of the target day, $\log RV_{i,t+1}$, an *ex-post* classification; nothing in this subsection should be read as a statement about states a forecaster could identify in advance, and Section 6.2, which repeats the exercise with both states defined at the origin, is where the forecasting question is settled. The univariate counterpart, which conditions on each state separately (Table E.1, Appendix E.2), is a diagnostic. Relative losses there are predominantly above one in every regime, and the largest penalties fall in the high-persistence regimes, where the benchmark is strongest and the augmented specifications add estimation error without information. A univariate regime pools firm-days across the other dimension and averages over a different set of contributing stocks, so it is not an aggregation of the joint cells.

Table 5 isolates the four bivariate regimes. In the *high-volatility, low-persistence* regime, three of the five specifications, HAR2PCA, HARPCA and HARmPCA, have relative losses below one under all three loss functions, between 0.988 and 0.993; HARLASSO is below one under MSE and MAE but not under QLIKE (1.004), and HARmPCAOpt is above one throughout (1.020, 1.005 and 1.030). For HAR2PCA the QLIKE ratio of 0.988 corresponds to a 1.2% reduction in average loss relative to HARPure, holding for 8 of the 13 contributing stocks. In the other three regimes the picture is one of losses or of no difference: every specification is above one in the *high-volatility, high-persistence* regime (1.004 to 1.056), HAR2PCA and HARPCA are within 0.003 of one in the *low-volatility, high-persistence*

TABLE 5: OUT-OF-SAMPLE PERFORMANCE: VOLATILITY–PERSISTENCE INTERACTION REGIMES. This table reports average relative loss ratios for the four bivariate interaction regimes defined by jointly conditioning on the volatility and persistence classifications described in Section 4.5. Ratios below 1.0 favour the sentiment-augmented model; ratios above 1.0 favour the HARPURE benchmark. Cell shading encodes the direction and magnitude of the relative performance: darker cells (< 0.990) indicate economically meaningful outperformance; lighter cells (0.990–0.999) indicate modest outperformance; unshaded cells (≥ 1.000) indicate underperformance. The high-volatility, low-persistence regime (bold rows) is the primary interaction of interest. Ratios are computed stock by stock and averaged across the stocks that have at least 20 firm-days; the contributing sets differ across the four regimes and are smaller than in Table E.1. Parentheses report the proportion of contributing stocks for which the extended model yields lower loss. Stars report the two-sided pooled Diebold–Mariano test of Section 4.5, which averages the loss differential across the stocks in the regime on each day and applies a single test with a Newey–West variance (five lags) to the daily series, so that cross-sectional dependence is absorbed; a starred cell with a ratio above 1.0 indicates significantly *worse* performance than the benchmark. Fisher's combination of the stock-level tests, the natural alternative, overstates significance under cross-sectional dependence (Appendix F) and is not used here. The proportions remain descriptive. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	HARPURE	HAR2PCA	HARPCA	HARLASSO	HARMPCA	HARMPCAOPT
		Rel. Loss (Proportion of stocks outperforming)				
Loss	Rel. Loss					
High volatility & high persistence						
MSE	1.0	1.025 (35.3%)	1.022 (35.3%)	1.029 (35.3%)	1.021 (35.3%)	1.056 (23.5%**)
MAE	1.0	1.009 (41.2%)	1.008 (35.3%)	1.010 (29.4%*)	1.004 (35.3%)	1.025 (41.2%**)
QLIKE	1.0	1.013 (35.3%)	1.009 (47.1%)	1.011 (41.2%)	1.006 (29.4%)	1.035 (23.5%)
High volatility & low persistence — primary interaction regime						
MSE	**1.0**	**0.990** (53.8%**)	0.991 (61.5%**)	0.999 (46.2%)	0.991 (46.2%**)	1.020 (53.8%)
MAE	**1.0**	**0.993** (61.5%***)	0.990 (69.2%***)	0.991 (53.8%)	0.993 (53.8%***)	1.005 (53.8%**)
QLIKE	**1.0**	**0.988** (61.5%)	0.992 (69.2%)	1.004 (46.2%)	0.992 (46.2%)	1.030 (46.2%)
Low volatility & high persistence						
MSE	1.0	1.000 (45.5%)	0.998 (50.0%)	1.021 (40.9%**)	1.013 (45.5%)	1.038 (31.8%**)
MAE	1.0	0.999 (59.1%)	0.997 (63.6%)	1.004 (50.0%)	1.003 (40.9%)	1.013 (36.4%)
QLIKE	1.0	0.999 (54.5%)	0.998 (54.5%)	1.015 (45.5%**)	1.010 (36.4%)	1.031 (31.8%**)
Low volatility & low persistence						
MSE	1.0	1.020 (35.7%)	1.018 (42.9%)	1.014 (57.1%)	1.033 (28.6%)	1.019 (64.3%)
MAE	1.0	1.001 (35.7%)	1.000 (50.0%)	0.992 (71.4%)	1.004 (42.9%)	0.988 (64.3%)
QLIKE	1.0	1.012 (28.6%)	1.011 (35.7%)	1.005 (57.1%)	1.022 (42.9%)	1.009 (71.4%)

regime while the other three lose, and no specification is below one under all three loss functions in the *low-volatility, low-persistence* regime. The gains are therefore not a property of low persistence as such, nor of high volatility as such (Table E.1); they appear only in the cell that combines a high realised volatility with low persistence, and they are small: a forecaster using HAR2PCA throughout would see no improvement on average, and the gain in the one favourable cell is of the order of one per cent of the loss in that cell. Whether the location survives a classification available at the forecast origin is the question of Section 6.2, and the answer is that it does not.

The specification that carries the in-sample analysis, HARmPCAopt, is the one specification that does not gain in the high-volatility, low-persistence regime (1.020, 1.005 and 1.030). The two facts are consistent: HARmPCAopt retains the largest number of media components and therefore has the most in-sample flexibility, which is why it shows the largest in-sample gains in Table 4 and the sharpest quantile pattern in Figures 1 and 2; in a 252-day rolling window the same flexibility costs it more in estimation error than any state-dependent signal returns, while the specifications that gain in this cell all enter fewer media terms. Figure D.1 (Appendix D.2) reports the quantile analysis for HAR2PCA: the same shape with smaller gains (median incremental fit 0.5% at $\tau = 0.975$, against 1.4% for HARmPCAopt; largest 16.3% for CSCO, against 25.2%). The *low-volatility, low-persistence* regime, the cell in which a "sparse-information" story would predict gains for the LASSO specification, shows none: no specification is below one under all three loss functions, so we do not claim that a sparse specification performs best in calm states, nor that media information is "dense" in some states and "sparse" in others. The rankings of specifications within a cell rest on 13 to 22 stocks and are descriptive.

A regime-specific loss reduction of about one per cent is modest, and it is not earned evenly: the cumulative squared-error comparison for Apple Inc. in Appendix E.3 (Figure E.1) ends above zero for all five specifications but is negative through 2004 and rises from 2005, and across the panel the pooled HAR2PCA MSE ratio in this cell is below one in 2005 (0.94), 2010 and 2011 (0.98), close to one in 2008 (1.00) and above one in 2009 (1.02), with most of the cell's firm-days falling in 2008–2009 (779 of 1,487) and 2004–2006 (427).

Taken at face value, Table 5 would support a conditional and modest version of the paper's claim: media-based factor summaries reduce forecast loss by about one per cent in firm-days that combine a high realised volatility with low persistence, and nowhere else. The next subsection shows that this reading does not survive two things that any user of a regime-conditional forecast comparison should ask for: a classification that uses only information available at the forecast origin, and a formal test that respects the cross-sectional dependence of the loss differentials.

6.2 Real-Time Classification and Formal Tests

Section 4.5 noted that the volatility classification in Table 5 uses the realised volatility of the day being forecast. Table 6 repeats the four-regime comparison with the volatility regime defined on the volatility observed at the forecast origin, $\log RV_{i,t}$, with the firm's full-sample percentiles (Panel A) and with trailing percentiles computed from the firm's history up to $t-1$ (Panel B); the persistence regime is unchanged and real-time in both. The high-volatility, low-persistence cell of Panel A has almost the same number of firm-days as its counterpart

in Table 5 (1,470 against 1,487) and the same thirteen contributing stocks, but only 53% of the firm-days are common to the two, and on it the gains disappear: the HAR2PCA ratios are 1.021, 1.002 and 1.010 under MSE, MAE and QLIKE, HARPCA is at 1.012, 0.994 and 1.003, and no specification is below one under all three loss functions. With trailing thresholds (Panel B) every cell is above one for every specification. The same holds across the real-time counterpart of the cut-off grid (Appendix E.5), where the HAR2PCA QLIKE ratio in this cell lies between 1.002 and 1.020 over nine cut-off combinations and the lowest value for any specification is 0.984, with an MSE ratio of 1.001. It also holds when the state is defined on the benchmark's own forecast of $\log RV_{i,t+1}$, so that "high volatility" means *expected* high volatility: 1.006 for HAR2PCA under both losses. There is, in short, no definition of a high-volatility state available at the forecast origin under which the media-augmented specifications show a gain that is consistent across loss functions or supported by the pooled tests of Table 7; the few ratios below one in Table 6 are isolated and small.

TABLE 6: OUT-OF-SAMPLE PERFORMANCE UNDER REAL-TIME VOLATILITY CLASSIFICATION. As Table 5, but with the volatility regime defined on the volatility observed at the forecast origin, $\log RV_{i,t}$, rather than on the realised target-day volatility $\log RV_{i,t+1}$. Panel A compares $\log RV_{i,t}$ with the firm's full-sample 90th and 10th percentiles; Panel B with the percentiles of the firm's expanding history up to $t-1$ (first 250 forecast days excluded). The persistence regime is unchanged (top and bottom cross-sectional deciles of the rolling ARFIMA $d_{i,t}$). Entries are cross-firm averages of stock-level relative losses with, in parentheses, the proportion of contributing stocks (at least 20 firm-days) with a ratio below one; no significance stars are shown because the combined test is descriptive (Section 4.5); pooled tests are in Table 7. The Panel A high-volatility, low-persistence cell shares 53% of its firm-days with the corresponding cell of Table 5; the Panel B cell shares 58%.

		Firm-days	Firms	HAR2PCA	HARPCA	HARLASSO	HARMPCA	HARMPCAOPT
Panel A: origin-day volatility, full-sample thresholds								
High volatility & high persistence	MSE	1,501	18	1.048 (28%)	1.039 (33%)	1.064 (17%)	1.048 (39%)	1.113 (22%)
	MAE			1.016 (28%)	1.012 (33%)	1.018 (22%)	1.012 (28%)	1.049 (22%)
	QLIKE			1.035 (33%)	1.025 (33%)	1.034 (28%)	1.024 (39%)	1.103 (22%)
High volatility & low persistence	MSE	1,470	13	1.021 (54%)	1.012 (46%)	1.037 (31%)	1.024 (46%)	1.080 (23%)
	MAE			1.002 (38%)	0.994 (54%)	1.006 (31%)	1.001 (46%)	1.030 (31%)
	QLIKE			1.010 (38%)	1.003 (54%)	1.034 (38%)	1.006 (54%)	1.085 (38%)
Low volatility & high persistence	MSE	1,357	23	1.003 (52%)	1.000 (48%)	1.038 (30%)	1.021 (30%)	1.069 (17%)
	MAE			1.002 (43%)	1.000 (43%)	1.021 (26%)	1.008 (43%)	1.040 (13%)
	QLIKE			1.000 (48%)	0.997 (52%)	1.036 (39%)	1.016 (30%)	1.069 (22%)
Low volatility & low persistence	MSE	1,070	13	1.027 (46%)	1.024 (38%)	1.029 (31%)	1.040 (23%)	1.080 (15%)
	MAE			1.006 (31%)	1.006 (38%)	1.015 (38%)	1.015 (38%)	1.028 (23%)
	QLIKE			1.029 (31%)	1.028 (23%)	1.034 (23%)	1.048 (23%)	1.097 (23%)
Panel B: origin-day volatility, trailing thresholds								
High volatility & high persistence	MSE	2,074	19	1.037 (21%)	1.028 (21%)	1.056 (32%)	1.037 (37%)	1.087 (21%)
	MAE			1.013 (21%)	1.009 (26%)	1.020 (32%)	1.011 (32%)	1.039 (16%)
	QLIKE			1.029 (26%)	1.020 (21%)	1.033 (37%)	1.023 (32%)	1.093 (21%)
High volatility & low persistence	MSE	910	10	1.041 (30%)	1.032 (30%)	1.049 (20%)	1.041 (20%)	1.063 (10%)
	MAE			1.015 (30%)	1.009 (50%)	1.019 (20%)	1.008 (40%)	1.028 (10%)
	QLIKE			1.013 (50%)	1.008 (50%)	1.026 (50%)	1.002 (50%)	1.052 (30%)
Low volatility & high persistence	MSE	1,586	23	1.021 (30%)	1.018 (35%)	1.043 (22%)	1.036 (30%)	1.114 (4%)
	MAE			1.013 (30%)	1.011 (35%)	1.026 (22%)	1.016 (30%)	1.061 (13%)
	QLIKE			1.026 (35%)	1.022 (35%)	1.044 (43%)	1.039 (35%)	1.119 (22%)
Low volatility & low persistence	MSE	1,825	15	1.030 (27%)	1.030 (33%)	1.044 (20%)	1.034 (40%)	1.082 (20%)
	MAE			1.007 (27%)	1.008 (33%)	1.023 (20%)	1.011 (27%)	1.027 (27%)
	QLIKE			1.030 (20%)	1.029 (33%)	1.036 (20%)	1.036 (33%)	1.075 (20%)

Table 7 reports the formal tests. Panel A applies the pooled Diebold–Mariano test, which averages the loss differential across the stocks in a regime on each day and so absorbs the cross-sectional dependence of the stock-level comparisons (Section 4.5), to the ex-post classification. In the high-volatility, low-persistence cell the pooled statistics for HAR2PCA, HARPCA and HARMPCA are negative and, under MSE, significant at the 5% level (-2.4, -2.3 and -2.1; the MAE statistics, omitted from the table, lie between -3.0 and -3.4); under QLIKE they are not (-0.9, -0.7 and -0.8). This is the one place in the paper where a test that

respects cross-sectional dependence still favours the augmented specifications, and it does so only under the outcome-based classification and only for the squared-error family of losses. Panel B applies the same test to the real-time classification: no statistic in any regime is significantly negative, the high-volatility, low-persistence QLIKE statistics are between −1.0 and −1.3 for the three factor specifications, and the significant statistics in the table are positive, that is, they identify regimes in which the augmented specifications are significantly worse. The Giacomini–White test of conditional predictive ability with the real-time regime indicator as instrument (Table E.2, Appendix E.4) rejects equal conditional accuracy for every specification under MSE, but the rejection is driven entirely by the firm-days outside the regime, on which the augmented specifications are significantly worse; inside the regime no mean loss differential differs significantly from zero. A threshold-free regression of the firm-day loss differential on persistence, origin-day volatility and their product (Appendix E.5, Table E.6) says the same thing in continuous form: no coefficient, and in particular no interaction coefficient, is significant for any specification or loss function.

TABLE 7: POOLED DIEBOLD–MARIANO TESTS BY REGIME. Each entry is the pooled Diebold–Mariano t-statistic for one specification, loss function and regime: the loss differential (augmented minus HARPURE) is averaged across the stocks in the regime on each day and a single Diebold–Mariano test with a Newey–West variance (five lags) is applied to the resulting daily series (616 to 882 days per regime, 3,269 unconditionally). Negative values favour the augmented specification; the tests are two-sided, so a significant positive value identifies a regime in which the augmented specification is *worse*. Cross-sectional dependence is absorbed by the daily averaging (Appendix F). Panel A uses the ex-post classification of Table 5; Panel B the real-time classification of Table 6, Panel A. The Giacomini–White conditional test with the real-time regime indicator is reported in Appendix E.4 (Table E.2). The entry marked n.c. is not computable: the unconditional QLIKE loss of HARMPCAOPT is dominated by a single firm-day (GE, 24 January 2011) on which its forecast of $\log RV$ was −49.9 against −8.7 for the benchmark, a failure of the kind that the stability diagnostics of Appendix C anticipate; that firm-day lies outside every regime cell and does not affect Tables E.1 and 5.

Specification	Loss	High vol. & high pers.	High vol. & low pers.	Low vol. & high pers.	Low vol. & low pers.	Unconditional
Panel A: ex-post classification (Table 5)						
HAR2PCA	MSE	+0.80	-2.44	+0.48	+1.62	+2.60
	QLIKE	-0.70	-0.94	+0.38	+1.47	+6.25
HARPCA	MSE	+0.91	-2.29	+0.43	+0.69	+11.7
	QLIKE	-0.79	-0.67	+0.32	+0.60	+6.87
HARLASSO	MSE	+1.41	-1.06	+2.18	-0.45	+19.4
	QLIKE	-0.16	-0.66	+1.98	-0.61	+9.51
HARMPCA	MSE	+0.92	-2.11	+1.58	+0.74	+16.4
	QLIKE	-0.70	-0.76	+1.36	+0.54	+8.03
HARMPCAOPT	MSE	+2.06	-0.82	+2.22	-0.11	+2.24
	QLIKE	+1.20	+0.50	+2.12	-0.31	n.c.
Panel B: real-time classification (Table 6, Panel A)						
HAR2PCA	MSE	+1.52	+0.07	+0.52	+2.65	+2.60
	QLIKE	+0.47	-1.04	+0.25	+1.80	+6.25
HARPCA	MSE	+1.49	-0.19	+0.56	+2.40	+11.7
	QLIKE	+0.38	-1.26	+0.22	+1.86	+6.87
HARLASSO	MSE	+1.94	+1.76	+2.83	+2.31	+19.4
	QLIKE	+1.09	+0.63	+1.66	+2.58	+9.51
HARMPCA	MSE	+1.55	+0.60	+2.89	+2.06	+16.4
	QLIKE	+0.28	-1.32	+2.27	+1.38	+8.03
HARMPCAOPT	MSE	+2.64	+1.99	+4.32	+4.04	+2.24
	QLIKE	+2.38	+0.79	+3.00	+2.51	n.c.

Why, then, does Table 5 show gains? Table 8 isolates the channel. Over low-persistence firm-days the three factor specifications forecast $\log RV_{i,t+1}$ higher than the benchmark by a small constant, about 0.004 for HAR2PCA and HARPCA and 0.003 for HARMPCA; HARLASSO and HARMPCAOPT forecast lower. The ex-post high-volatility cell selects days on which the benchmark under-forecasts by 0.79 log units on average, the ex-post low-volatility cell days on which it over-forecasts by 0.68. A forecast equal to the benchmark

plus the constant c on low-persistence firm-days, which contains no information beyond the benchmark, reproduces the sign pattern of Table 5 for every specification: for HAR2PCA it gives 0.993 in the high-volatility, low-persistence cell (for all thirteen stocks) and 1.009 in the low-volatility, low-persistence cell (for none), against the actual 0.988 and 1.012, while a forecast equal to the benchmark plus independent noise of the same dispersion is above one in every cell. More than half of the headline gain is therefore the arithmetic of selecting on the outcome; the remainder is small, and whether it registers as significant depends on the loss function (Table 7, Panel A: yes under MSE and MAE, no under QLIKE), which is not the behaviour of a reliable effect. The in-sample quantile evidence of Section 5 is not affected by this argument, since conditional-quantile fit is its object, but it does not translate into robust out-of-sample forecast gains for the augmented models in any state that can be identified in real time.

TABLE 8: THE LEVEL-SHIFT CHANNEL BEHIND THE EX-POST REGIME PATTERN. For each specification, c is the mean difference between the augmented and the HARPURE forecast of $\log RV_{i,t+1}$ over low-persistence firm-days (and the share of those firm-days on which the augmented forecast is the higher of the two). The "actual" columns repeat the cross-firm average relative losses of Table 5; the "placebo" columns replace the augmented forecast by the benchmark forecast plus the constant c on low-persistence firm-days and nothing elsewhere, a forecast that contains no information beyond the benchmark. The last row replaces the augmented forecast by the benchmark plus independent noise with the standard deviation of the HAR2PCA–HARPURE forecast difference. Because the ex-post high-volatility cell selects days on which the benchmark under-forecasts (mean error +0.79 in log units) and the low-volatility cell days on which it over-forecasts (mean error −0.68), a forecast shifted upward by c wins in the first and loses in the second whatever its information content.

			High vol. & low pers.		Low vol. & low pers.		All low persistence	
Specification	c	Share > 0	actual	placebo	actual	placebo	actual	placebo
HAR2PCA	+0.0041	53%	0.988	0.993	1.012	1.009	1.017	0.997
HARPCA	+0.0040	52%	0.992	0.993	1.011	1.009	1.013	0.997
HARLASSO	-0.0012	50%	1.004	1.002	1.005	0.997	1.030	1.001
HARMPCA	+0.0027	51%	0.992	0.995	1.022	1.006	1.027	0.998
HARMPCAOPT	-0.0040	49%	1.030	1.007	1.009	0.991	1.086	1.003
Benchmark + noise (s.d. 0.098)	0	50%		1.022		1.026		1.017

The channel can be stated exactly. Let $e_b = \log RV_{t+1} - \hat{y}^b_{t+1}$ be the benchmark's error and let the competing forecast be $\hat{y}^b_{t+1} + c$ for a constant c. Then the squared-error differential is $(e_b - c)^2 - e_b^2 = c^2 - 2c\,e_b$, and on any set of firm-days S,

$$\frac{E\big[(e_b - c)^2 \mid S\big]}{E\big[e_b^2 \mid S\big]} \;=\; 1 + \frac{c^2 - 2c\,E[e_b \mid S]}{E[e_b^2 \mid S]}. \tag{11}$$

A shifted forecast with no information therefore beats the benchmark on exactly those sets on which the benchmark's mean error has the sign of c, by an amount proportional to that mean error. Selecting on the realised outcome makes $E[e_b \mid S]$ large by construction (+0.79 in the top decile of realised volatility among low-persistence firm-days, −0.68 in the bottom decile); with $c = 0.0041$ and $E[e_b^2 \mid S] = 1.28$, Eq. (11) gives 0.9949 for the high-volatility cell and 1.0091 for the low-volatility cell, against 0.994 and 1.010 for the placebo in Table 8. Appendix E.5 applies the same comparison to sets selected in other ways (Table E.3): selecting a period identified ex post, the NBER recession, or the origin-day volatility leaves the mean benchmark error an order of magnitude smaller (+0.04 and +0.02) and the placebo within 0.001 of one, so period-based and real-time conditioning are exposed to the same channel

only to a negligible degree. The level difference itself is documented in Table E.4: on low-persistence firm-days the three factor specifications over-forecast by slightly more than the benchmark, so that their upward shift is, on average, in the wrong direction. Varying the cut-offs of the ex-post classification leaves the high-volatility, low-persistence ratios of the factor specifications below one at most combinations, as the channel predicts, and correcting QLIKE for retransformation bias narrows the gains further (Appendix E.5, Table E.5).

The conclusion of this subsection is negative. Augmenting a daily re-estimated HAR model with firm-level media predictors does not robustly improve one-day-ahead forecasts of realised volatility for DJIA constituents over 2004–2016, on average, in any univariate regime, or in any volatility–persistence regime that can be identified at the forecast origin; in most states it worsens them by one to three per cent. The regime-conditional gains reported in Table 5 arise from conditioning on the realised outcome, which favours any forecast that is shifted in the direction of the realised surprise. Under that classification the pooled tests of Table 7, Panel A, still reject equal accuracy under MSE and MAE for HAR2PCA, HARPCA and HARmPCA (t-statistics between -2.1 and -3.4) but not under QLIKE, so what the pooled tests remove is the breadth of the evidence rather than every star; what removes the gain is the real-time classification, under which no statistic in Panel B is significantly negative. A gain that is visible only after the outcome is known is not an advantage a forecaster could have used, whatever its significance level, and that is the sense in which the pattern is an artefact.

6.3 Model Confidence Sets, Encompassing and Combination

The evidence so far compares each augmented specification with the benchmark one at a time. Three further questions arise: whether any of the six augmented specifications can be distinguished from the benchmark once the search over specifications is accounted for; whether the augmented forecasts carry information that the benchmark does not, even if they use it badly; and whether the negative result is stable over time and across reasonable variations of the design. Table 9 addresses the first. The model confidence set (Hansen et al., 2011), computed stock by stock over the seven specifications, contains HARPure at the 90% level for every one of the 33 stocks under both loss functions, and the augmented specifications are eliminated for between 18 and 91% of the stocks under MSE and between 18 and 73% under QLIKE, the kitchen-sink and the richest factor specification most often; on the pooled daily losses HARPure is the last model standing under both loss functions. The test for superior predictive ability of Hansen (2005), which asks whether the best of the augmented specifications beats the benchmark after allowing for the fact that the best of six was chosen, has a consistent p-value of one, stock by stock and pooled, because no augmented specification has even a positive mean loss differential; within the high-volatility, low-persistence cell the pooled SPA p-values are 0.54 (real-time) and 0.64 (ex post) under MSE and 0.57 and 0.68 under QLIKE. The reality-check argument of White (2000) therefore cuts in the usual direction: there is nothing to correct, because nothing is significant to begin with.

Whether the unconditional result hides periods in which the augmented specifications did better, the "pockets of predictability" question that Odendahl et al. (2023) formalise and that the fluctuation test of Giacomini and Rossi (2010) answers without specifying the state,

TABLE 9: MODEL CONFIDENCE SETS AND THE TEST FOR SUPERIOR PREDICTIVE ABILITY. Panel A reports, for the model confidence set (MCS) of Hansen et al. (2011) computed stock by stock over the seven specifications (HARPURE, the five augmented specifications of Table 5 and the kitchen-sink HARSENT), the share of the 33 stocks for which each specification is in the 90% and 75% MCS and the median MCS p-value; the $T_{\max}$ statistic with a circular block bootstrap (block length 15, 1,000 draws) is used. Panel B reports the MCS p-values on the pooled daily losses (averaged across stocks) unconditionally and within the high-volatility, low-persistence cell under the real-time and the ex-post classification (2,000 draws); HARMPCAOPT and HARSENT are excluded from the pooled QLIKE sets because single forecast failures (Table 7) make their pooled QLIKE losses uninformative. Panel C reports the consistent p-value of the test for superior predictive ability (SPA) of Hansen (2005) for the null that no augmented specification beats HARPURE, stock by stock (share of stocks rejecting at 10% and 5%, median p) and pooled. HV & LP denotes the high-volatility, low-persistence cell under the real-time (Table 6, Panel A) or the ex-post (Table 5) classification.

	MSE			QLIKE		
Panel A: stock-level MCS						
Specification	In 90% MCS	In 75% MCS	Median p	In 90% MCS	In 75% MCS	Median p
HARPURE	100%	100%	1.00	100%	100%	1.00
HAR2PCA	42%	21%	0.09	82%	45%	0.23
HARPCA	45%	21%	0.09	82%	55%	0.27
HARLASSO	15%	6%	0.03	52%	27%	0.10
HARMPCA	27%	12%	0.06	67%	33%	0.17
HARMPCAOPT	12%	6%	0.00	30%	12%	0.07
HARSENT	9%	3%	0.00	27%	12%	0.03
Panel B: pooled MCS p-values						
Specification	Uncond.	HV & LP (real-time)	HV & LP (ex post)	Uncond.	HV & LP (real-time)	HV & LP (ex post)
HARPURE	1.00	0.86	0.22	1.00	0.35	0.71
HAR2PCA	0.12	0.86	0.77	0.00	0.52	1.00
HARPCA	0.12	1.00	0.77	0.00	0.74	0.84
HARLASSO	0.12	0.17	0.77	0.00	0.10	1.00
HARMPCA	0.12	0.57	1.00	0.00	1.00	1.00
HARMPCAOPT	0.12	0.17	0.77	–	–	–
HARSENT	0.02	0.17	0.01	–	–	–
Panel C: SPA test, H_0: no augmented specification beats HARPURE						
	Share $p < 0.10$	Share $p < 0.05$	Median p	Share $p < 0.10$	Share $p < 0.05$	Median p
Stock by stock	0%	0%	1.00	0%	0%	1.00
	Uncond.	HV & LP (real-time)	HV & LP (ex post)	Uncond.	HV & LP (real-time)	HV & LP (ex post)
Pooled p-value	1.00	0.54	0.64	1.00	0.57	0.68

is examined in Appendix E.6 (Table E.7): the rolling standardised loss differential never falls below zero by a significant margin for any specification under either loss function, in windows of 30% or of 10% of the sample, and the years in which the ex-post regime cell showed year-by-year gains do not stand out once all firm-days are included. There are no pockets.

The encompassing regressions in Panel A of Table 10 give the one result in this section that is not simply negative. Regressing the benchmark's error on the difference between the augmented and the benchmark forecast yields a slope between 0.06 and 0.16 unconditionally, with t-statistics between 2.4 and 7.0; on low-persistence firm-days the slope is about 0.2, and in the real-time high-volatility, low-persistence cell it is between 0.23 and 0.48 with t-statistics above 2.4. The benchmark does not encompass the augmented forecasts: their deviation from the benchmark is correlated with the benchmark's error, and the optimal linear combination would put a weight of roughly one tenth on it unconditionally and nearly one half in the real-time high-volatility, low-persistence cell. That the augmented forecasts nevertheless lose to the benchmark at weight one is the familiar result that a predictor with modest content and substantial estimation noise is worth including only at a fraction of its estimated weight.

Whether the fraction can be chosen in real time is the question of Panel B, which was specified once before any result was seen: the combination weight is the no-intercept least-squares slope on the preceding 252 forecast days, truncated to $[0, 1]$. Unconditionally the combination is slightly worse than the benchmark under both loss functions (pooled Diebold–Mariano statistics between +1.0 and +3.6); the estimated weight is zero on about 40% of firm-days and averages 0.12 to 0.23. In the real-time high-volatility, low-persistence cell the combination is better than the benchmark under QLIKE by about 2%, with pooled statistics between −2.0 and −2.2 for four of the five specifications, and by about 1% under MSE without significance. That is one cell, one loss function, significance at the 5% level, in an exercise run once, and against an unconditional loss. It is consistent with the encompassing evidence and it is the only real-time gain in the paper; it is not strong enough to carry a claim, and we do not build on it.

TABLE 10: FORECAST ENCOMPASSING AND A PRE-SPECIFIED ROLLING COMBINATION. Panel A reports the slope λ in the encompassing regression $\log RV_{i,t+1} - \hat{y}^{b}_{i,t+1} = a + \lambda\,(\hat{y}^{m}_{i,t+1} - \hat{y}^{b}_{i,t+1}) + u_{i,t+1}$, where $\hat{y}^{\,b}$ is the HARPURE forecast and $\hat{y}^{\,m}$ the augmented forecast, estimated on the firm-day panel with standard errors clustered by day (five Newey–West lags); $\lambda = 0$ means that the benchmark encompasses the augmented forecast, and λ is the weight on the augmented forecast in the optimal linear combination. Panel B evaluates the combination $\hat{y}^{c}_{i,t+1} = \hat{y}^{b}_{i,t+1} + w_{i,t}(\hat{y}^{m}_{i,t+1} - \hat{y}^{b}_{i,t+1})$ with $w_{i,t}$ the no-intercept least-squares slope on the preceding 252 forecast days, truncated to $[0, 1]$ (so the first 252 days of each stock are lost); entries are the pooled relative loss of the combination against the benchmark with the pooled Diebold–Mariano statistic in parentheses. The exercise was specified once, before any result was seen, and is reported in full. HV & LP denotes the high-volatility, low-persistence cell under the real-time (Table 6, Panel A) or the ex-post (Table 5) classification.

Panel A: encompassing slope λ (t-statistic)

Specification	Unconditional	Low persistence	HV & LP (real-time)	HV & LP (ex post)
HAR2PCA	0.100 (+2.93)	0.172 (+2.98)	0.441 (+3.66)	0.206 (+1.61)
HARPCA	0.162 (+7.00)	0.224 (+3.85)	0.478 (+4.04)	0.173 (+1.39)
HARLASSO	0.112 (+6.61)	0.210 (+5.20)	0.234 (+2.40)	0.354 (+3.01)
HARMPCA	0.132 (+6.91)	0.202 (+4.02)	0.391 (+3.13)	0.294 (+2.27)
HARMPCAOPT	0.056 (+2.35)	0.180 (+6.01)	0.295 (+3.74)	0.296 (+2.93)

Panel B: rolling combination, relative loss (pooled DM)

Specification	Mean w (share $w = 0$)	Unconditional MSE / QLIKE	Low persistence MSE	HV & LP (real-time) MSE / QLIKE
HAR2PCA	0.22 (43%)	1.0021 (+2.95) / 1.0027 (+1.37)	1.0008 (+0.53)	0.9895 (-1.43) / 0.9784 (-2.04)
HARPCA	0.23 (42%)	1.0019 (+2.77) / 1.0026 (+1.60)	1.0007 (+0.41)	0.9913 (-0.96) / 0.9790 (-1.96)
HARLASSO	0.16 (39%)	1.0018 (+3.63) / 1.0012 (+1.38)	0.9998 (-0.13)	0.9941 (-1.68) / 0.9865 (-2.09)
HARMPCA	0.18 (43%)	1.0018 (+2.64) / 1.0023 (+1.04)	1.0005 (+0.26)	0.9948 (-0.53) / 0.9769 (-2.24)
HARMPCAOPT	0.12 (38%)	1.0024 (+1.74) / 1.0011 (+1.19)	0.9979 (-1.13)	0.9897 (-1.38) / 0.9834 (-1.46)

The robustness checks collected in Appendix E.6 (Table E.8) do not alter this picture: the real-time high-volatility, low-persistence cell shows no significant gain in either half of the sample, in the NBER recession or outside it; replacing the ARFIMA d by the sum of the rolling HAR coefficients as the persistence measure changes nothing; and sign-randomisation p-values for the pooled daily differentials, which assume nothing about the distribution of the losses, agree with the pooled Diebold–Mariano test cell by cell.

6.4 What Regime-Conditional Evaluation Can and Cannot Show

The evidence in Table 5 is obtained by sorting rolling-window forecast losses, ex post, into regimes, and a reader may ask what such a design can show. Appendix F applies the design

of Section 4.5, unchanged, to simulated panels of 33 firms whose log volatility follows a HAR process with Markov-switching persistence, under four data-generating processes that differ only in how an exogenous predictor relates to the persistence state. Four results matter here. First, when the predictor carries the same information in every state, the augmented model is worse than the benchmark in every regime, and the high-volatility, low-persistence cell is the worst cell rather than the best (below one in 8% of replications): a real-time regime comparison does not manufacture gains. Second, when the predictor's content rises in the low-persistence state, the design reproduces the empirical signature at comparable magnitudes (a relative QLIKE loss of 0.979 in that cell, below one in 86% of replications, against 0.988 for HAR2PCA in the data), insensitively to the cut-offs and to trailing rather than full-sample thresholds; a predictor that merely tracks the state produces the opposite pattern. Third, the Fisher-combined test over-rejects under cross-sectional dependence of the size plausible for index constituents (about 15% at the nominal 5% level), and the proportion of stocks with a ratio below one, read as a test, over-rejects far more; the pooled test of Table 7 is correctly sized. Fourth, when the simulated panels are sorted on the realised target-day volatility, as Table 5 is, the comparison rewards whatever component of the difference between the two forecasts is correlated with the realised surprise: nothing when the augmented forecast differs from the benchmark only by estimation noise, the predictor's information in every cell when it is state-dependent, and a split between high- and low-outcome cells when the predictor shifts the level of the forecast without improving it on average. The DJIA pattern is of the last kind, with the level difference identified directly by the placebo of Table 8, and the configuration observed on the DJIA panel requires the level difference to persist on the days the real-time flag identifies (Appendix F.3). The simulation therefore rules out the objection that a real-time regime comparison manufactures gains, and confirms that an outcome-based one can.

7 Conclusion

This paper asks a precise question: when does media sentiment predict volatility? For firm-level sentiment and attention indices and the daily realised volatility of DJIA constituents over 2004–2016, the answer is that no sentiment-augmented model does so robustly in any state that can be identified when the forecast is made; the only real-time gain, from a pre-specified combination of the augmented and benchmark forecasts in one state under one loss function, is small and is reported rather than built on. Augmenting a daily re-estimated HAR model with media-based predictors worsens one-day-ahead forecasts on average and in most states, by one to four per cent of the benchmark's loss; the benchmark is in the model confidence set of every stock; no augmented specification beats it after allowance for the search over specifications; and no window of the evaluation period shows otherwise. The in-sample evidence that motivated the question, incremental conditional-quantile fit in the upper tail that is largest for the least persistent stocks, is a description of in-sample fit, and it does not carry over. The paper's in-sample results are a different matter, and the two have to be kept apart. The 27 predictors constructed in Section 3 from the vendor's two raw series per channel, organised by the direction, salience and instability of coverage and by cross-channel disagreement, raise the in-sample conditional-quantile fit of a HAR model by up to a quarter in the upper tail for the least persistent stocks, and their importance rotates over time in

TABLE 11: SUMMARY OF THE EVIDENCE BY SAMPLE AND OBJECT. Each row is one result of Sections 5 and 6. "Object" is what the result measures: in-sample fit (adjusted R^2, quantile pseudo-R^2 or variable importance computed on the estimation sample) or out-of-sample forecast loss (one-day-ahead, 252-day rolling window re-estimated daily). "State" says how the market state is defined where the result is conditional: none, real-time (information at the forecast origin) or ex post (the realised target-day volatility). The last column says whether the result holds under the real-time classification of Section 6.2, which is the only one a forecaster could act on.

Result	Where	Object	State	Holds in real time?
Panel A: In-sample (estimation-sample) evidence, Section 5				
Average fit gain of 0.1–0.6% of R^2	Table 4	Fit	None	Not applicable
Upper-quantile fit gain, up to 25% at $\tau = 0.975$ for the least persistent stocks	Figs. 1, 2, D.1	Fit	Conditional quantile	Not applicable
Rotation of predictor-family importance over time; decay with window length	Figs. D.2, D.3 (App. D.3)	Importance	None	Not applicable
Kitchen-sink specification unstable; factor summaries stable	Appendix C	Estimation diagnostics	None	Yes: forecast failures in Table 7
Panel B: Out-of-sample evidence, Section 6				
No unconditional forecast gain; losses of 1–4%	Tables E.1, 7, 9	Forecast loss	None	Yes
Losses in both persistence regimes; no gain in either volatility regime	Table E.1	Forecast loss	Real-time (persistence); ex post (volatility)	Yes (Table 6)
Gain of about 1% in the high-volatility, low-persistence cell	Table 5	Forecast loss	Ex post	No: pooled test rejects under MSE/MAE but not QLIKE ex post (Table 7); cell at or above one in real time (Table 6); placebo reproduces it (Table 8)
No gain at any cut-off, threshold, subperiod or persistence measure	Tables 6, E.5, E.8 (App. E)	Forecast loss	Real-time	Yes (negative)
Benchmark in every stock's 90% MCS; SPA $p = 1$; no favourable window	Tables 9, E.7 (App. E.6)	Forecast loss	None	Yes (negative)
Benchmark does not encompass the augmented forecasts ($\lambda \approx 0.1$; about 0.45 in the real-time cell)	Table 10, Panel A	Forecast information	None; real-time	Yes
Rolling combination: worse unconditionally; about 2% QLIKE gain in the real-time cell	Table 10, Panel B	Forecast loss	Real-time	Yes, in one cell under one loss function (5% level)

a way that the rolling-window diagnostics make visible. Entering all 27 directly produces severe multicollinearity, unstable coefficients and, on at least two firm-days, forecasts that fail outright; principal-component summaries within families are stable. The framework is vendor-agnostic, fully documented, and in our view the right way to extract what commercial media indices contain. What this paper adds is the finding that, extracted this way, the information describes the conditional distribution of volatility in sample without improving its one-day-ahead forecast out of sample, which is a stronger and more useful negative result than one obtained with a cruder representation of the same data.

The paper's second finding concerns how such questions are answered. A regime-conditional evaluation that sorts forecast errors by the realised volatility of the day being forecast displays precisely the pattern the state-dependence hypothesis predicts: gains of about one per cent where high realised volatility meets low persistence, for most stocks, under every loss function. The pattern is produced by the sort. Selecting on an extreme outcome selects on the sign of every model's error, and a forecast shifted in that direction wins without knowing anything: a placebo built from the augmented specifications' 0.4% upward shift on low-persistence days reproduces the pattern, Eq. (11) predicts the placebo's relative loss, the pattern disappears under a classification available at the forecast origin, and the Monte Carlo study of Appendix F shows that a real-time classification does not manufacture gains while an outcome-based one rewards whatever component of a forecast difference is correlated with the realised surprise. We suspect that the device is in wider use than its properties warrant, and we offer the real-time classification, the placebo and the proposition as the checks to run.

One result points the other way. The benchmark does not encompass the augmented forecasts: their deviation from the benchmark is correlated with the benchmark's error, with an encompassing weight of about one tenth unconditionally and nearly one half in the real-time high-volatility, low-persistence state, and a rolling combination with that weight estimated on the preceding year, specified once and run once, is worse than the benchmark unconditionally and better by about two per cent, under QLIKE only and at the five per cent level only, in that one state. This is consistent with media predictors that carry some information about next-day volatility in low-persistence, high-volatility episodes, too little and too noisily measured to be used at full weight; it is not a result we build on, but it is the natural starting point for further work on these predictors.

For practitioners the implication is twofold: firm-level media sentiment of the kind studied here is not worth adding to a well-fitted daily HAR model as a predictor of next-day realised volatility, and a demonstration that it helps "on high-volatility days" should be read with the definition of those days in hand, because if the days were identified by the volatility that was realised, the demonstration shows nothing a forecaster could have used. Three limitations define the next step. The benchmark contains only lagged volatility, and the market-level predictors that are known to improve HAR forecasts would, if anything, leave less for media signals, but the comparison should be made. The rolling window of 252 days is shorter than the two and a half to four years that Chassot and Audrino (2026) find optimal; a longer window strengthens the benchmark and is unlikely to change the real-time negative result, but it may change the encompassing and combination results, which depend on how precisely the augmented models are estimated. And the sample ends in 2016 and covers 33 large firms: the methodological results do not depend on this, the substantive ones do. The broader message is modest. A daily re-estimated HAR model is hard to improve upon, and hardest in

exactly the states in which every forecast is poor; the states in which narrative information might help can be defined in advance or after the fact, only the first kind of definition can show that it does, and under that definition, for this data, it does not.

References

Andersen, T. G., Bollerslev, T., Diebold, F. X., & Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, *71*(2), 579–625.

Antweiler, W., & Frank, M. Z. (2004). Is All That Talk Just Noise? The Information Content of Internet Stock Message Boards. *The Journal of Finance*, *59*(3), 1259–1294.

Audrino, F., Sigrist, F., & Ballinari, D. (2020). The impact of sentiment and attention measures on stock market volatility. *International Journal of Forecasting*, *36*(2), 334–357.

Bai, J., & Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, *70*(1), 191–221.

Ballinari, D., Audrino, F., & Sigrist, F. (2022). When does attention matter? The effect of investor attention on stock market volatility around news releases. *International Review of Financial Analysis*, *82*, 102185.

Barndorff-Nielsen, O. E., & Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society. Series B. Statistical Methodology*, *64*(2), 253–280.

Becker, B. J. (1994). Combining significance levels. In H. Cooper & L. V. Hedges (Eds.), *The handbook of research synthesis* (pp. 215–230). Russell Sage Foundation.

Behrendt, S., & Schmidt, A. (2018). The Twitter myth revisited: Intraday investor sentiment, Twitter activity and individual-level stock return volatility. *Journal of Banking & Finance*, *96*, 355–367.

Belsley, D. A., Kuh, E., & Welsch, R. E. (1980). *Regression Diagnostics: Identifying Influential Data and Sources of Collinearity*. John Wiley & Sons.

Bodilsen, S. T., & Lunde, A. (2025). Exploiting News Analytics for Volatility Forecasting. *Journal of Applied Econometrics*, *40*(1), 18–36.

Bollen, J., Mao, H., & Zeng, X. (2011). Twitter mood predicts the stock market. *Journal of Computational Science*, *2*(1), 1–8.

Boudoukh, J., Feldman, R., Kogan, S., & Richardson, M. (2019). Information, trading, and volatility: Evidence from firm-specific news. *The Review of Financial Studies*, *32*(3), 992–1033.

Brauneis, A., & Şahiner, M. (2026). Crypto Volatility Forecasting: Mounting a HAR, Sentiment, and Machine Learning Horserace. *Asia-Pacific Financial Markets*, *33*, 379–411.

Burnham, K. P., & Anderson, D. R. (2002). *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach* (2nd). Springer-Verlag New York.

Calomiris, C. W., & Mamaysky, H. (2019). How news and its context drive risk and returns around the world. *Journal of Financial Economics*, *133*(2), 299–336.

Chassot, J., & Audrino, F. (2026). HARd to beat: The overlooked impact of rolling windows in the era of machine learning. *International Journal of Forecasting*, *42*(2), 330–343.

Chen, H., De, P., Hu, Y., & Hwang, B.-H. (2014). Wisdom of Crowds: The Value of Stock Opinions Transmitted Through Social Media. *The Review of Financial Studies*, *27*(5), 1367–1403.

Chen, Y., & Mai, W. (2024). Investor attention and environmental performance of Chinese high-tech companies: the moderating effects of media attention and coverage sentiment. *Humanities and Social Sciences Communications*, *11*, 1020.

Clark, T. E., & West, K. D. (2007). Approximately Normal Tests for Equal Predictive Accuracy in Nested Models. *Journal of Econometrics*, *138*(1), 291–311.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, *7*(2), 174–196.

Corsi, F., Audrino, F., & Renò, R. (2012). HAR Modeling for Realized Volatility Forecasting. In L. Bauwens, C. Hafner, & S. Laurent (Eds.), *Handbook of volatility models and their applications* (pp. 363–382). John Wiley & Sons.

Da, Z., Engelberg, J., & Gao, P. (2011). In Search of Attention. *The Journal of Finance*, *66*(5), 1461–1499.

Da, Z., Engelberg, J., & Gao, P. (2015). The sum of all FEARS: Investor sentiment and asset prices. *The Review of Financial Studies*, *28*(1), 1–32.

Deveikyte, J., Geman, H., Piccari, C., & Provetti, A. (2022). A sentiment analysis approach to the prediction of market volatility. *Frontiers in Artificial Intelligence*, *5*, 836809.

Diebold, F. X., & Mariano, R. S. (1995). Comparing Predictive Accuracy. *Journal of Business & Economic Statistics*, *13*(3), 253–263.

Dutta, A., Sihvonen, J., Park, D., Lucey, B., & Uddin, G. S. (2025). Impact of News and Social Media Sentiments on Rare Earth Investments. *Resources Policy*, *107*, 105655.

Engle, R. F., Hansen, M. K., Karagozoglu, A. K., & Lunde, A. (2021). News and idiosyncratic volatility: The public information processing hypothesis. *Journal of Financial Econometrics*, *19*(1), 1–38.

Epstein, L. G., & Schneider, M. (2008). Ambiguity, Information Quality, and Asset Pricing. *The Journal of Finance*, *63*(1), 197–242.

Fan, J., Ke, Y., & Wang, K. (2020). Factor-adjusted regularized model selection. *Journal of Econometrics*, *216*(1), 71–85.

Gan, B., Alexeev, V., Bird, R., & Yeung, D. (2020). Sensitivity to sentiment: News vs social media. *International Review of Financial Analysis*, *67*, 101390.

García, D. (2013). Sentiment during Recessions. *The Journal of Finance*, *68*(3), 1267–1300.

Giacomini, R., & Rossi, B. (2010). Forecast comparisons in unstable environments. *Journal of Applied Econometrics*, *25*(4), 595–620.

Giacomini, R., & White, H. (2006). Tests of conditional predictive ability. *Econometrica*, *74*(6), 1545–1578.

Glasserman, P., & Mamaysky, H. (2019). Does unusual news forecast market stress? *Journal of Financial and Quantitative Analysis*, *54*(5), 1937–1974.

Gonçalves, S., & Meddahi, N. (2011). Box–Cox transforms for realized volatility. *Journal of Econometrics*, *160*(1), 129–144.

Granger, C. W. J., & Newbold, P. (1976). Forecasting transformed series. *Journal of the Royal Statistical Society: Series B (Methodological)*, *38*(2), 189–203.

Gray, A., Modell, A., Rubin-Delanchy, P., & Whiteley, N. (2024). Hierarchical clustering with dot products recovers hidden tree structure [NeurIPS 2023 Spotlight]. *Advances in Neural Information Processing Systems*.

Groß-Klußmann, A., & Hautsch, N. (2011). When machines read the news: Using automated text analytics to quantify high frequency news-implied market reactions. *Journal of Empirical Finance*, *18*(2), 321–340.

Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The model confidence set. *Econometrica*, *79*(2), 453–497.

Hansen, P. R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, *23*(4), 365–380.

Hirshleifer, D., Peng, L., & Wang, Q. (2024). News Diffusion in Social Networks and Stock Market Reactions. *The Review of Financial Studies*, *38*(3), 883–937.

Izzeldin, M., Hassan, M. K., Pappas, V., & Tsionas, M. (2019). Forecasting realised volatility using ARFIMA and HAR models. *Quantitative Finance*, *19*(10), 1627–1638.

Jiang, Z., Lu, Y., & Zhang, L. (2025). Integrating sentiment information for risk prediction: the case of crude oil futures market in China. *Empirical Economics*, *68*, 1677–1718.

Jiao, P., Veiga, A., & Walther, A. (2020). Social media, news media and the stock market. *Journal of Economic Behavior & Organization*, *176*, 63–90.

Jolliffe, I. T. (2002). *Principal Component Analysis* (2nd). Springer.

Koenker, R., & Bassett, G. (1978). Regression Quantiles. *Econometrica*, *46*(1), 33–50.

Koenker, R., & Machado, J. A. F. (1999). Goodness of fit and related inference processes for quantile regression. *Journal of the American Statistical Association*, *94*(448), 1296–1310.

Kutner, M. H., Nachtsheim, C., & Neter, J. (2004). *Applied Linear Regression Models*. McGraw-Hill/Irwin.

Lei, B., & Song, Y. (2023). The impact of contagion effects of media reports, investors' sentiment and attention on the stock market based on HAR-RV model. *International Journal of Financial Engineering (IJFE)*, *10*(2), 1–54.

Liu, L., Patton, A., & Sheppard, K. (2015). Does anything beat 5-minute RV? A comparison of realized measures across multiple asset classes. *Journal of Econometrics*, *187*(1), 293–311.

Odendahl, F., Rossi, B., & Sekhposyan, T. (2023). Evaluating forecast performance with state dependence. *Journal of Econometrics*, *237*(2), 105220.

Oti, E. U., & Olusola, M. O. (2024). Overview of Agglomerative Hierarchical Clustering Methods. *British Journal of Computer, Networking and Information Technology*, *7*(2), 14–23.

Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, *160*(1), 246–256.

Patton, A. J., & Sheppard, K. (2009). Evaluating Volatility and Correlation Forecasts. In T. Mikosch, J.-P. Kreiß, R. A. Davis, & T. G. Andersen (Eds.), *Handbook of financial time series* (pp. 801–838). Springer, Berlin, Heidelberg.

Peterson, R. L. (2016). Appendix A: Understanding the Thomson Reuters MarketPsych Indices. In *Trading on Sentiment: The Power of Minds Over Markets* (pp. 297–315). John Wiley & Sons, Ltd.

Rahimikia, E., & Poon, S.-H. (2026). Alternative data for realised volatility forecasting: Limit order book and news stories [Published online 8 June 2026; volume and pages to be added at proof stage]. *International Journal of Finance & Economics*.

Rousseeuw, P. J. (1987). Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis. *Journal of Computational and Applied Mathematics*, *20*, 53–65.

Stock, J. H., & Watson, M. W. (2002). Forecasting using principal components from a large number of predictors. *Journal of the American Statistical Association*, *97*(460), 1167–1179.

Sun, L., Najand, M., & Shen, J. (2016). Stock return predictability and investor sentiment: A high-frequency perspective. *Journal of Banking & Finance*, *73*, 147–164.

Tetlock, P. C. (2007). Giving Content to Investor Sentiment: The Role of Media in the Stock Market. *The Journal of Finance*, *62*(3), 1139–1168.

Van Kervel, V., & Menkveld, A. J. (2019). High-Frequency Trading around Large Institutional Orders. *The Journal of Finance*, *74*(3), 1091–1137.

Varneskov, R. T., & Perron, P. (2018). Combining long memory and level shifts in modelling and forecasting the volatility of asset returns. *Quantitative Finance*, *18*(3), 371–393.

West, K. D. (1996). Asymptotic Inference about Predictive Ability. *Econometrica*, *64*(5), 1067–1084.

White, H. (2000). A reality check for data snooping. *Econometrica*, *68*(5), 1097–1126.

Yang, Y. (2005). Can the strengths of AIC and BIC be shared? *Biometrika*, *92*(4), 937–950.

Zhang, C., Zhang, Y., Cucuringu, M., & Qian, Z. (2023). Volatility Forecasting with Machine Learning and Intraday Commonality. *Journal of Financial Econometrics*, *22*(2), 492–530.

Appendix

Contents

A Media Signal Construction **2**
- A.1 Preprocessing of Buzz variables . . . 2
- A.2 Media-Based Predictor Definitions . . . 4
- A.3 Descriptive Statistics for Constructed Predictors . . . 11
- A.4 Empirical Characterisation of Media Predictors . . . 11
  - A.4.1 Data Quality . . . 11
  - A.4.2 Time-Series Dynamics . . . 17
  - A.4.3 Cross-Sectional Distributions . . . 18
  - A.4.4 Pairwise Correlation Structure . . . 19
  - A.4.5 Cross-Channel Disagreement Dynamics . . . 19
  - A.4.6 News versus Social Media Channel Comparison . . . 21

B Dimension Reduction: Technical Details **23**
- B.1 One-Stage PCA . . . 23
- B.2 Two-Stage PCA . . . 23
- B.3 Illustrative PCA Analysis: Apple Inc. (AAPL) . . . 24

C Stability tests **34**

D Additional In-Sample Results **35**
- D.1 Conditional Correlations Between Media Metrics and Index Volatility . . . 35
- D.2 Quantile Variation in Incremental Fit for HAR2PCA . . . 35
- D.3 Predictor Importance: Static and Rolling . . . 36

E Additional Out-of-Sample Results **37**
- E.1 Evaluation Details . . . 37
- E.2 Univariate Regime Diagnostics . . . 41
- E.3 Cumulative Forecast Performance in the Interaction Regime . . . 42
- E.4 Giacomini–White Conditional Test . . . 43
- E.5 Period-Based Conditioning, Forecast Bias, Cut-Offs and the Loss-Differential Regression . . . 44
- E.6 Fluctuation Test and Robustness . . . 46

F Monte Carlo Assessment of the Regime-Conditional Evaluation Design **48**
- F.1 Design . . . 48
- F.2 Results . . . 50
- F.3 Outcome-Based Classification . . . 53
- F.4 Size of the Cross-Sectional Summaries . . . 54
- F.5 What the Study Does and Does Not Show . . . 55

A Media Signal Construction

This section provides technical detail supporting the media signal construction described in Section 3. We first document the empirical justification for the log-transformation applied to raw Buzz variables (Section A.1), then provide complete definitions and economic interpretations for all 27 constructed predictors (Section A.2), and finally report their descriptive statistics (Section A.3).

A.1 Preprocessing of Buzz variables

A key empirical feature of Buzz is that, like many count-based media attention measures, it exhibits a heavy-tailed distribution with substantial skewness and heteroskedasticity. Such properties are undesirable in linear forecasting models because they allow a small number of extreme attention days to dominate estimation and may distort the relative contribution of media-based predictors.

To stabilise the variance and assess an appropriate transformation, we examine the Box–Cox family of power transformations:

$$Y(\lambda) = \begin{cases} \frac{((1+X)^{\lambda}-1)}{\lambda}, & \text{for } \lambda \neq 0, \\ \ln(1+X), & \text{for } \lambda = 0, \end{cases} \tag{A.1}$$

where $X \geq 0$ denotes the raw Buzz variable, the constant 1 is added to accommodate zero-attention days, and the $\lambda = 0$ case is the continuous limit of the upper branch as $\lambda \to 0$.

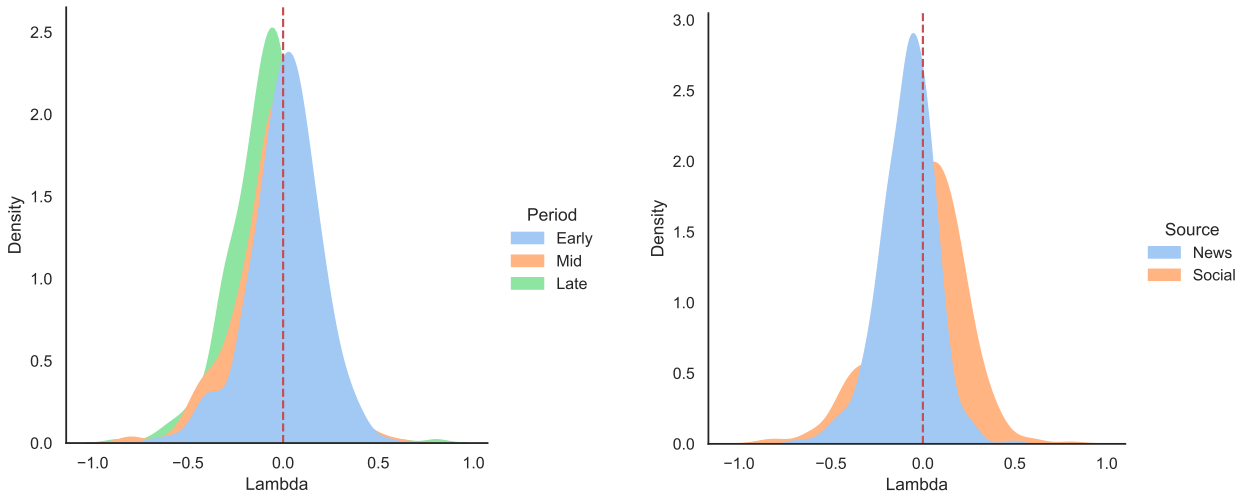
Because media attention evolves over time and the scale of coverage changes substantially across the sample, particularly with the rapid growth of social media activity in the later years, we estimate the Box–Cox parameter λ using rolling one-year windows for each stock and media source. This rolling approach allows the transformation to adapt to changing media environments rather than imposing a single static transformation over the entire sample.

Table A.1 summarises the distribution of the estimated λ values across time periods and media sources. The results strongly support a log-linear specification. Across all periods, the mean optimal λ remains close to zero, indicating that the simple logarithmic transformation is a robust and parsimonious choice. This conclusion also holds across sources, despite some differences in scale and dispersion between News and Social channels.

Group	N	Mean	Std	Min	25%	Median	75%	Max
Panel A: By Period								
Early	378	0.010	0.183	−0.674	−0.087	0.026	0.126	0.544
Mid	350	−0.030	0.206	−0.809	−0.145	−0.022	0.105	0.595
Late	360	−0.106	0.182	−0.870	−0.202	−0.088	0.003	0.804
Panel B: By Media Source								
News	544	−0.084	0.149	−0.691	−0.174	−0.070	0.012	0.492
Social	544	0.001	0.227	−0.870	−0.106	0.033	0.152	0.804

TABLE A.1: DISTRIBUTION OF ESTIMATED OPTIMAL BOX–COX PARAMETERS FOR BUZZ VARIABLES. This table reports summary statistics for the estimated optimal Box–Cox transformation parameter λ applied to the raw Buzz variables. Buzz measures the daily intensity of firm-specific media attention and exhibits substantial right skewness and heavy tails. To assess an appropriate transformation, λ is estimated separately using rolling one-year windows for each stock and media source. Panel A groups the estimated parameters by sample period, while Panel B groups them by media source (News versus Social). Values of λ close to zero indicate that the logarithmic transformation, $\ln(1 + \text{Buzz})$, is approximately optimal. Across all groupings, the estimated parameters cluster tightly around zero, providing strong empirical support for the use of a log-transformation in the main analysis.

Figure A.1 visualises the distribution of the estimated optimal Box–Cox parameters. Panel A shows that the estimated values remain tightly centred around zero across subsample periods, indicating strong temporal stability in the appropriate transformation. Panel B shows that this conclusion also holds across media sources, despite differences in scale and dispersion between News and Social attention. Based on these diagnostics, we apply the logarithmic transformation $\ln(1 + \text{Buzz})$ throughout the main analysis. This choice is empirically appropriate, robust, and avoids introducing unnecessary complexity into the forecasting design.



PANEL A: Distribution of estimated λ by sample period

PANEL B: Distribution of estimated λ by media source

FIGURE A.1: DISTRIBUTION OF ESTIMATED OPTIMAL BOX–COX PARAMETERS (λ) FOR BUZZ VARIABLES. The figure plots the empirical distribution of the estimated transformation parameter λ obtained from rolling one-year Box–Cox estimations applied to firm-level Buzz series. Panel (a) groups estimates by sample period, while Panel (b) groups them by media source (News versus Social). Values of λ close to zero indicate that a logarithmic transformation, $\ln(1 + \text{Buzz})$, is approximately optimal. Across all groupings, the estimated parameters cluster tightly around zero, indicating that the log transformation provides a robust and stable approximation.

These preprocessing steps ensure that the raw TRMI variables are placed on a stable and interpretable scale before the construction of the transformed media predictors used in the main analysis.

A.2 Media-Based Predictor Definitions

Raw sentiment and buzz variables are often too coarse to capture the mechanisms through which media coverage affects volatility. A neutral average sentiment score may conceal a sharp reversal in tone, while a moderately negative signal may carry little economic relevance if media attention is negligible. To address these limitations, we transform the four raw TRMI series ($sent^{\mathrm{N}}$, $sent^{\mathrm{S}}$, $buzz^{\mathrm{N}}$, and $buzz^{\mathrm{S}}$) into 27 predictors designed to capture distinct informational dimensions of the firm-specific media environment. The 27 features span overlapping categories (features may belong to more than one group simultaneously):

- 11 features are derived exclusively from social media,
- 11 features are derived exclusively from news media,
- 5 features are calculated by combining data from both media channels,
- 8 features focus on "sentiment" only,
- 9 features focus on "buzz" only,
- 10 features are calculated using both sentiment and buzz metrics, and
- 10 transformed features are based on raw data.

Table A.2 lists the 27 predictors by family and channel with a one-line description of each. Table A.3 provides the complete construction formula and economic rationale for each predictor, organised into the four families defined in Section 3.1. All features are subsequently standardised within the estimation window prior to dimension reduction.

Table A.2: The 27 Media-Based Predictors by Informational Family. All predictors are transformations of the four raw daily TRMI series, sentiment and buzz for the news (N) and social (S) channels. $c \in \{\mathrm{N}, \mathrm{S}\}$ denotes a predictor constructed for each channel. Relative buzz (RB) is the buzz-weighted three-day mean of sentiment; MCS is the media coverage sentiment score, a non-linear balance of positive against negative sentiment days over a weekly (w) or monthly (m) window; D denotes an absolute cross-channel difference and the superscript $\star$ its polarity-magnitude version. Formulae, imputation rules and economic rationales are in Appendix A.2; descriptive statistics in Appendix A.3.

Family	Predictor	Construction	Channel
Direction (6)	$sent^{c}$	raw daily sentiment, $[-1, 1]$	N, S
	MCS^{c}_{w}	media coverage sentiment, weekly window	N, S
	MCS^{c}_{m}	media coverage sentiment, monthly window	N, S
Salience (10)	$buzz^{c}$	raw daily buzz (volume of scored references)	N, S
	$\ln buzz^{c}$	log buzz, $\ln(1 + buzz^{c})$	N, S
	RB^{c}	relative buzz: buzz-weighted three-day mean of sentiment	N, S
	$\ln RB^{c}$	log relative buzz	N, S
	sgn^{c}	signed log buzz: $\mathrm{sgn}(sent^{c}) \ln(1 + buzz^{c})$	N, S
Instability (6)	$\Delta \ln buzz^{c}$	day-on-day log change in buzz	N, S
	$\Delta^{2} \ln buzz^{c}$	log change in log buzz	N, S
	$\Delta \ln RB^{c}$	log change in relative buzz	N, S
Disagreement (5)	D^{Sent}	$\lvert sent^{\mathrm{N}} - sent^{\mathrm{S}} \rvert$	both
	$D^{\mathrm{Sent}\star}$	$\big\lvert \lvert sent^{\mathrm{N}} \rvert - \lvert sent^{\mathrm{S}} \rvert \big\rvert$	both
	D^{Buzz}	$\lvert \widehat{buzz}^{\mathrm{N}} - \widehat{buzz}^{\mathrm{S}} \rvert$, buzz relative to its trailing five-day mean	both
	D^{RB}	$\lvert RB^{\mathrm{N}} - RB^{\mathrm{S}} \rvert$	both
	$D^{\mathrm{RB}\star}$	$\big\lvert \lvert RB^{\mathrm{N}} \rvert - \lvert RB^{\mathrm{S}} \rvert \big\rvert$	both

TABLE A.3: MEDIA-BASED PREDICTOR DEFINITIONS. This table reports all 27 media-based predictors constructed from the Thomson Reuters MarketPsych Indices (TRMI) dataset. The predictors are organised into four feature families reflecting *Direction* (coverage tone), *Salience* (media attention), *Instability* (narrative dynamics, rapid changes) and Cross-Channel *Disagreement* (conflicting signals). Superscripts N and S denote the **News** and **Social** media channels, respectively; $c \in \{\mathrm{N}, \mathrm{S}\}$ indicates that an identical formula applies to both channels. Subscripts i, t denote scores for firm i on day t. All features are subsequently standardised within the estimation window prior to dimension reduction.

Variable (shorthand)	**Construction (formula)**	**Economic rationale**	**Channel**
Panel A: Direction — whether the tone of coverage is favourable or unfavourable			
`sentiment_News` ($sent^{\mathrm{N}}_{i,t}$)	Raw daily sentiment score from the news channel, $sent^{\mathrm{N}}_{i,t} \in [-1, 1]$. Positive (negative) values indicate net positive (negative) tone.	The prevailing narrative tone in traditional media reflects the balance of optimistic and pessimistic information, which may anchor expectations and affect investor sentiment towards the firm.	News
`sentiment_Social` ($sent^{\mathrm{S}}_{i,t}$)	Raw daily sentiment score from the social media channel, $sent^{\mathrm{S}}_{i,t} \in [-1, 1]$ (same construction as $sent^{\mathrm{N}}_{i,t}$).	Social media sentiment captures the retail investor crowd's aggregate assessment, which may reflect faster-moving or more speculative narrative shifts than traditional news.	Social
`MCS_News_weekly` `MCS_News_monthly` (MCS^{N}_{w}, MCS^{N}_{m})	Media Coverage Sentiment based on counts of days with positive and negative sentiment over a weekly (w) or monthly (m) rolling window, where n^+ (n^-) denotes the number of days with net positive (negative) sentiment: $$MCS = \begin{cases} \dfrac{(n^+)^2 - n^+ n^-}{(n^+ + n^-)^2}, & n^+ > n^- \\ \dfrac{n^+ n^- - (n^-)^2}{(n^+ + n^-)^2}, & n^+ < n^- \\ 0, & n^+ = n^- \end{cases}$$ $MCS \in [-1, 1]$; values closer to $+1$ (-1) indicate predominantly positive (negative) trends in media coverage.	Aggregating the count of positive versus negative days over a rolling window smooths day-to-day noise and captures the *direction* of accumulated media narrative; the nonlinear weighting rewards dominant tone and penalises mixed coverage. MCS definition is adapted from Chen and Mai (2024) to isolate the directional consistency of sentiment independent of daily magnitude.	News
`MCS_Social_weekly` `MCS_Social_monthly` (MCS^{S}_{w}, MCS^{S}_{m})	Same formula applied to the social media channel with weekly and monthly rolling windows.	Social media MCS tracks the directional balance of user-generated content, which may evolve at a different pace and with different persistence than the equivalent news-based signal.	Social
Panel B: Salience — whether sentiment is accompanied by unusually high media attention			

Continued on next page…

(Table A.3 continued)

Variable	Construction (formula)	Economic rationale	Channel
`buzz_News` ($buzz^{\mathrm{N}}_{i,t}$)	Raw daily article count from the news channel, $buzz^{\mathrm{N}}_{i,t} \geq 0$.	Buzz captures the intensity of media attention independently of sentiment direction. Elevated coverage, even without a directional signal, may increase investor attention and uncertainty.	News
`buzz_Social` ($buzz^{\mathrm{S}}_{i,t}$)	Raw daily post/article count from the social media channel, $buzz^{\mathrm{S}}_{i,t} \geq 0$.	Social media buzz is particularly responsive to firm-specific events and investor discussion, and may lead traditional media during periods of rapid information diffusion.	Social
`buzz_News_log` ($\ln buzz^{\mathrm{N}}_{i,t}$)	$\ln\big(buzz^{\mathrm{N}}_{i,t} + 1\big)$. Justification for ln(.) transformation is detailed in Section A.1.	The heavy-tailed distribution of raw buzz distorts linear models. Log-transformation stabilises variance and renders the scale of attention approximately symmetric; the +1 offset accommodates zero-attention days.	News
`buzz_Social_log` ($\ln buzz^{\mathrm{S}}_{i,t}$)	$\ln\big(buzz^{\mathrm{S}}_{i,t} + 1\big)$. Justification for ln(.) transformation is detailed in Section A.1.	Same motivation as `buzz_News_log` applied to the social media channel.	Social

Continued on next page...

(Table A.3 continued)

Variable	Construction (formula)	Economic rationale	Channel
`rela_buzz_News` ($RB^{\mathrm{N}}_{i,t}$)	Buzz-weighted 3-day rolling mean of news sentiment computed on EWM-smoothed series ($\alpha = \frac{1}{2}$). For each series $x \in \{sent^{\mathrm{N}}, buzz^{\mathrm{N}}\}$, the smoothed value is defined recursively as: $$\widetilde{x}_{i,t} = \begin{cases} \alpha\, x_{i,t} + (1-\alpha)\, \widetilde{x}_{i,t-1} & \text{if } x_{i,t} \text{ observed} \\ \tilde{x}_{i,t-1} & \text{if } x_{i,t} \text{ missing} \end{cases}$$ so that on silent (no-coverage) days the smoother carries forward a recency-weighted memory of past values rather than a raw carry-forward. Each weight decays geometrically at rate $(1-\alpha)$ with no normalisation correction, so that the influence of observation $x_{i,t-k}$ on $\widetilde{x}_{i,t}$ is proportional to $(1-\alpha)^k = 2^{-k}$. The buzz-weighted (EWM-imputed) mean is then: $$RB^{\mathrm{N}}_{i,t} = \begin{cases} \dfrac{\sum_{\tau=0}^{2} \widetilde{buzz}^{\mathrm{N}}_{i,t-\tau}\, \widetilde{sent}^{\mathrm{N}}_{i,t-\tau}}{\sum_{\tau=0}^{2} \widetilde{buzz}^{\mathrm{N}}_{i,t-\tau}} & \text{if } B^{\mathrm{N}}_{i,t} > 0 \\ 0 & \text{otherwise} \end{cases}$$ where $B^{\mathrm{N}}_{i,t} = \sum_{\tau=0}^{2} \widetilde{buzz}^{\mathrm{N}}_{i,t-\tau}$ is the three-day sum of smoothed buzz. The zero-denominator case arises when all three smoothed buzz values are zero, indicating a sustained silent period in which no news coverage has occurred within the EWM memory horizon; in this case $RB^{\mathrm{N}}_{i,t}$ is set to zero, reflecting the absence of any sentiment signal.	Sentiment is economically relevant primarily when it is widely disseminated. Buzz-weighting assigns greater influence to days with high coverage, capturing the interaction between narrative tone and media salience. EWM imputation with $\alpha = \frac{1}{2}$ preserves recent dynamics while avoiding stale carry-forwards on silent days: the influence of a past observation decays geometrically at rate 2^{-k} after k days, so that values older than a few days contribute negligibly to the smoothed series.	News
`rela_buzz_Social` ($RB^{\mathrm{S}}_{i,t}$)	Same formula as $RB^{\mathrm{N}}_{i,t}$ using social-channel EWM-imputed buzz and sentiment ($\widetilde{buzz}^{\mathrm{S}}$, $\widetilde{sent}^{\mathrm{S}}$).	Same motivation as $RB^{\mathrm{N}}_{i,t}$ for the social media channel.	Social

Continued on next page…

(Table A.3 continued)

Variable	Construction (formula)	Economic rationale	Channel
`rela_buzz_News_log` $(\ln RB^{\mathrm{N}}_{i,t})$	$\ln\big(RB^{\mathrm{N}}_{i,t}+1\big)$ after clipping to $[0,\infty)$.	Log-transforms relative buzz to reduce the influence of extreme attention-weighted sentiment values and improve distributional regularity.	News
`rela_buzz_Social_log` $(\ln RB^{\mathrm{S}}_{i,t})$	$\ln\big(RB^{\mathrm{S}}_{i,t}+1\big)$ after clipping to $[0,\infty)$.	Same motivation as `rela_buzz_News_log` for social media.	Social
`signed_buzz_News_log` $(sgn^{\mathrm{N}}_{i,t})$	$\mathrm{sign}(sent^{\mathrm{N}}_{i,t}) \times \ln\big(buzz^{\mathrm{N}}_{i,t}+1\big)$.	Combines the direction of sentiment with the log-scale magnitude of media attention. Captures directional attention effects: whether unusually high coverage coincides with positive or negative tone. Complements RB^{N} by abstracting from sentiment magnitude and focusing on the direction of attention flow.	News
`signed_buzz_Social_log` $(sgn^{\mathrm{S}}_{i,t})$	$\mathrm{sign}(sent^{\mathrm{S}}_{i,t}) \times \ln\big(buzz^{\mathrm{S}}_{i,t}+1\big)$.	Same motivation as `signed_buzz_News_log` for social media.	Social
Panel C: Instability — whether tone or attention is changing rapidly or inconsistently over time			
`log_change_buzz_News` $(\Delta \ln buzz^{\mathrm{N}}_{i,t})$	$\ln\left(\frac{buzz^{\mathrm{N}}_{i,t}+1}{buzz^{\mathrm{N}}_{i,t-1}+1}\right)$	Day-over-day growth in news attention captures sudden surges or collapses in media coverage, reflecting episodic events or fading narratives. Rapid changes in attention signal informational instability and may presage higher volatility.	News
`log_change_buzz_Social` $(\Delta \ln buzz^{\mathrm{S}}_{i,t})$	$\ln\left(\frac{buzz^{\mathrm{S}}_{i,t}+1}{buzz^{\mathrm{S}}_{i,t-1}+1}\right)$	Same motivation as $\Delta \ln buzz^{\mathrm{N}}_{i,t}$ for social media. Social buzz changes may be more volatile and faster-moving than news, making this measure a sensitive indicator of emerging narrative shocks.	Social
`log_change_buzz_News_log` $(\Delta^2 \ln buzz^{\mathrm{N}}_{i,t})$	$\ln\left(\frac{\ln(buzz^{\mathrm{N}}_{i,t}+1)+1}{\ln(buzz^{\mathrm{N}}_{i,t-1}+1)+1}\right)$	A second-order log-change that measures the growth rate of already log-transformed buzz. This further compresses extreme observations and captures proportional changes in media attention on a stabilised scale, reducing the influence of outlier days.	News
`log_change_buzz_Social_log` $(\Delta^2 \ln buzz^{\mathrm{S}}_{i,t})$	$\ln\left(\frac{\ln(buzz^{\mathrm{S}}_{i,t}+1)+1}{\ln(buzz^{\mathrm{S}}_{i,t-1}+1)+1}\right)$	Same motivation as `log_change_buzz_News_log` for the social media channel.	Social

Continued on next page…

(Table A.3 continued)

Variable	Construction (formula)	Economic rationale	Channel
`log_change_rela_buzz_News_log` ($\Delta \ln RB^{\mathrm{N}}_{i,t}$)	$\ln\left(\frac{\ln(RB^{\mathrm{N}}_{i,t}+1)+1}{\ln(RB^{\mathrm{N}}_{i,t-1}+1)+1}\right)$	Captures day-over-day changes in attention-weighted news sentiment (on a log scale). This measure reflects sudden shifts in the joint distribution of tone and coverage, indicating that previously salient narratives are reversing or fading.	News
`log_change_rela_buzz_Social_log` ($\Delta \ln RB^{\mathrm{S}}_{i,t}$)	$\ln\left(\frac{\ln(RB^{\mathrm{S}}_{i,t}+1)+1}{\ln(RB^{\mathrm{S}}_{i,t-1}+1)+1}\right)$	Same motivation as `log_change_rela_buzz_News_log` applied to the social media channel.	Social
Panel D: Disagreement — whether different media channels convey conflicting signals			
`D_sentiment` ($D^{\mathrm{Sent}}_{i,t}$)	$\left\lvert sent^{\mathrm{N}}_{i,t} - sent^{\mathrm{S}}_{i,t}\right\rvert$	Measures the absolute divergence in narrative tone across traditional and social media. When news and social sentiment disagree, the informational environment is ambiguous, potentially amplifying uncertainty and volatility.	Both
`D_sentiment_polar` ($D^{\mathrm{Sent}\star}_{i,t}$)	$\left\lvert\lvert sent^{\mathrm{N}}_{i,t}\rvert - \lvert sent^{\mathrm{S}}_{i,t}\rvert\right\rvert$	Measures disagreement in sentiment *intensity* (polarity magnitude) rather than direction. Captures situations where one channel displays strong conviction (positive or negative) while the other remains muted, reflecting differences in the strength of narratives across media ecosystems.	Both

Continued on next page…

(Table A.3 continued)

Variable	Construction (formula)	Economic rationale	Channel
`D_buzz` ($D^{\text{Buzz}}_{i,t}$)	Normalised buzz for each channel: $$\widehat{buzz}^{c}_{i,t} = \frac{buzz^{c}_{i,t}}{\frac{1}{5}\sum_{\tau=1}^{5} buzz^{c}_{i,t-\tau}}, \quad c \in \{\text{N}, \text{S}\}$$ where the ratio is set to zero whenever the denominator vanishes. Disagreement: $$D^{\text{Buzz}}_{i,t} = \left| \widehat{buzz}^{\text{N}}_{i,t} - \widehat{buzz}^{\text{S}}_{i,t} \right|$$ The zero-denominator case arises when a firm records no coverage in either the news or social media channel across all five preceding days; in this case $\widehat{buzz}^{c}_{i,t}$ is set to zero, so that $D^{\text{Buzz}}_{i,t}$ reflects only the non-silent channel's normalised attention, or equals zero if both channels are silent.	Captures whether news and social media are allocating attention to the same firm at similar relative intensities. A large gap in normalised attention indicates that the two ecosystems are reacting differently to the same underlying events, reflecting cross-channel information asymmetries. When one channel is silent and the other is active, the disagreement measure takes the value of the active channel's normalised buzz, capturing the most extreme form of cross-channel divergence.	Both
`D_rela_buzz` ($D^{\text{RB}}_{i,t}$)	$\left\lvert RB^{\text{N}}_{i,t} - RB^{\text{S}}_{i,t} \right\rvert$	Measures cross-channel disagreement in attention-weighted sentiment. Combines disagreement in both tone and coverage salience, capturing situations where the joint sentiment-attention signal diverges across media ecosystems.	Both
`D_rela_buzz_polar` ($D^{\text{RB}\star}_{i,t}$)	$\left\lvert \lvert RB^{\text{N}}_{i,t}\rvert - \lvert RB^{\text{S}}_{i,t}\rvert \right\rvert$	Measures disagreement in the *magnitude* of attention-weighted sentiment across channels. Complements $D^{\text{RB}}_{i,t}$ by capturing differences in the strength of the sentiment-attention interaction, rather than its direction.	Both

A.3 Descriptive Statistics for Constructed Predictors

Table A.4 reports summary statistics for all 27 media-based predictors pooled across the full sample of 151,580 firm-day observations. Several patterns are worth noting. Raw buzz variables exhibit extreme right skewness (skewness exceeding 10 for both channels), confirming the distributional motivation for the log-transformation documented in Section A.1; the log-transformed series are approximately symmetric. Instability measures are centred near zero with near-zero skewness, consistent with their construction as log-changes. Disagreement measures are strictly non-negative by construction and right-skewed, reflecting that cross-channel divergence is episodic rather than persistent. Direction and Salience measures show modest unconditional means, indicating that the media environment is approximately balanced on average, with informative variation concentrated in the tails. This table further illustrates why subsequent standardisation and dimension reduction are necessary before these predictors can be incorporated into the forecasting models. To facilitate interpretation, the variables are reported using the same conceptual grouping as in Table A.3.

A.4 Empirical Characterisation of Media Predictors

We characterise the empirical properties of the 27 media-based predictors defined in Appendix A.2 and used throughout the main analysis. Our aim is to verify that the predictor set is of high quality and that its distributional and time-series properties are consistent with the economic interpretations assigned to each informational family in Section 3.1. Here, we also document the structural features: strong within-family correlation, cross-channel heterogeneity, and high dimensionality. This motivates us to resort to dimension-reduction strategy of Section 3.2.

We proceed through six diagnostics: data quality (Section A.4.1), time-series dynamics (Section A.4.2), marginal distributions (Section A.4.3), pairwise correlation structure (Section A.4.4), cross-channel disagreement dynamics (Section A.4.5), and a direct news-versus-social comparison (Section A.4.6). All figures are presented for AAPL as a representative example; results for all other DJIA constituents are qualitatively similar and available upon request.

A.4.1 Data Quality

Prior to the substantive analysis, each media-based predictor is subjected to a systematic data quality assessment covering eight diagnostic dimensions. Figure A.2 presents this assessment for AAPL, chosen as a representative illustration; an identical procedure is applied to every constituent of the DJIA sample and the findings reported below are qualitatively robust across firms. The diagnostics are designed to assess not only data completeness but also the distributional properties and temporal behaviour of each predictor, all of which bear directly on the reliability of subsequent forecasting models. The eight columns are described below.

Missing values (% missing). The first column reports the percentage of firm-day observations for which the feature takes a missing value. Missing data in media-based predictors typically arise from days with no textual coverage of a given firm, from gaps in the TRMI feed, or from the construction of lag-based features at the start of the sample. A high incidence

TABLE A.4: Summary Statistics of Media-Based Predictors (N = 151,580 firm-day observations)

Feature	Mean	Std	Median	Min	Max	Skew
Panel A: Direction — raw sentiment and MCS						
sentiment_News	0.015	0.255	0.003	−1.000	1.000	0.157
sentiment_Social	−0.007	0.190	−0.029	−1.000	1.000	0.750
MCS_sentiment_News_weekly	0.022	0.601	0.120	−1.000	1.000	−0.005
MCS_sentiment_News_monthly	0.023	0.408	0.000	−1.000	1.000	0.172
MCS_sentiment_Social_weekly	−0.171	0.583	−0.120	−1.000	1.000	0.169
MCS_sentiment_Social_monthly	−0.164	0.421	−0.107	−1.000	1.000	−0.223
Panel B: Salience — buzz, relative buzz, and signed buzz						
buzz_News	1114.077	2923.883	273.500	0.100	141943.200	10.929
buzz_Social	833.184	2202.012	231.500	0.200	119884.100	11.230
buzz_News_log	5.400	2.036	5.615	0.095	11.863	−0.224
buzz_Social_log	5.352	1.785	5.449	0.182	11.694	−0.224
rela_buzz_News	0.013	0.196	0.005	−1.000	1.000	0.214
rela_buzz_Social	−0.009	0.139	−0.027	−1.000	1.000	0.988
rela_buzz_News_log	0.071	0.105	0.005	0.000	0.693	1.911
rela_buzz_Social_log	0.040	0.080	0.000	0.000	0.693	3.006
signed_buzz_News_log	0.198	5.755	0.865	−11.858	11.863	−0.054
signed_buzz_Social_log	−1.356	5.457	−3.800	−11.694	11.621	0.308
Panel C: Instability — log-changes in buzz and relative buzz						
log_change_buzz_News	0.001	1.117	0.000	−6.837	6.292	0.121
log_change_buzz_Social	0.000	0.862	0.000	−5.435	5.760	0.043
log_change_buzz_News_log	0.000	0.237	0.000	−1.636	1.551	0.013
log_change_buzz_Social_log	0.000	0.193	0.000	−1.520	1.532	−0.029
log_change_rela_buzz_News_log	0.000	0.066	0.000	−0.527	0.527	0.104
log_change_rela_buzz_Social_log	0.000	0.051	0.000	−0.527	0.523	0.148
Panel D: Disagreement — cross-channel divergence in sentiment and buzz						
D_sentiment	0.210	0.214	0.144	0.000	2.000	2.172
D_sentiment_polar	0.142	0.157	0.092	0.000	1.000	2.349
D_buzz	0.561	0.540	0.407	0.000	4.662	1.956
D_rela_buzz	0.153	0.154	0.110	0.000	2.000	2.429
D_rela_buzz_polar	0.106	0.114	0.071	0.000	1.000	2.635

of missingness would raise concerns about the representativeness of the available observations and could introduce selection bias if absences are concentrated in low-attention periods. As seen in Figure A.2, the vast majority of features record no missing values. The only exceptions are the six log-change features, each of which records a single missing observation corresponding to the first trading day in the sample, where no lagged value is available to form the daily difference. This represents 0.03% of observations and is addressed by a sequential forward-fill followed by backward-fill prior to estimation, ensuring that isolated interior gaps are filled from the preceding observation while the single leading missing value at the start of the sample is filled from the first available subsequent observation.

Structural zeros (% zeros). The second column reports the percentage of observations that equal exactly zero. Structural zeros are economically distinct from missing values: they indicate days on which the feature is observed but carries no signal, such as days with precisely neutral sentiment or no day-over-day change in attention. A high proportion of zeros

implies a sparse signal that activates only around salient events and behaves differently from a continuously informative predictor. The incidence of structural zeros is heterogeneous across features. Raw sentiment variables and the MCS measures record between 0.2% and 7.7% zeros, reflecting days of precisely neutral coverage. The log-change measures of relative buzz exhibit substantially higher proportions (approximately 35% and 47% for the news and social variants respectively), reflecting the episodic nature of meaningful day-over-day shifts in attention-weighted sentiment.

Skewness. The third column reports the sample skewness coefficient,

$$\hat{\gamma}_1 = \frac{\frac{1}{T}\sum_{t=1}^{T}(x_t - \bar{x})^3}{\left[\frac{1}{T}\sum_{t=1}^{T}(x_t - \bar{x})^2\right]^{3/2}}, \tag{A.2}$$

where $\bar{x}$ denotes the sample mean nd T the number of trading days in the estimation sample (T = 3,633 for AAPL). Strong positive skewness indicates a heavy right tail driven by episodic spikes, which is characteristic of raw buzz and disagreement variables and further motivates the log-transformations applied during feature construction. Skewness values close to zero indicate approximately symmetric distributions suitable for linear models without further transformation.

Outlier incidence (% beyond 3σ). The fourth column reports the percentage of observations falling more than three standard deviations from the sample mean,

$$\widehat{\text{Out}} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{1}[|x_t - \bar{x}| > 3\hat{\sigma}] \times 100, \tag{A.3}$$

where $\hat{\sigma}$ is the sample standard deviation.

Under Gaussianity this quantity equals approximately 0.27%; values substantially above this threshold indicate heavy tails or episodic extreme observations that could unduly influence linear estimators. Two distinct mechanisms drive elevated outlier incidence across the feature set, and it is important not to conflate them. For **raw buzz** variables, the high outlier incidence reflects the heavy-tailed, right-skewed distribution of media attention volumes, which is characterised by rare but very large spikes around firm-specific events; this is precisely the distributional feature that motivates the log-transformation applied during feature construction. For **raw sentiment** variables, however, the interpretation is qualitatively different. Sentiment is bounded within $[-1, 1]$ by construction, so heavy tails and scale cannot account for elevated outlier incidence. Instead, a high percentage beyond $3\hat{\sigma}$ for sentiment reflects days of unusually strong narrative consensus (either strongly positive or strongly negative tone) relative to the typically moderate average sentiment level. This distinction matters for model specification: buzz outliers are a distributional artefact addressed by transformation, whereas sentiment outliers may carry genuine informational content about extreme narrative episodes and **should not be winsorised or discarded** without further investigation.

First-order autocorrelation, AR(1). The fifth column reports the sample first-order autocorrelation coefficient,

$$\hat{\rho}_1 = \frac{\sum_{t=2}^{T}(x_t - \bar{x})(x_{t-1} - \bar{x})}{\sum_{t=1}^{T}(x_t - \bar{x})^2}. \tag{A.4}$$

Persistence is economically informative because highly autocorrelated predictors carry information about the medium-term informational environment, whereas near-zero autocorrelation indicates transient, event-driven signals.

Direction features such as raw sentiment are expected to display moderate persistence, while Instability features (log-change variables) should exhibit near-zero autocorrelation by construction. The MCS variables warrant separate comment: because they are constructed as rolling-window counts of positive and negative sentiment days over 5-day and 22-day windows respectively, consecutive values share all but one observation from the previous window. This mechanical overlap induces very high persistence by construction (particularly for the monthly variant, which recycles 21 out of 22 daily observations) so an AR(1) coefficient close to unity for these variables reflects the overlapping window design rather than any genuine economic persistence in narrative tone. This distinction is important for forecasting model specification: the high autocorrelation of MCS should not be interpreted as evidence of slow-moving sentiment dynamics but rather as a smoothing artefact, and first-differencing or detrending may be warranted before including these variables in levels. Diverging colours in Figure A.2 distinguish features with positive persistence (red) from those that are mean-reverting (blue).

Proportion of days with no change (% no change). The sixth column reports the percentage of consecutive observation pairs with identical values,

$$\widehat{\mathrm{NC}} = \frac{1}{T-1}\sum_{t=2}^{T}\mathbf{1}[x_t = x_{t-1}] \times 100. \tag{A.5}$$

This diagnostic is complementary to the zero count: a feature may be non-zero but constant over consecutive days, indicating that it carries no new information on those days. A high proportion of unchanged observations suggests a feature that is effectively flat for extended periods, which reduces its value as a high-frequency forecasting signal.

The MCS variables display particularly high values on this diagnostic, with the proportion of unchanged days ranging from approximately 53% to 62% depending on the channel and window length. This is again a direct consequence of the overlapping rolling-window construction: when the entering and exiting days carry the same sentiment classification, the counts of positive and negative days (and therefore the MCS score) remain identical to the previous day. The monthly variant is affected more severely than the weekly variant because its longer window makes it less sensitive to any single day's coverage. Taken together with the near-unit AR(1) documented above, the high no-change incidence confirms that the MCS variables behave more like slow-moving smoothed indicators than daily signals, and their informational contribution at the daily forecasting horizon should be interpreted accordingly.

Interquartile range (IQR). The seventh column reports the sample interquartile range, $\widehat{\mathrm{IQR}} = \hat{Q}_{0.75} - \hat{Q}_{0.25}$, where $\hat{Q}_p$ denotes the p-th sample quantile. The IQR measures the

typical day-to-day dispersion of the feature in a manner that is robust to the extreme values that inflate the standard deviation for skewed distributions. A very small IQR relative to the overall range flags features that are near-constant on most days with rare large spikes, a pattern that characterises several buzz and disagreement variables and is relevant for model stability.

Signal-to-noise ratio (SNR). The eighth column reports a robust signal-to-noise ratio defined as the ratio of the sample median to the interquartile range,

$$\widehat{\mathrm{SNR}} = \begin{cases} \dfrac{\tilde{x}}{\widehat{\mathrm{IQR}}} & \text{if } \widehat{\mathrm{IQR}} > 0 \\ 0 & \text{otherwise} \end{cases} \tag{A.6}$$

where $\tilde{x}$ denotes the sample median. A zero IQR indicates that more than half of observations take an identical value, implying negligible day-to-day variation; in this case the SNR is set to zero, reflecting the absence of a meaningful signal relative to dispersion. Features with a high absolute SNR carry a large typical signal relative to their noise, whereas near-zero SNR indicates that the median is small relative to the dispersion, characteristic of variables that fluctuate symmetrically around zero with limited directional content. Diverging colours distinguish features with positive SNR (net positive median, red) from those with negative SNR (net negative median, blue).

Two findings from this column are particularly noteworthy. First, the log-transformed buzz variables display substantially higher SNR than their raw counterparts for both channels, confirming that the log-transformation is not merely a variance-stabilising device but also materially improves the signal quality of the attention measure. The raw buzz variables for both news and social media exhibit relatively low SNR, reflecting the fact that their distributions are dominated by extreme observations that inflate the IQR without proportionally raising the median. After log-transformation, the SNR rises markedly, indicating that the transformed series carries a more consistent and reliable level of attention signal relative to its day-to-day variation.

Second, a very notable asymmetry emerges between the two media channels: the SNR of $\ln buzz^{\mathrm{S}}$ (approximately 6.30) is substantially larger than that of $\ln buzz^{\mathrm{N}}$ (approximately 3.87), despite both variables being constructed identically. This difference indicates that social media attention is considerably more concentrated and directionally consistent relative to its dispersion than traditional news coverage, suggesting that the social media channel carries a more pronounced and stable informational signal at the daily horizon. This asymmetry provides additional empirical justification for treating the two channels as distinct predictors rather than pooling or averaging them, and is consistent with the view that social media reflects a more focused and persistent form of investor attention than the broader and more heterogeneous news flow.

Our overall assessment using the eight diagnostics in Figure A.2 confirms that the 27 features are of high quality and that their distributional properties are broadly consistent with the economic interpretation assigned to each informational panel, while also revealing several features that warrant careful treatment in the subsequent analysis. *Direction* features display moderate persistence and near-symmetric distributions after transformation. A notable exception within this panel is the MCS variables, which exhibit near-unit AR(1) coefficients and

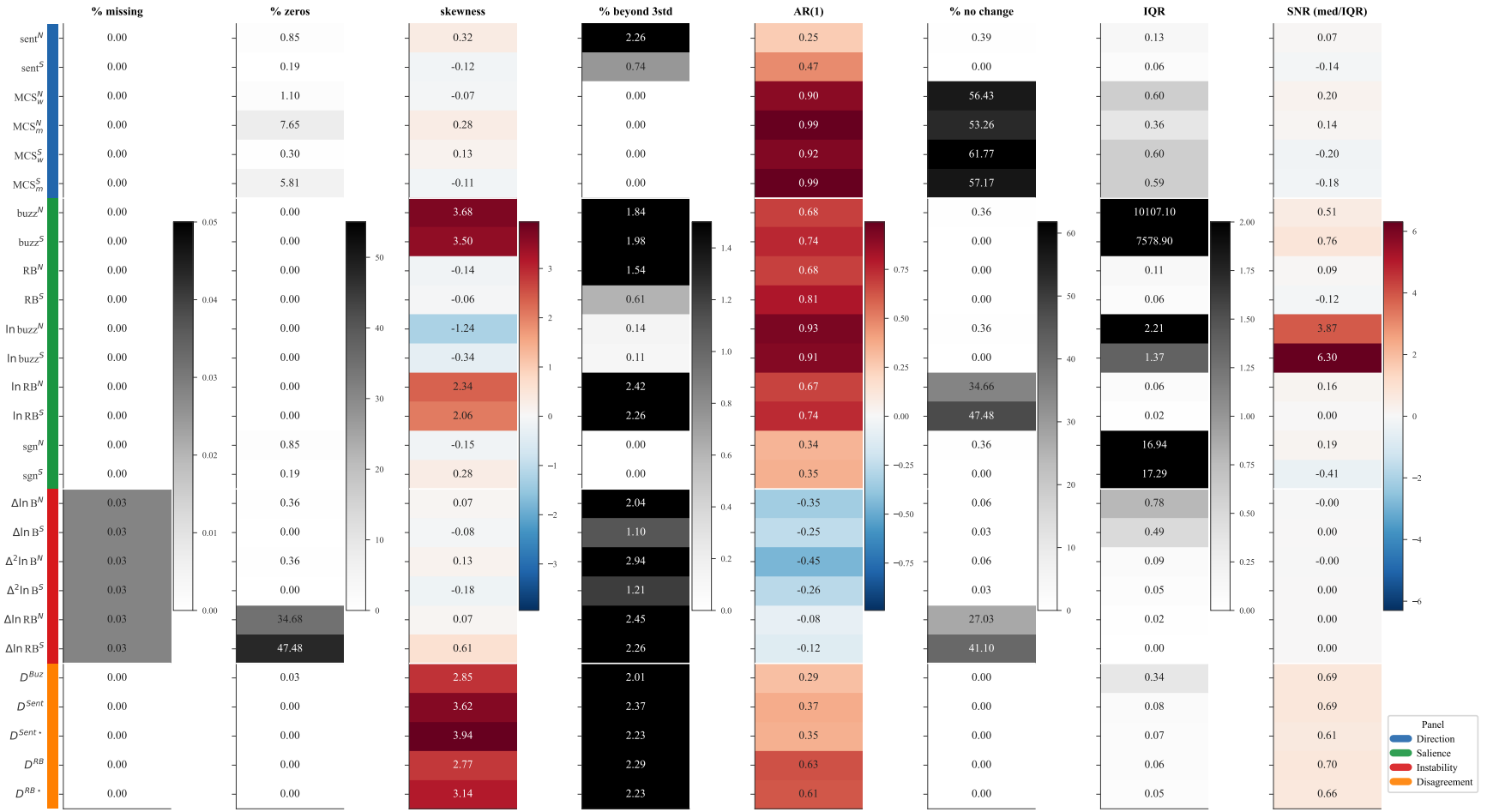


FIGURE A.2: DATA QUALITY DIAGNOSTICS FOR THE 27 MEDIA-BASED PREDICTORS. Each row corresponds to one of the 27 features defined in Table A.3, ordered by informational panel as indicated by the colour strip on the left margin: **Direction** (blue), **Salience** (green), **Instability** (red), and **Disagreement** (orange). Each column reports one diagnostic statistic, colour-coded by intensity. **% missing**: percentage of firm-day observations with missing values. **% zeros**: percentage of observations equal to exactly zero. **Skewness**: sample skewness coefficient $\hat{\gamma}_1$. **% beyond 3σ**: percentage of observations more than three standard deviations from the mean. **AR(1)**: first-order autocorrelation coefficient $\hat{\rho}_1$. **% no change**: percentage of consecutive observation pairs with identical values. **IQR**: interquartile range $\hat{Q}_{0.75} - \hat{Q}_{0.25}$. **SNR**: robust signal-to-noise ratio $\tilde{x}/\widehat{\text{IQR}}$ indicates positive (negative) median relative to dispersion. Analysis conducted for AAPL ($N = 3{,}633$ trading days, 2002–2016); diagnostics for all other DJIA constituents are qualitatively similar and are omitted for brevity.

no-change incidence exceeding 50%. These properties arise simply mechanically from their overlapping rolling-window construction rather than from genuine economic persistence, but it does suggest that these variables are better interpreted as slow-moving smoothed indicators than as daily signals. *Salience* features exhibit strong right skewness in raw form, motivating the log-transforms applied during construction; after transformation, skewness is substantially reduced and the SNR improves markedly, confirming that the log-transform improves signal quality beyond mere variance stabilisation. A notable asymmetry between channels is also apparent: the SNR of log-transformed social media buzz is substantially higher than its news counterpart, providing additional empirical support for treating the two channels as distinct predictors. *Instability* features display negative first-order autocorrelation across all six variables, consistent with the mean-reverting dynamics expected of first-difference constructions where large daily changes tend to be followed by partial reversals. The SNR is effectively zero across the entire panel, confirming that these variables are symmetric signals with no persistent directional bias. *Disagreement* features display a consistent and distinctive profile. Being non-negative by construction, they exhibit strong right skewness with virtually no zero incidence or unchanged days, indicating that some degree of cross-channel conflict is present on almost every trading day. Autocorrelation is moderate and positive, suggesting that elevated disagreement tends to persist over several consecutive days. The SNR is consistently positive and stable across the panel, which supports their interpretation as persistent informational state variables rather than transient shock indicators.

No feature displays a problematic incidence of missing values; the handful of missing observations introduced by the log-change construction at the start of the sample are addressed through a sequential forward-fill followed by backward-fill, ensuring robustness to both interior gaps and leading missing values. These diagnostics are conducted separately for each of the DJIA constituents; results are qualitatively similar across firms and are omitted for brevity.

A.4.2 Time-Series Dynamics

Figure A.3 plots the time series of all 27 standardised features over the full sample period (July 2002 to December 2016), organised into the four informational panels introduced in Section 3: Direction, Salience, Instability, and Disagreement. Within each panel, features are standardised to unit variance for visual comparability across variables with different natural scales.

Several patterns are noteworthy. Direction features (Panel A) exhibit moderate persistence with occasional sharp reversals, consistent with the episodic nature of narrative shifts in media coverage. Salience features (Panel B) display pronounced spikes concentrated around firm-specific events, product launches, and earnings announcements, and are characterised by strong right skewness in their raw distributions. The log-transformation visibly compresses these extremes. Instability features (Panel C) are considerably less persistent than sentiment levels, fluctuating around zero with symmetric distributions, consistent with their interpretation as first-difference or growth-rate measures. Disagreement features (Panel D) are strictly non-negative and exhibit sustained elevated periods, particularly during the global financial crisis (2008–2009) and the period of heightened market volatility in 2011, suggesting that cross-channel informational disagreement is time-varying and state-dependent.



FIGURE A.3: STANDARDISED FEATURE TIME SERIES BY INFORMATIONAL DIMENSION. Each panel plots the standardised (zero mean, unit variance) daily time series of all features belonging to the corresponding informational dimension over the full sample period (July 2002 to December 2016). **Panel A** (Direction): six sentiment and MCS variables capturing the tone of media coverage. **Panel B** (Salience): ten buzz, relative buzz, log-buzz, and signed-buzz variables capturing the intensity of media attention and its interaction with sentiment direction. **Panel C** (Instability): six log-change variables measuring day-over-day dynamics in attention and attention-weighted sentiment. **Panel D** (Disagreement): five cross-channel divergence variables measuring the gap between news and social media in tone, polarity, and buzz. Within each panel, individual feature series are distinguished by monochromatic shading from dark to light within the panel's colour. All series are standardised within the full sample before plotting. Analysis conducted for AAPL; results for other DJIA constituents are qualitatively similar and omitted for brevity.

A.4.3 Cross-Sectional Distributions

Figure A.4 displays the marginal distributions of all 27 standardised features using a combined violin and box-plot representation. Features are ordered by informational panel and labelled using the shorthand notation introduced in Section 3.

The distribution plots confirm several features of the data that are consequential for the design of the forecasting models. Raw buzz features ($buzz^{\mathrm{N}}$, $buzz^{\mathrm{S}}$) are extremely right-skewed with heavy tails, consistent with the count-like nature of media volume; their log-transformed counterparts are substantially more symmetric. Relative buzz features ($\mathtt{RB}^{\mathrm{N}}$, $\mathtt{RB}^{\mathrm{S}}$) are bounded within $[-1, 1]$ and exhibit moderate skewness. Instability features are near-symmetric and leptokurtic, with excess kurtosis driven by episodic bursts. Disagreement features are bounded below at zero and strongly right-skewed, with most mass concentrated near zero and occasional extreme values during high-volatility periods. These distributional properties collectively motivate the standardisation step applied prior to PCA and forecasting

model estimation.

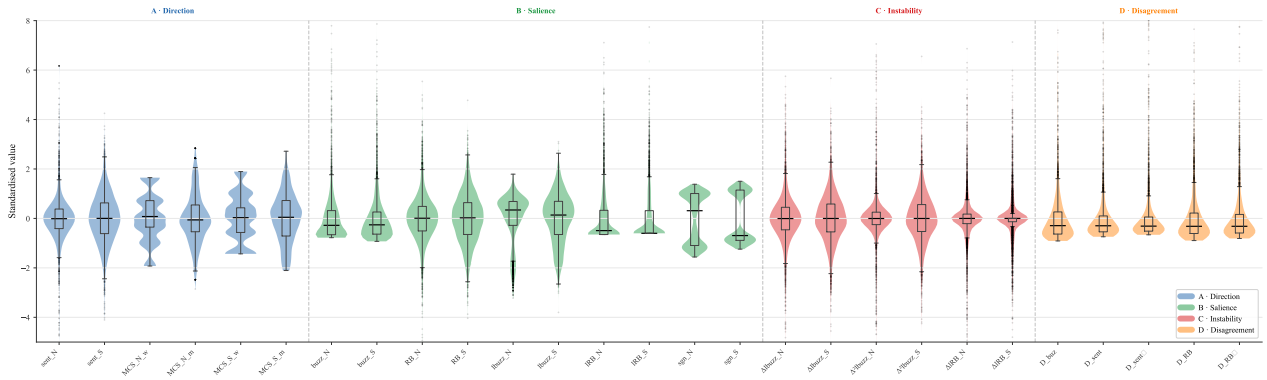


FIGURE A.4: MARGINAL DISTRIBUTIONS OF 27 MEDIA-BASED FEATURES. The figure displays, for each of the 27 features used in the paper, the marginal distribution of standardised daily values using a combined violin and box plot. The violin (shaded region) traces a kernel density estimate of the empirical distribution; the box shows the interquartile range, whiskers extend to 1.5 times the IQR, and individual outliers are plotted as small grey dots. Features are ordered left-to-right by informational panel: **Direction** (6 features), **Salience** (10 features), **Instability** (6 features), and **Disagreement** (5 features), separated by dashed vertical dividers. All values are standardised to zero mean and unit variance. Analysis conducted for AAPL ($N = 3{,}633$ trading days, 2002–2016); results for other DJIA constituents are qualitatively similar and omitted for brevity.

A.4.4 Pairwise Correlation Structure

Figure A.5 presents the lower-triangular correlogram of pairwise Pearson correlations for all 27 features. Coloured boundary boxes highlight the four informational panels.

The correlation structure reveals several patterns that are directly relevant to the choice of dimension-reduction strategy. Within the Salience panel, raw and log-transformed buzz variables are very strongly correlated (typically $r > 0.95$), as are the news and social media variants of the same transformation. Similarly, within the Direction panel, raw sentiment and MCS measures share substantial common variation. Across panels, correlations are generally weaker, though non-trivial dependence exists between Direction and Salience features through the relative-buzz variables, which embed both tone and attention. Disagreement features are largely orthogonal to Direction and Salience variables, consistent with their construction as absolute cross-channel differences rather than directional signals.

This pattern of strong within-panel and moderate cross-panel correlation motivates the two-stage PCA procedure described in Section 3.2 and Appendix B, which preserves the panel-level informational structure before extracting a final low-dimensional factor representation.

A.4.5 Cross-Channel Disagreement Dynamics

Figure A.6 isolates the five Disagreement features and plots their daily values alongside a 60-day rolling mean over the full sample. The disagrement variables capture the gap between traditional news and social media in sentiment tone, sentiment intensity (polarity), buzz volume, and attention-weighted sentiment.

All five series display clear time variation with episodes of sustained elevated disagreement, suggesting that the degree of cross-channel informational conflict is itself a state variable. Disagreement peaks cluster around well-known periods of market stress and heightened un-

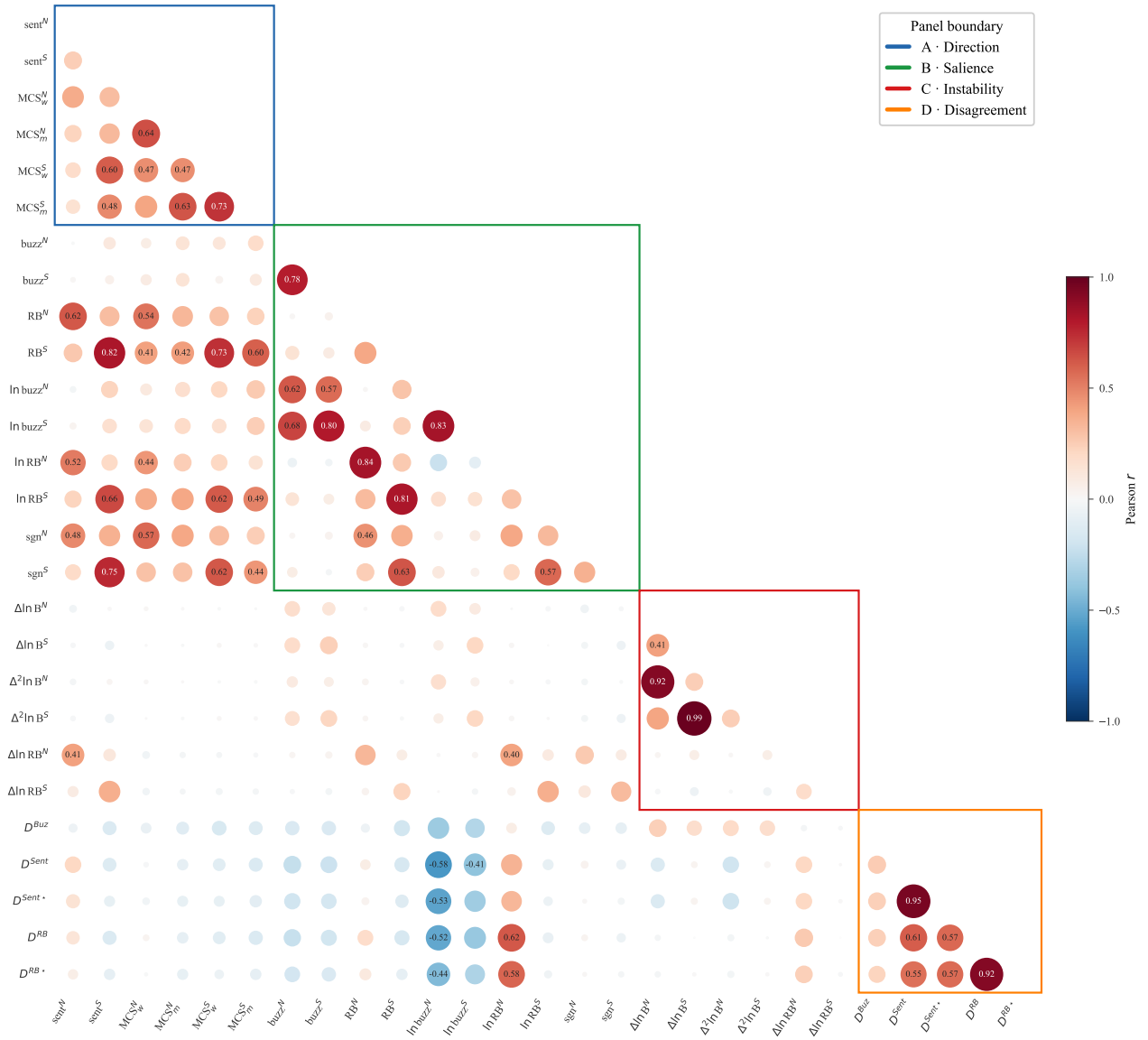


FIGURE A.5: CORRELOGRAM OF PAIRWISE CORRELATIONS AMONG 27 MEDIA-BASED FEATURES. Each cell in the lower triangle displays a filled circle whose *colour* encodes the sign and magnitude of the Pearson correlation coefficient (r) and whose *radius* is proportional to $|r|^{0.5}$, with red (blue) indicating strong positive (negative) dependence and white indicating near-zero dependence; the colour scale spans $[-1, 1]$. Correlation values are annotated inside circles where $|r| > 0.4$. The diagonal is omitted. Features are ordered by informational panel (i.e., Direction, Salience, Instability, Disagreement) and coloured boundary boxes identify the four panel blocks (see legend). The strong within-panel correlations motivate the two-stage PCA procedure used in the main analysis (see Section 3.2 and Appendix B). Analysis conducted for AAPL; results for other DJIA constituents are qualitatively similar and omitted for brevity.

certainty, including the 2008–2009 global financial crisis, the 2010 flash crash, and the European sovereign debt episode of 2011. This time-variation is economically significant: if volatility responds not only to the content of media narratives but also to their cross-channel coherence, then disagreement variables may provide incremental forecasting power beyond sentiment and attention variables alone. The evidence in Section 5 is consistent with this hypothesis.

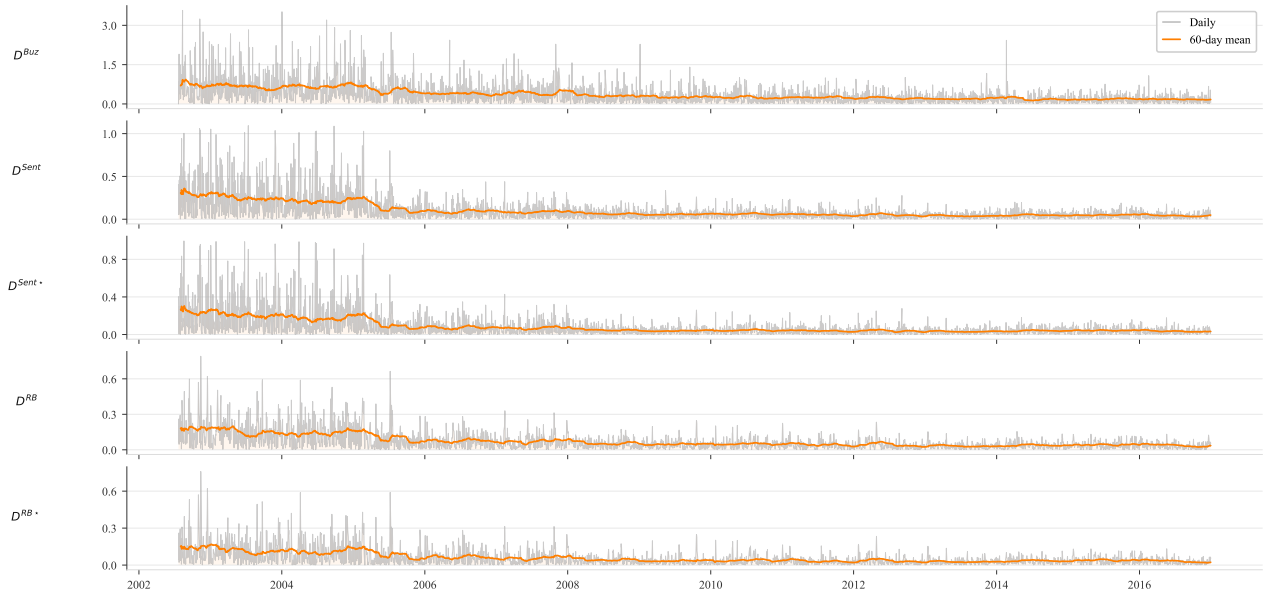


FIGURE A.6: CROSS-CHANNEL DISAGREEMENT FEATURES OVER TIME. The figure plots the daily values (grey line) and 60-day rolling mean (coloured line) of the five Disagreement features over the full sample period (July 2002 to December 2016). From top to bottom: D^{Sent} (absolute difference in raw sentiment between news and social media); $D^{\text{Sent}\star}$ (absolute difference in sentiment polarity magnitudes); D^{Buz} (absolute difference in normalised buzz volumes); D^{RB} (absolute difference in attention-weighted sentiment); $D^{\text{RB}\star}$ (absolute difference in attention-weighted sentiment magnitudes). All variables are non-negative by construction. Sustained elevated disagreement is visible around the 2008–2009 global financial crisis and other high-uncertainty episodes, suggesting that cross-channel informational conflict is state-dependent and time-varying. Analysis conducted for AAPL; results for other DJIA constituents are qualitatively similar and omitted for brevity.

A.4.6 News versus Social Media Channel Comparison

The distinction between channels is economically relevant because the two media environments process and transmit information differently. Traditional news is more curated and institutionally filtered, reflecting the views of professional analysts and journalists (Tetlock, 2007), whereas social media reacts faster, is more fragmented, and better captures shifts in retail investor attention and emerging narratives (Bollen et al., 2011; Da et al., 2011). Importantly, the two channels do not merely differ in speed or source: Gan et al. (2020) show directly that news and social media sentiment have distinct effects on asset prices, and Antweiler and Frank (2004) and Chen et al. (2014) establish that user-generated content carries information about volatility and returns that is not subsumed by traditional news flow. Pooling or averaging across channels would therefore discard economically meaningful cross-channel variation, including the disagreement signals documented in Section A.4.5.

Figure A.7 examines this distinction directly by plotting the standardised distributions of news and social media variants of all 11 matched feature pairs side by side, with the Pearson correlation coefficient between the two channels annotated in each panel.

The results confirm that news and social media carry related but non-redundant information. Correlations range from moderately high (for buzz and log-buzz, where both channels co-move with overall market activity) to low or near-zero (for log-change features, which capture idiosyncratic timing differences in how the two channels respond to new information). The distributions also differ systematically: social media sentiment tends to be slightly more negatively skewed than its news counterpart, and social media buzz exhibits greater dispersion.

These differences justify retaining news and social media as separate channels in the predictor construction rather than pooling or averaging them.

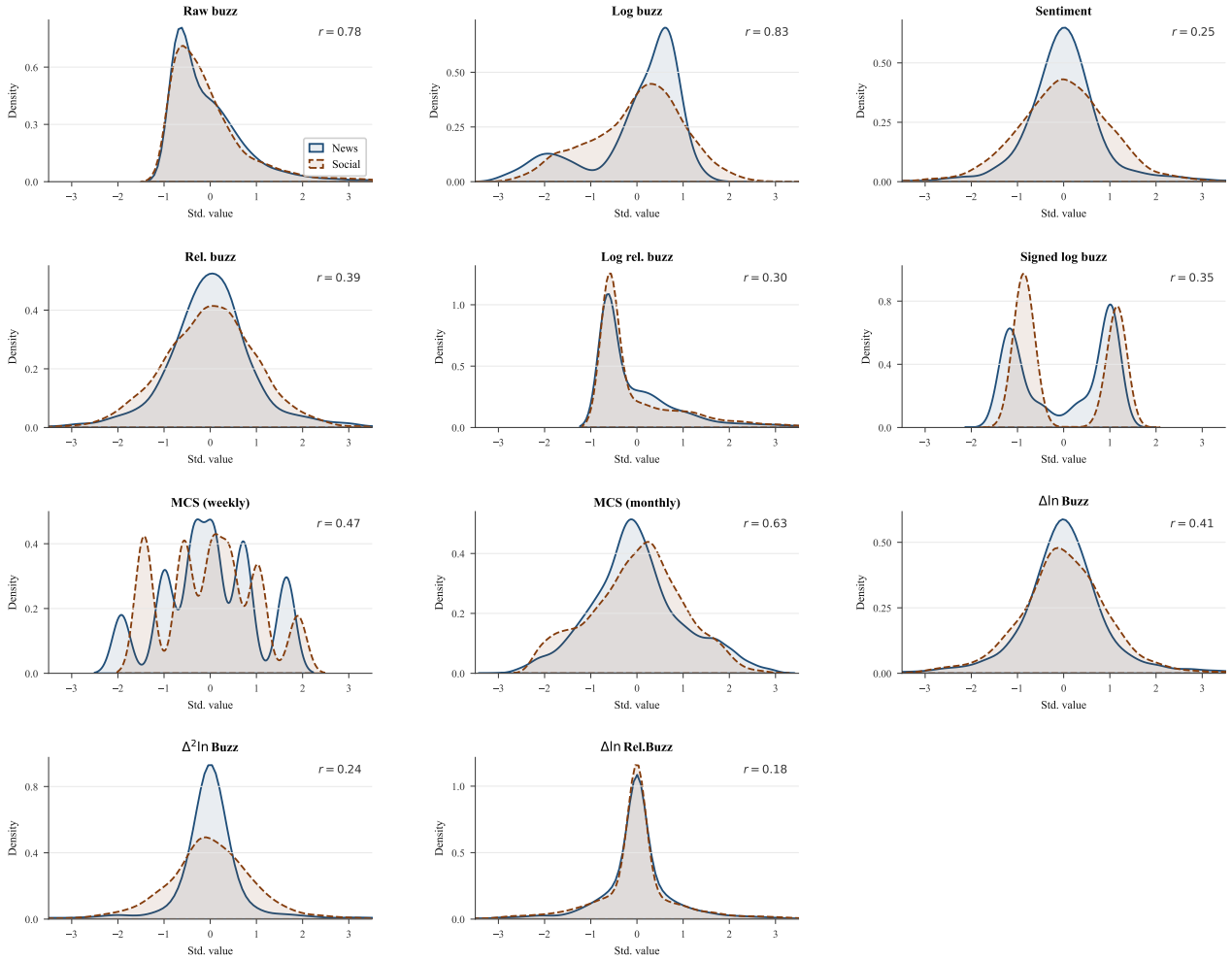


FIGURE A.7: NEWS VERSUS SOCIAL MEDIA CHANNEL DISTRIBUTIONS. Each sub-panel plots the kernel density estimate of the standardised daily values for the news-channel (navy, solid) and social-media-channel (sienna, dashed) variants of the same underlying variable. The Pearson correlation coefficient between the two channels is annotated in the upper-right corner of each sub-panel. The 11 pairs shown correspond to the matched News/Social variants of: raw buzz, log buzz, raw sentiment, relative buzz, log relative buzz, signed log buzz, weekly MCS, monthly MCS, Δ lnBuzz, Δ^2 lnBuzz, and Δ lnRel.Buzz. All values are standardised to zero mean and unit variance within the sample. The consistently imperfect correlations and distributional differences between channels justify retaining news and social media as distinct information sources rather than pooling or averaging them. Analysis conducted for AAPL; results for other DJIA constituents are qualitatively similar and omitted for brevity.

B Dimension Reduction: Technical Details

This section provides the formal description of the two PCA strategies described in Section 3.2, followed by an illustrative analysis for Apple Inc. (AAPL). Results for all other DJIA constituents are qualitatively similar and available upon request.

B.1 One-Stage PCA

Let $\mathbf{X}_{i,t} \in \mathbb{R}^N$ denote the vector of $N = 27$ standardised media predictors for firm i on day t, where standardisation is applied within the estimation window to avoid look-ahead bias. PCA extracts the principal component scores via:

$$\mathbf{F}_{i,t} = \mathbf{X}_{i,t}\,\mathbf{W}, \tag{B.1}$$

where the columns of $\mathbf{W} \in \mathbb{R}^{N\times N}$ are the eigenvectors of the sample covariance matrix of $\mathbf{X}_{i,t}$, ordered by descending eigenvalue. We retain a fixed number K of leading components, chosen once so that the cumulative variance share

$$\overline{\frac{\sum_{j=1}^{K}\lambda_j}{\sum_{j=1}^{N}\lambda_j}} \approx 0.80, \tag{B.2}$$

where λ_j is the j-th eigenvalue and the bar denotes the average across estimation windows ($K = 4$); the number is not re-selected window by window, so that the dimension of the media component is the same in every window. This yields the global component set $\{F_{1,i,t}, \ldots, F_{K,i,t}\}$, which enters the HARPCA forecasting model. This approach is attractive because it is agnostic about which specific transformed variables matter most and instead lets the data identify the dominant latent structure.

B.2 Two-Stage PCA

Stage 1: Within-cluster compression. The N predictors are partitioned into G clusters $\{\mathbf{X}_{i,t}^{(g)}\}_{g=1}^{G}$, either via HAC on one-stage PCA loadings (data-driven) or via the six channel-by-signal groups of Section 3.2 (news sentiment, social sentiment, news attention, social attention, relative sentiment, disagreement) (manual). Within each cluster g, PCA is applied independently:

$$\mathbf{F}_{i,t}^{(g)} = \mathbf{X}_{i,t}^{(g)}\,\mathbf{W}^{(g)}, \tag{B.3}$$

where $\mathbf{W}^{(g)}$ contains the eigenvectors of the within-cluster covariance matrix. For HAR2PCA and HARmPCA, only the single leading component $F_{1,i,t}^{(g)}$ is retained. For HARmPCAopt, the number of retained components n_g is fixed per group, chosen so that the retained components explain about 80% or more of within-group variance on average (91% over the full sample).

Stage 2: Cross-cluster combination. The retained cluster summaries, $\mathbf{Z}_{i,t} = [F_{1,i,t}^{(1)}, \ldots, F_{1,i,t}^{(G)}]$, form the exogenous input vector $\mathbf{F}_t$ that enters the forecasting models directly without a further compression step. This preserves the cluster-level economic interpretation of each component rather than further blending them into undifferentiated global factors.

Data-driven cluster construction. For HAR2PCA, the G clusters are obtained by applying HAC with cosine distance to the one-stage PCA loading matrix $\mathbf{W}$. Clustering on loadings rather than raw data ensures that the partitions reflect systematic co-movement in the predictor space. The number of clusters G is selected using the silhouette criterion (Rousseeuw, 1987). Cosine distance is used because it captures directional alignment among loading vectors while remaining invariant to magnitude differences, which is appropriate when predictors are standardised (Gray et al., 2024).

All standardisation, covariance estimation, and PCA decompositions are performed within the rolling 252-day estimation window, using only information available at the time of forecasting. No transformation or component extraction step crosses the window boundary.

B.3 Illustrative PCA Analysis: Apple Inc. (AAPL)

The following figures characterise the PCA output for AAPL and motivate the design choices described above.

Explained variance and component selection. Figure B.1 presents the scree plot and cumulative explained variance for the first 15 principal components. The predictor space is intrinsically low-dimensional: the first three components account for 51.6% of total variance, and the cumulative curve crosses the 80% threshold at PC9 (82.6%). The scree plot declines steadily without a single dominant elbow, consistent with a predictor set in which information is distributed across several correlated but distinct informational dimensions rather than concentrated in one or two factors. This gradual decay further motivates retaining up to PC9 in the one-stage specification and confirms that discarding the trailing components does not sacrifice the bulk of common variation.

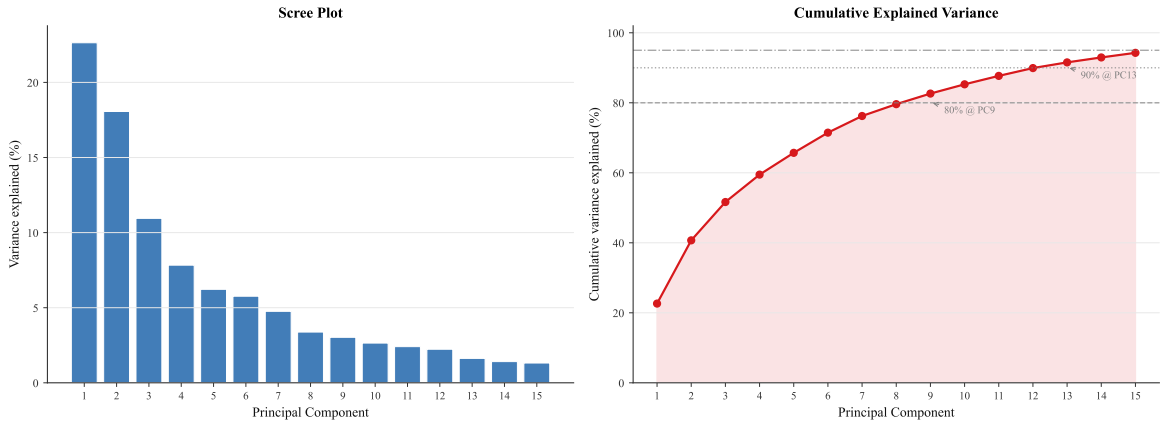


FIGURE B.1: PCA SCREE PLOT AND CUMULATIVE EXPLAINED VARIANCE (AAPL). The left panel displays the share of total variance explained by each of the first 15 principal components extracted from the full 27-dimensional standardised predictor covariance matrix. The right panel plots cumulative explained variance with reference lines at the 80%, 90%, and 95% thresholds annotated with the corresponding component count. The steady, gradual decline of the scree plot without a single sharp elbow indicates that predictive information is distributed across several correlated dimensions rather than concentrated in one dominant factor. Analysis conducted for AAPL; results for other DJIA constituents are qualitatively similar and omitted for brevity.

Loading structure and economic interpretation. Figure B.3 presents the loading matrix for the first nine principal components (explaining 82.6% of total variance). The loadings

reveal a latent structure that partially, but not perfectly, aligns with the four-family taxonomy. PC1 (22.6%) is dominated by social media sentiment and salience variables (RB^{S}, $sent^{\mathrm{S}}$, MCS_w^{S}, MCS_m^{S}, and $\ln RB^{\mathrm{S}}$ carry the largest loadings) alongside their news counterparts (MCS_w^{N}, MCS_m^{N}, sgn^{N}). Rather than capturing a single family, PC1 reflects the *common social-media-driven narrative signal*: days when social sentiment is strong and media engagement is elevated across both channels.

PC2 (18.1%) is most heavily loaded by Disagreement variables: D^{RB}, D^{Sent}, $D^{\mathrm{RB}\star}$, and $D^{\mathrm{Sent}\star}$ all carry loadings of approximately 0.31–0.34, while log-transformed news buzz ($\ln buzz^{\mathrm{N}}$, $\ln buzz^{\mathrm{S}}$) loads negatively. PC2 therefore captures the *tension between cross-channel divergence and news attention*: days of high cross-channel disagreement tend to coincide with lower news volume, and vice versa.

PC3 (10.9%) is cleanly dominated by Instability variables ($\Delta \ln buzz^{\mathrm{S}}$, $\Delta^2 \ln buzz^{\mathrm{S}}$, $\Delta \ln buzz^{\mathrm{N}}$, and $\Delta^2 \ln buzz^{\mathrm{N}}$) making it a pure *day-over-day change in media attention* factor. PC4 through PC6 capture progressively more specific contrasts: PC4 separates raw buzz levels (Salience, positive) from log-change buzz (Instability, negative); PC6 loads on $\Delta \ln RB^{\mathrm{S}}$ and $\Delta \ln RB^{\mathrm{N}}$ positively against monthly MCS variables negatively, contrasting short-run attention dynamics with longer-horizon narrative direction.

Two findings from this loading structure are economically consequential. First, the Disagreement family is not relegated to later components: it emerges prominently in PC2, the second most important dimension of the predictor space, indicating that cross-channel informational conflict is a primary rather than residual source of variation. Second, the leading components do not map one-to-one onto the four families, which vindicates the two-stage PCA design: by compressing each family separately before combining, the two-stage procedure preserves economically distinct channels that a single global extraction conflates.

Biplot geometry. Figures B.2 and B.4 present two-dimensional biplots for the PC1–PC2 and PC1–PC3 planes, confirming the loading structure documented in Figure B.3. In the PC1–PC2 plane (Figure B.2), social media sentiment and salience arrows (RB^{S}, $sent^{\mathrm{S}}$, MCS_w^{S}, MCS_m^{S}) cluster along the positive PC1 axis, consistent with PC1 capturing the common social-channel narrative signal. Disagreement arrows (D^{RB}, D^{Sent}, $D^{\mathrm{RB}\star}$, $D^{\mathrm{Sent}\star}$) extend prominently along the positive PC2 axis, confirming that PC2 is primarily a cross-channel divergence factor rather than a Direction factor. News log-buzz arrows ($\ln buzz^{\mathrm{N}}$, $\ln buzz^{\mathrm{S}}$) point in the negative PC2 direction, capturing the inverse relationship between news attention volume and cross-channel disagreement documented in the loading matrix. In the PC1–PC3 plane (Figure B.4), Instability arrows ($\Delta \ln buzz^{\mathrm{S}}$, $\Delta^2 \ln buzz^{\mathrm{S}}$, $\Delta \ln buzz^{\mathrm{N}}$, $\Delta^2 \ln buzz^{\mathrm{N}}$) emerge cleanly along the PC3 axis, largely orthogonal to the PC1 social-channel signal. In this projection, Disagreement arrows appear shorter, as their primary variation is captured by PC2, which is not shown in this plane.

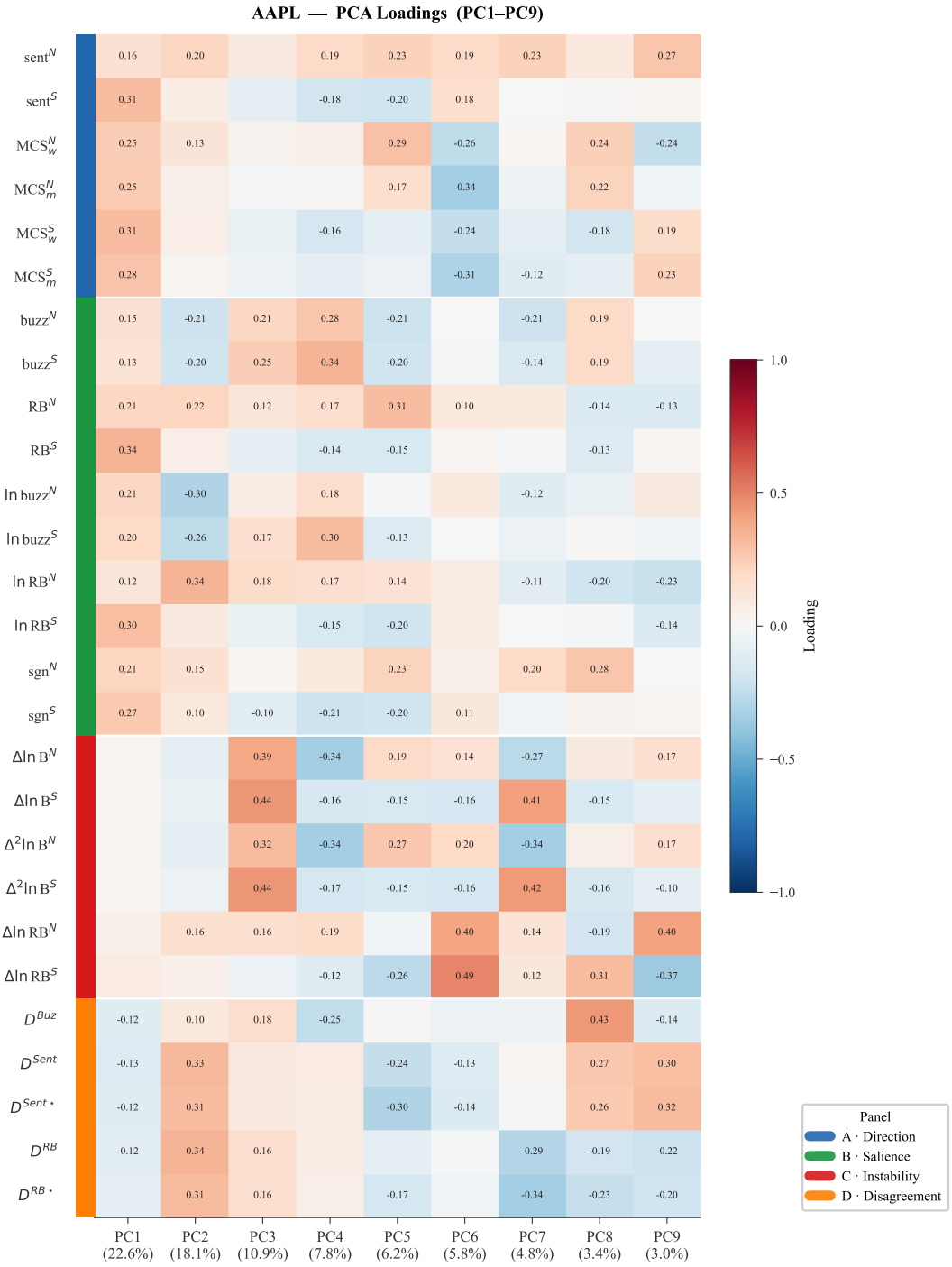


FIGURE B.3: PCA LOADING MATRIX, PC1–PC9 (AAPL). Colour encodes loading magnitude and sign: red (blue) indicates a large positive (negative) loading; white indicates near-zero contribution. The percentage of total variance explained by each component is reported in parentheses on the x-axis. Loading values are annotated where $|\text{loading}| > 0.10$. The colour strip along the left margin identifies the informational family of each feature: **Direction** (blue), **Salience** (green), **Instability** (red), and **Disagreement** (orange). PC1 captures a common social-media narrative signal, mixing Direction and Salience from the Social channel. PC2 is dominated by Disagreement variables contrasted against news attention volume. PC3 is a clean Instability factor (day-over-day changes in buzz). Higher components capture progressively finer Salience-vs-Instability contrasts and channel-specific variation. The loading structure motivates the two-stage PCA: the four families are not cleanly separated by global PCA, confirming that within-family compression before cross-family aggregation better preserves economic structure. Analysis conducted for AAPL; results for other DJIA constituents are qualitatively similar and omitted for brevity.

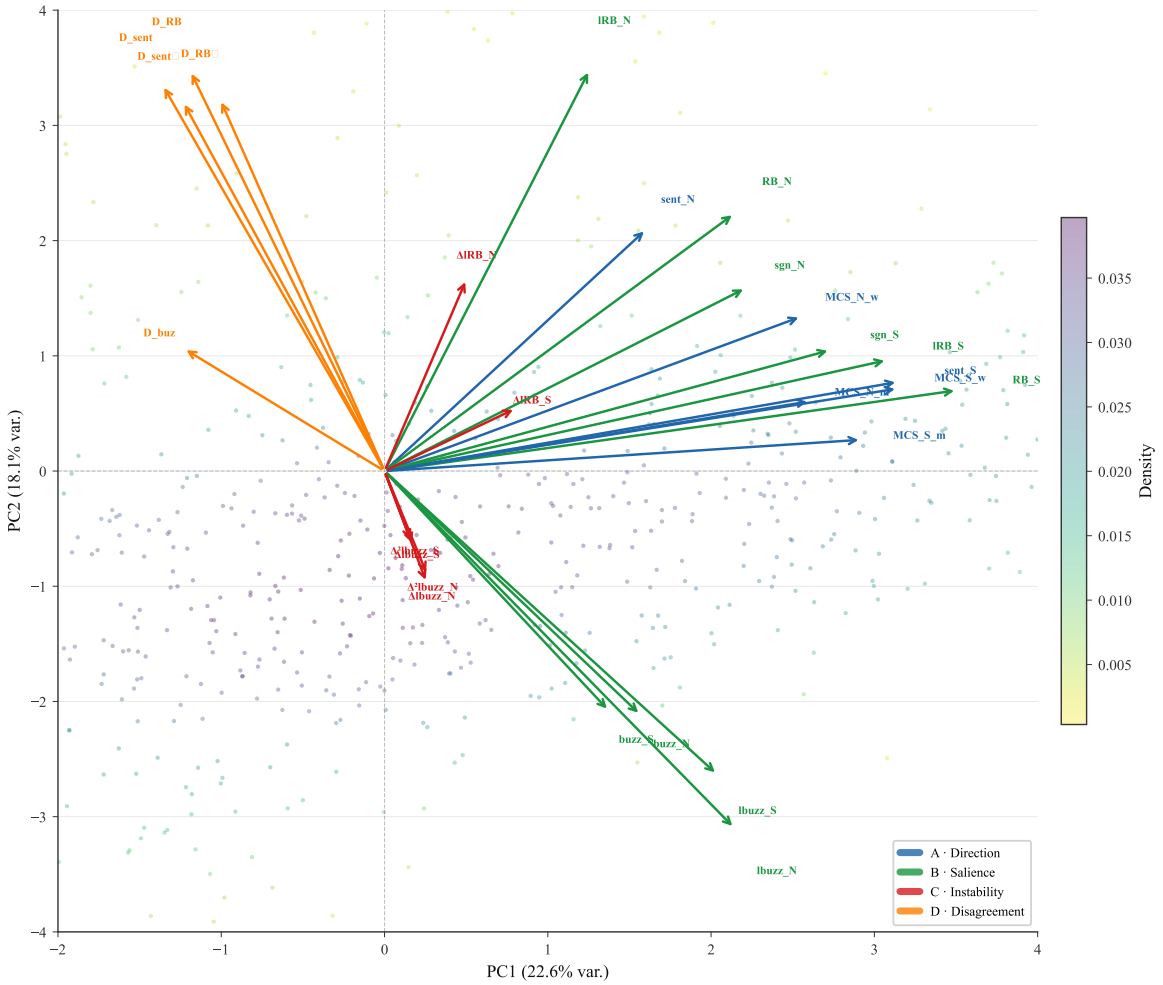


FIGURE B.2: TWO-DIMENSIONAL PCA BIPLOT: PC1 × PC2 (AAPL). The biplot displays the joint projection of all 3,633 firm-day observations (point cloud, kernel density colouring) and the loading vectors of the 27 media features (arrows) onto the plane spanned by the first two principal components. The percentage of total predictor variance explained by each component is reported on the axis label. Arrow colour identifies the informational family: **Direction** (blue), **Salience** (green), **Instability** (red), and **Disagreement** (orange). Arrow length is proportional to loading magnitude. Dashed reference lines mark the origin. Social media sentiment and salience features cluster along the positive PC1 axis; Disagreement features extend along the positive PC2 axis, with news log-buzz loading negatively on PC2, reflecting the inverse relationship between news attention volume and cross-channel divergence. Analysis conducted for AAPL; results for other DJIA constituents are omitted for brevity.

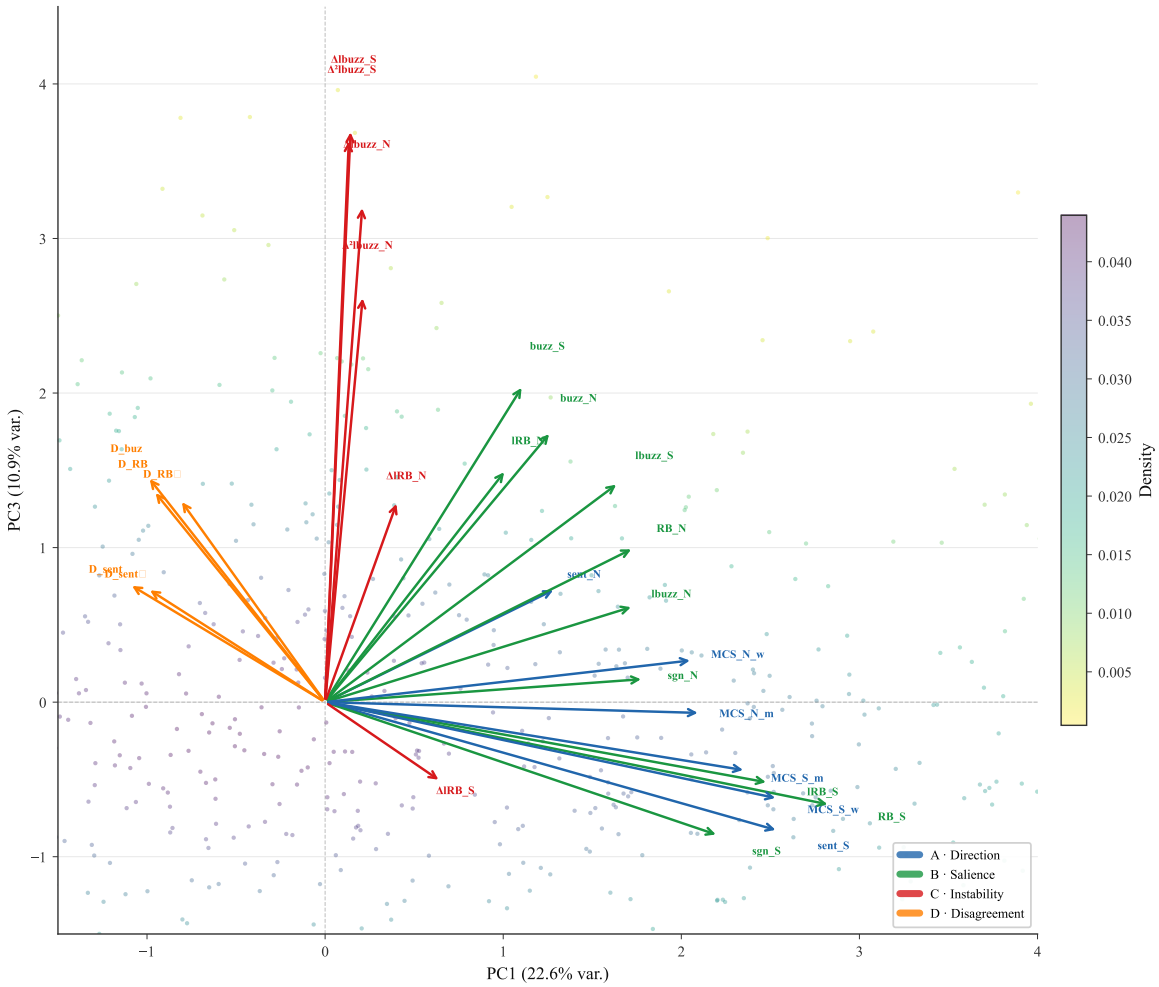


FIGURE B.4: TWO-DIMENSIONAL PCA BIPLOT: PC1 × PC3 (AAPL). As Figure B.2, projecting onto the PC1–PC3 plane. Instability features emerge as the primary drivers of PC3, with day-over-day log-change buzz arrows ($\Delta \ln buzz^{N}$, $\Delta \ln buzz^{S}$, $\Delta^2 \ln buzz^{N}$, $\Delta^2 \ln buzz^{S}$) pointing cleanly along the PC3 axis and largely orthogonal to the social-channel signal on PC1. Disagreement arrows appear short in this projection because their dominant variation is captured by PC2, which is not displayed here. Analysis conducted for AAPL; results for other DJIA constituents are omitted for brevity.

Three-dimensional loading geometry. Figure B.5 embeds all 27 loading vectors simultaneously in the PC1–PC2–PC3 space to characterise the joint geometry. The three-dimensional representation reveals that the informational families do not occupy fully distinct regions: social Direction and Salience features overlap considerably along PC1, reflecting their shared sensitivity to the overall social media narrative signal. What the 3D biplot clarifies, however, is the role of the two remaining families: Disagreement arrows extend along the PC2 axis (visible in the PC1–PC2 face of the 3D space), while Instability arrows extend along PC3. Disagreement arrows appear near the origin only in the PC1–PC3 projection (the plane in which neither PC2, their dominant axis, nor the features themselves are well represented). This geometry confirms that the three leading global components collectively account for Direction-Salience overlap (PC1), cross-channel Disagreement (PC2), and temporal Instability (PC3), while blending Direction and Salience into a single social-channel factor rather than separating them. This is precisely the limitation that motivates the two-stage PCA: compressing each family separately before aggregation preserves the Direction–Salience dis-

tinction that global extraction collapses.

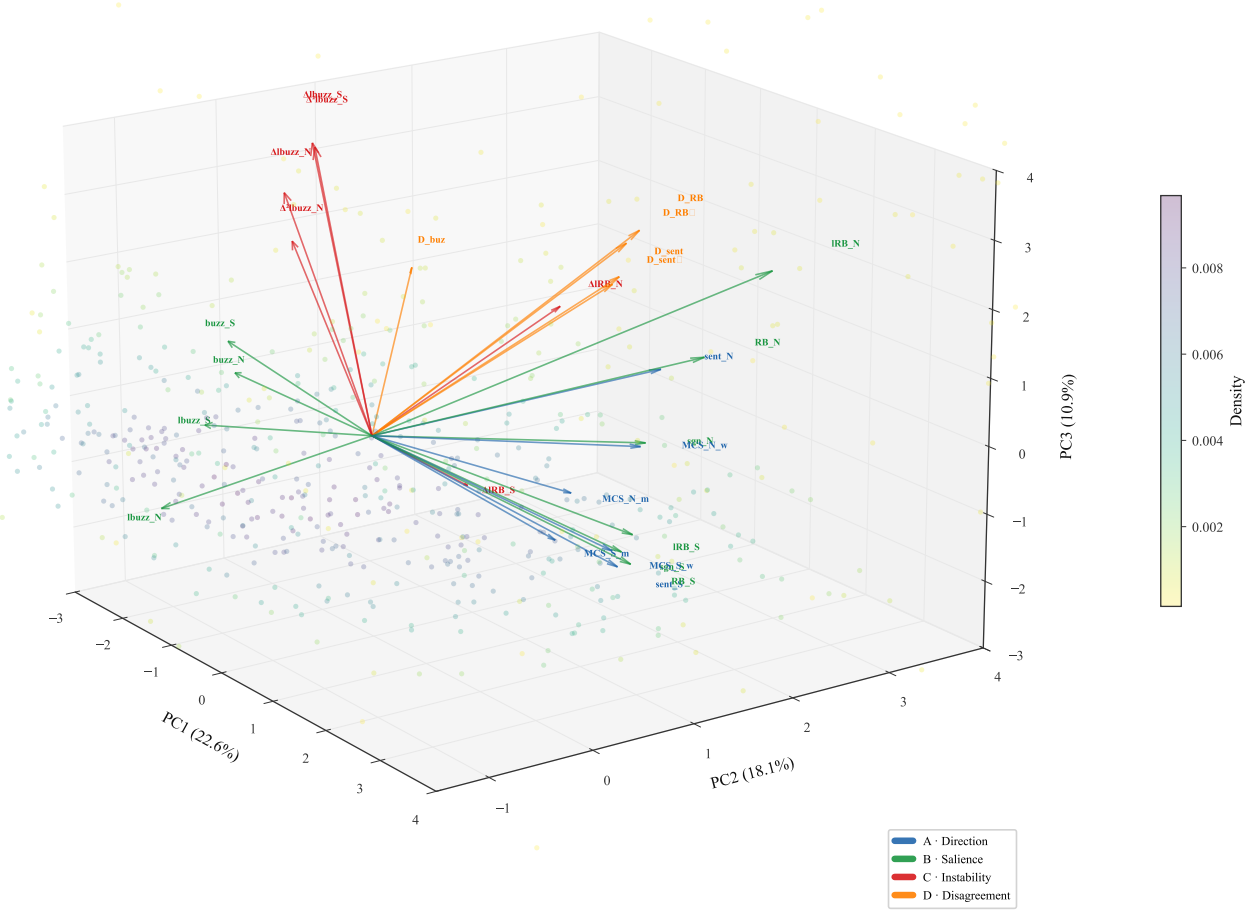


FIGURE B.5: THREE-DIMENSIONAL PCA BIPLOT: PC1 × PC2 × PC3 (AAPL). The plot displays the joint projection of all 3,633 firm-day observations (point cloud, kernel density colouring) and the loading vectors of the 27 media features (arrows) onto the plane spanned by the first three principal components. The percentage of total predictor variance explained by each component is reported on the axis label. Arrow colour identifies the informational family: **Direction** (blue), **Salience** (green), **Instability** (red), and **Disagreement** (orange). Arrow length is proportional to loading magnitude. Social Direction and Salience arrows overlap along PC1; Disagreement arrows extend along PC2; Instability arrows extend along PC3. The partial overlap between Direction and Salience in the PC1 dimension motivates the two-stage PCA, which compresses each family separately to preserve the distinction that global extraction conflates.

Two-stage PCA: within-block structure. Figure B.6 examines what each family's leading within-block component $F_{1,t}^{(g)}$ actually captures, for $g \in \{\text{Direction, Salience, Instability, Disagreement}\}$. The results reveal that the four families are not only economically distinct in design but also empirically distinct in their internal structure. The *Direction* PC1 (52.6% of within-block variance) is driven primarily by the rolling Media Coverage Sentiment variables (MCS_w^{S}, MCS_m^{S}, MCS_m^{N}, and MCS_w^{N} carry loadings of 0.41–0.47) with raw sentiment scores playing a secondary role. This confirms that the dominant directional signal for AAPL comes from accumulated narrative tone over weekly and monthly horizons rather than from daily sentiment fluctuations. The *Salience* PC1 (35.1%) is more diffuse, loading broadly across all

ten features; log-transformed social buzz ($\ln buzz^{S}$, 0.40) and relative social buzz (RB^{S}, 0.38) lead, but the remaining features contribute substantially, reflecting the high internal heterogeneity of the Salience family (a finding that explains why Salience has the lowest within-block coherence of the four families). The *Instability* PC1 (43.5%) is tightly concentrated on four first-difference buzz variables ($\Delta \ln buzz^{S}$, $\Delta \ln buzz^{N}$, $\Delta^2 \ln buzz^{S}$, $\Delta^2 \ln buzz^{N}$, loadings 0.45–0.52), while the two relative-buzz change variables ($\Delta \ln RB^{N}$, $\Delta \ln RB^{S}$) contribute negligibly (0.05, −0.03), suggesting that raw attention volatility rather than sentiment-weighted attention volatility drives the Instability signal. The *Disagreement* PC1 (63.8%, the highest of any family) loads almost equally on the four sentiment and relative-buzz disagreement measures (D^{Sent}, D^{RB}, $D^{Sent\star}$, $D^{RB\star}$, loadings 0.48–0.49), with D^{Buzz} contributing less (0.21). The tight clustering of loadings and the high within-block explained variance confirm that the five Disagreement features measure a single coherent latent construct (i.e., cross-channel informational conflict) from slightly different angles.

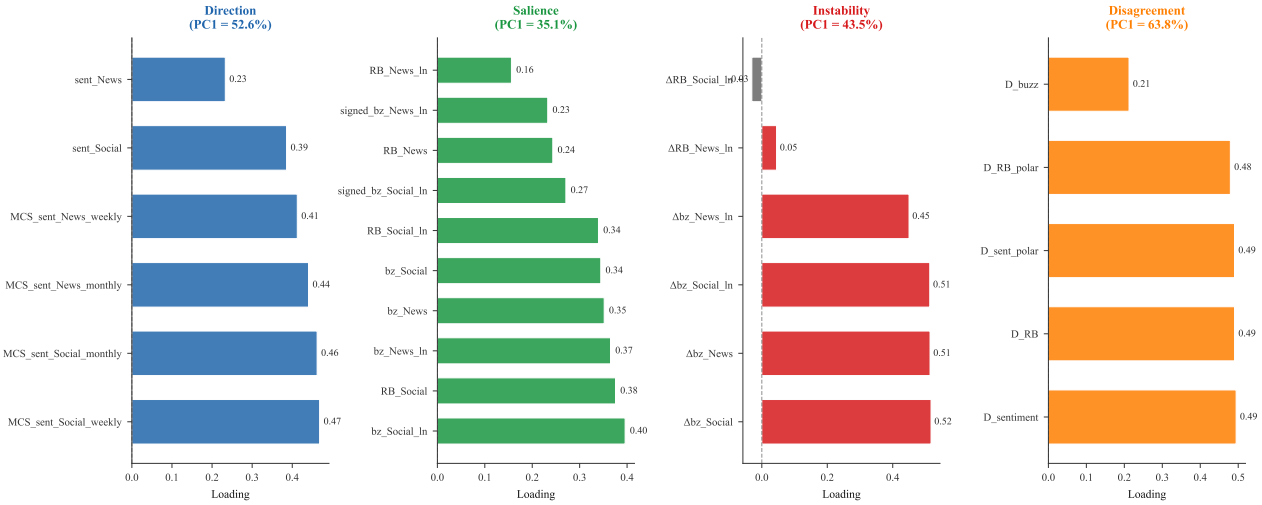


FIGURE B.6: STAGE 1 WITHIN-BLOCK PC1 LOADINGS BY FAMILY (AAPL). Each panel displays the loadings of the leading within-block principal component on the member features of that family, sorted by absolute loading. The percentage of within-block variance explained by PC1 is reported in the panel title. **Direction** (52.6%) is dominated by rolling MCS variables; **Salience** (35.1%) loads diffusely across all ten features; **Instability** (43.5%) concentrates on raw first-difference buzz variables; **Disagreement** (63.8%) loads near-equally on sentiment and relative-buzz divergence measures, confirming internal coherence. Analysis conducted for AAPL; results for other DJIA constituents are qualitatively similar and omitted for brevity.

Two-stage PCA: temporal dynamics of family summaries. Figure B.7 plots the daily scores on each family's leading Stage 1 component over the full sample, with 60-day rolling means. The four components display sharply different persistence profiles that validate their economic interpretations. The *Direction* summary is the most persistent (AR(1) = 0.92), with a rolling mean that drifts slowly and a pronounced negative shift during the GFC (period mean −1.25 vs pre-GFC mean −0.65), reflecting a sustained deterioration in narrative tone surrounding the crisis. The *Salience* summary is similarly persistent (AR(1) = 0.84) and displays a secular upward trend in the post-GFC period (post-GFC mean +1.13 vs pre-GFC −1.33), consistent with the growth of social media coverage throughout the sample. The *Instability* summary, by contrast, is near-white-noise (AR(1) = −0.27), with a sign change rate of 0.57 per day and a flat rolling mean throughout the sample, exactly the behaviour expected from a first-difference-type factor. Most notable is the *Disagreement* summary

(AR(1) = 0.61), which exhibits strong right skewness (skewness = 2.80, excess kurtosis = 10.08) and a pronounced structural break: the rolling mean is strongly positive in the early sample (pre-GFC mean +1.17) but turns negative and remains so after the GFC (post-GFC mean −0.73). This suggests a long-run decline in cross-channel informational conflict for AAPL, possibly reflecting the maturing of social media as an information channel whose sentiment gradually converged with that of traditional news over the sample period.

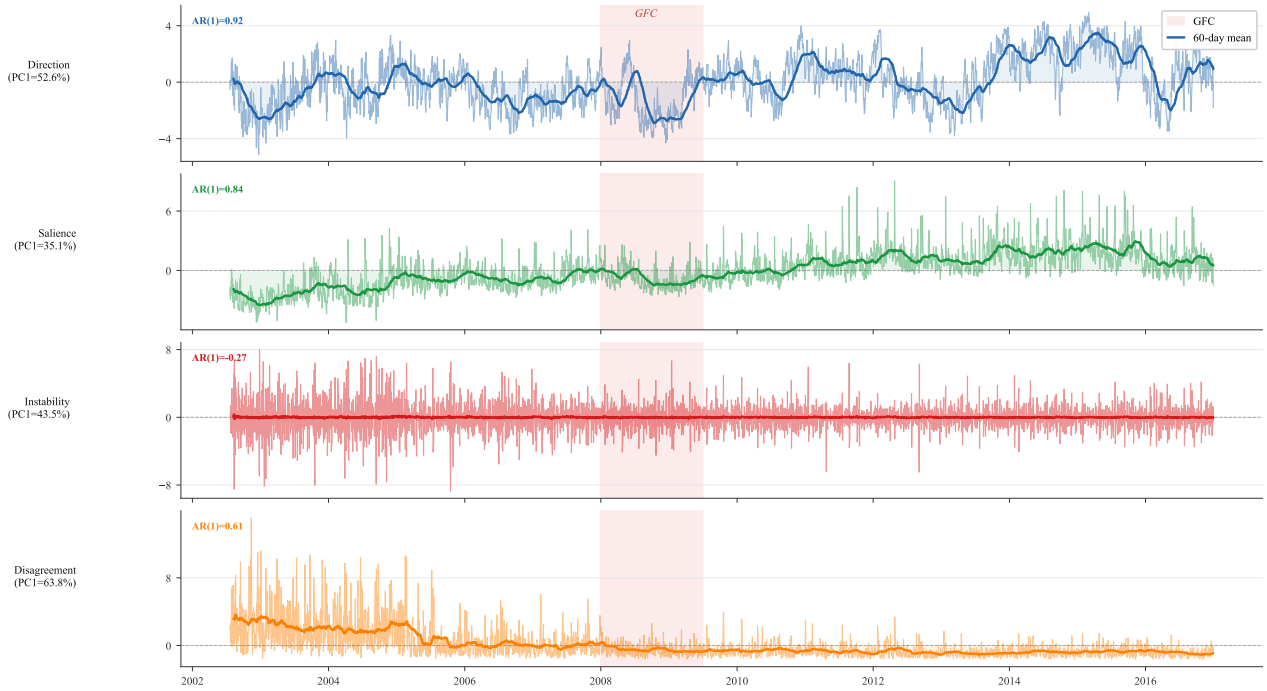


FIGURE B.7: STAGE 1 FAMILY SUMMARY SCORE TIME SERIES (AAPL). Each panel plots the daily score on the leading within-block PC1 for one of the four informational families (faint line) together with a 60-day rolling mean (bold line). The shaded region marks the Global Financial Crisis (January 2008 to June 2009). AR(1) autocorrelation coefficients are annotated. **Direction** (AR(1) = 0.92) and **Salience** (AR(1) = 0.84) are highly persistent with distinct GFC shifts; **Instability** (AR(1) = −0.27) is near-white-noise with a flat rolling mean; **Disagreement** (AR(1) = 0.61) displays a structural break, with the rolling mean shifting from positive pre-GFC to persistently negative post-GFC.

Two-stage PCA: parsimony and variance coverage. A key practical question is whether the two-stage procedure offers any compression efficiency relative to one-stage PCA. Figure B.8 provides a direct comparison. The two-stage curve rises far more steeply: three Stage 2 components already cover 92.4% of the variance in the four family summaries, whereas one-stage PCA requires nine components to reach 80% and thirteen to reach 90% of the full 27-dimensional predictor space. At the 80% threshold, the two-stage approach requires six fewer components than one-stage PCA; at 90%, ten fewer. This compression gain is not merely a mathematical artefact of reducing the input dimension from 27 to 4. It reflects the fact that the block structure concentrates common variation within each family before aggregation, making the cross-family Stage 2 PCA operate on a cleaner, lower-noise signal. The efficiency advantage has a direct practical benefit: fewer components in the rolling-window forecasting framework means fewer parameters to estimate within each 252-day window, which reduces estimation noise and improves out-of-sample stability.

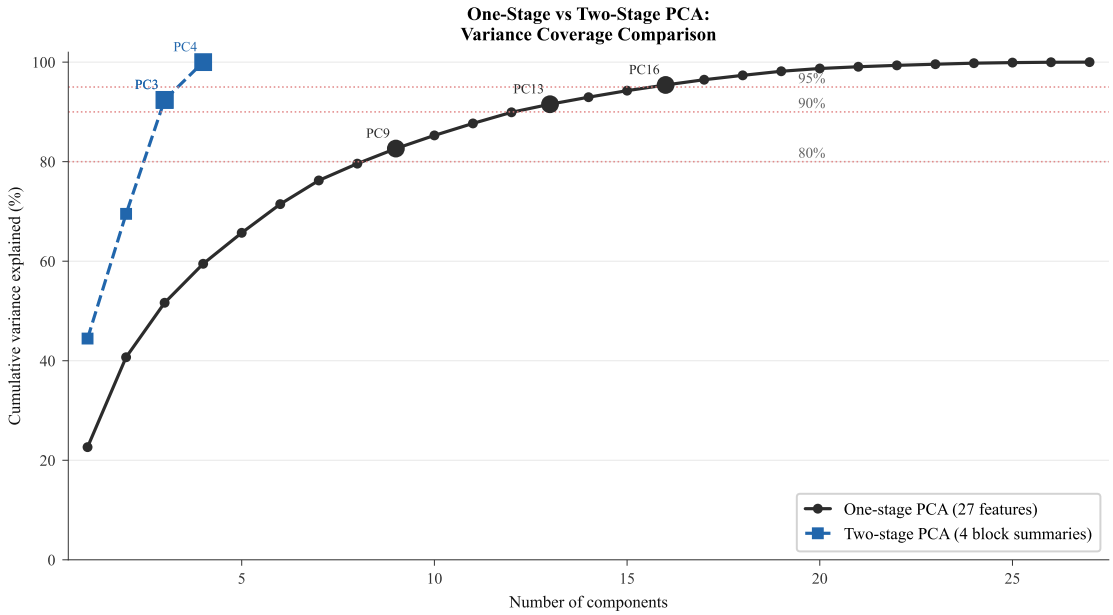


FIGURE B.8: ONE-STAGE VS TWO-STAGE PCA: VARIANCE COVERAGE COMPARISON (AAPL). Cumulative variance explained as a function of the number of retained components under one-stage PCA applied to all 27 standardised features (black circles, solid) and two-stage PCA applied to the four family-level block summaries (blue squares, dashed). Reference lines mark the 80%, 90%, and 95% thresholds. The two-stage curve reaches 92.4% with three components; one-stage PCA requires nine components to cross 80% and thirteen to cross 90%. The compression efficiency of the hierarchical approach reduces the number of parameters estimated in rolling-window forecasting, improving out-of-sample stability.

Figure B.9 consolidates the two-stage procedure, reporting within-block coherence (left panel) and the cross-block Stage 2 variance decomposition (right panel).

Within-block coherence varies meaningfully across families. Disagreement is the most internally coherent (63.8% of within-block variance explained by its leading component $F_{1,t}^{(g)}$), reflecting the near-equal loading of its five constituent features on a single latent construct, cross-channel informational conflict. Direction is moderately coherent (52.6%), driven primarily by the rolling MCS variables as documented in Figure B.6. Instability follows (43.5%), concentrated on raw first-difference buzz variables. Salience is the least coherent (35.1%), consistent with the greater internal heterogeneity of its ten constituent features, which span raw counts, log-transforms, and signed and relative variants of the same underlying attention signal.

The right panel reveals the cross-block structure. We estimate pairwise Pearson correlations between the four Stage 1 family summaries and find that Direction and Salience are the only strongly correlated pair ($r = 0.66$), consistent with the observation from the global loading analysis that these two families partially overlap in the PC1 dimension. Instability is nearly orthogonal to all other families ($|r| \leq 0.13$), and Disagreement is weakly negatively correlated with Salience ($r = -0.30$) and Direction ($r = -0.11$), reflecting the tendency for cross-channel conflict to decline when sentiment is strong and attention is elevated. Stage 2 PC1 (44.5% of Stage 2 variance) loads positively on Direction and Salience and negatively on Disagreement, representing a *coherent narrative* factor: days when tone is favourable and attention is elevated tend to coincide with low cross-channel conflict. Stage 2 PC2 (25.1%) loads almost entirely on Instability (0.98), confirming that attention volatility is orthogonal to

the coherent narrative dimension. Together, three Stage 2 components span 92.4% of family summary variance, six fewer than the nine one-stage components required to reach 80% of the full 27-dimensional space. This compression efficiency comes without sacrificing economic structure: the four informational dimensions of narrative direction, media salience, attention instability, and cross-channel disagreement remain separable in the Stage 2 decomposition, whereas a single global extraction step would partially collapse them.

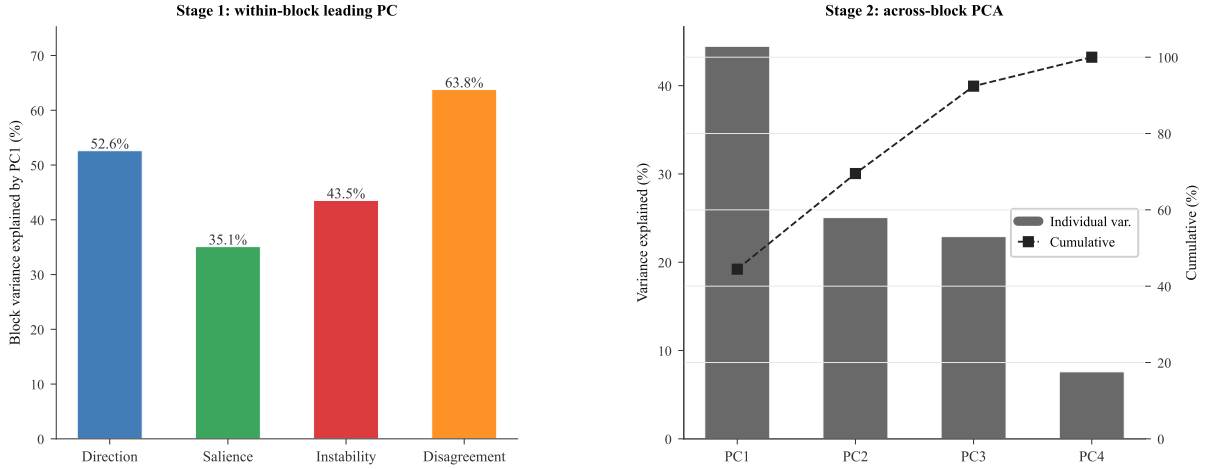


FIGURE B.9: TWO-STAGE PCA SUMMARY (AAPL). *Left panel*: share of within-family variance explained by the leading Stage 1 component $F_{1,t}^{(g)}$ for each of the four informational families. Within-block coherence ranges from 35.1% (Salience) to 63.8% (Disagreement). *Right panel*: individual (bars) and cumulative (squares, dashed) variance explained by Stage 2 principal components extracted from the matrix of four family-level summaries. Three components suffice for 92.4% coverage. Stage 2 PC1 (44.5%) represents a coherent narrative factor (Direction and Salience positive, Disagreement negative); Stage 2 PC2 (25.1%) is a pure Instability factor (loading 0.98). Analysis conducted for AAPL; results for other DJIA constituents are qualitatively similar and omitted for brevity.

C Stability tests

To assess numerical stability and multicollinearity, we analyze diagnostics for both the full sample and a rolling-window scheme. We contrast the full feature set strategy (denoted as HARSent) against the dimensionality-reduced strategy (denoted as HARmPCAopt). Note that these model specifications are formally defined in Section 4.2. The results are summarized in Table C.1 and Figure C.1.

TABLE C.1: Stability Diagnostics for Full-Sample and Rolling-Window Analyses. This table reports variance inflation factors (VIF), condition numbers (CN), and coefficient standard errors. Panel A presents full-sample statistics. Panel B summarizes the distribution of these metrics across a three-month rolling window. The thresholds for severe multicollinearity are based on standard econometric literature: VIF > 10 (Kutner et al., 2004), CN > 30 (Belsley et al., 1980). Additionally, we report the frequency of coefficient standard errors exceeding 1 to illustrate the resulting parameter instability.

PANEL A: Full-Sample Diagnostics

	HARSent	HARmPCAopt
VIF median	4.494	1.556
VIF mean	339.521	2.649
Condition number median	22.013	8.257
Condition number mean	133.512	8.380
Coefficient std. error median	0.017	0.010
Coefficient std. error mean	0.059	0.013

PANEL B: Rolling-Window Diagnostics

	HARSent	HARmPCAopt
VIF median	9.170	2.430
% of windows with VIF > 10	47.9%	2.5%
Condition number median	495.390	7.120
% of windows with CN > 30	95.2%	3.1%
Coefficient std. error median	0.270	0.098
% of windows with Std. Err. > 1	26.2%	2.2%

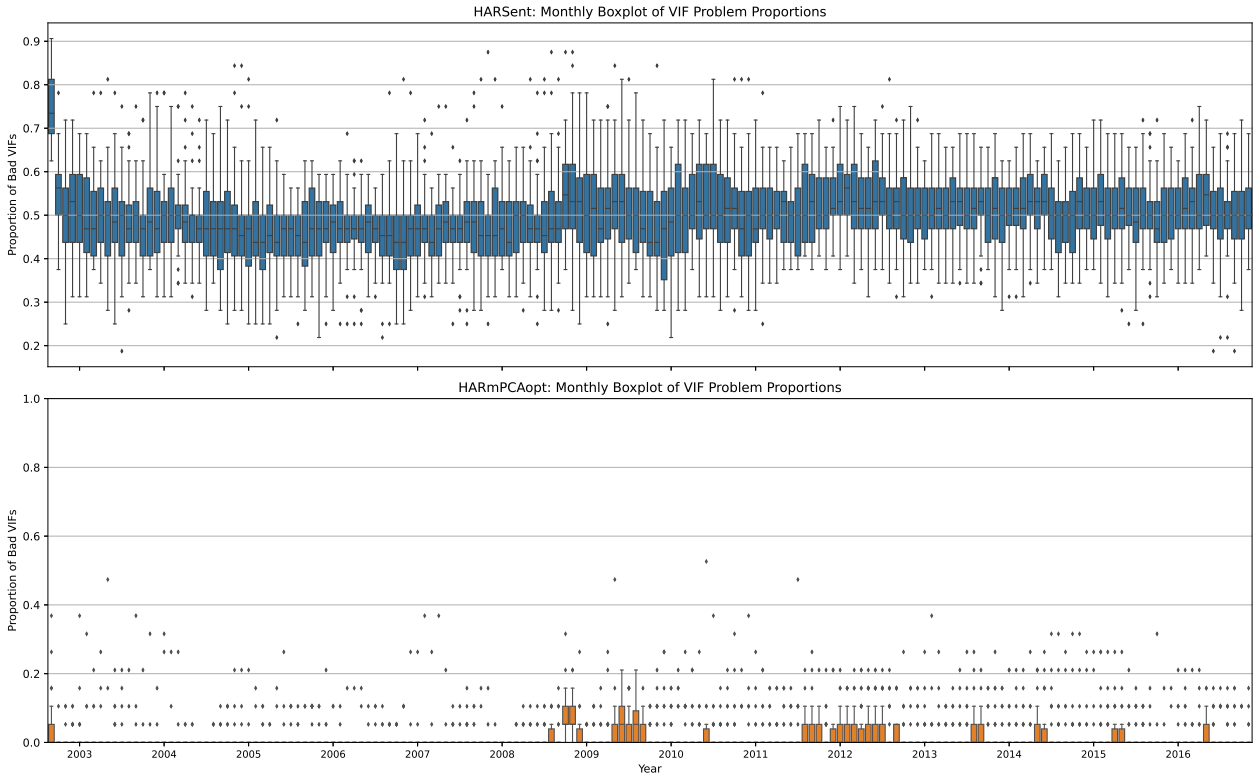


FIGURE C.1: Distribution of VIFs from Rolling-Window Estimation. This figure displays boxplots of the proportion of predictors exhibiting severe multicollinearity within each three-month estimation window. A problematic hit is defined as any feature having a VIF > 10 within a given window. The top panel corresponds to the HARSent specification; the bottom panel corresponds to the HARmPCAopt specification.

Our stability diagnostics reveal a critical flaw in the kitchen sink strategy. As reported in

Table C.1, the HARSent specification exhibits severe multicollinearity across both estimation schemes. In the full sample, the mean VIF is 339.5, compared to just 2.6 for the PCA-based strategy. This instability is even more pervasive in the rolling-window analysis: HARSent violates the VIF threshold in nearly half of all estimation windows (47.9%), whereas the PCA specification does so in only 2.5% of cases. Figure C.1 corroborates this instability, illustrating that the proportion of problematic VIFs in the HARSent model is not only high but fluctuates erratically over time. This multicollinearity manifests in inflated standard errors (frequently exceeding 1), rendering the raw model's point estimates unreliable. In sharp contrast, the PCA-based strategy reduces these diagnostics to safe levels, effectively restoring parameter stability. Consequently, we retain HARSent solely as a cautionary benchmark; our substantive analysis relies on the HAR variants using PCA, which are designed to resolve this instability.

D Additional In-Sample Results

D.1 Conditional Correlations Between Media Metrics and Index Volatility

Table D.1 reports the conditional correlations referred to in Section 4.3. They are computed on the DJIA index rather than on the constituents, condition on the level of attention or on the sign of tone, and do not control for lagged volatility; their role is to motivate a specification in which the contribution of the media predictors is allowed to vary across the conditional distribution of volatility, not to measure forecast value.

Table D.1 provides direct motivating evidence for this concern. The conditional correlations reported there reveal that the relationship between sentiment and volatility is fundamentally non-linear. Panel A documents a striking amplification pattern: the correlation between news sentiment and future volatility is negligible under low media attention ($\rho = -0.075$) but rises sharply to -0.308 under high attention, with an analogous pattern in social media (-0.029 to -0.280). Panel B reveals further asymmetry in the role of buzz: for news, high attention is associated with lower future volatility when tone is positive ($\rho = -0.192$), whereas for social media, high attention predicts higher future volatility specifically when tone is negative ($\rho = 0.148$), consistent with a fear-contagion dynamic. The timing pattern is less uniform than the amplification pattern. In the medium- and high-attention rows of Panel A, and for negative social-media tone in Panel B, the magnitudes rise from the lagged $(t-1)$ to the leading $(t+1)$ horizon; for positive tone they peak at t for news (-0.136, -0.195, -0.192) and decline at $t+1$ for social media (-0.120, -0.121, -0.090). The media metrics are therefore contemporaneous with index volatility and, in the high-attention states, at most slightly ahead of it. These are unconditional, index-level, in-sample correlations that do not control for lagged volatility; they motivate the specification below and say nothing yet about forecast value. An OLS model would average these distinct regimes together, obscuring precisely the dynamics we seek to identify.

D.2 Quantile Variation in Incremental Fit for HAR2PCA

Figure D.1 repeats the quantile analysis of Figure 1 for HAR2PCA, the two-stage specification that enters the fewest media terms and that carries the out-of-sample comparison in

TABLE D.1: CONDITIONAL CORRELATIONS BETWEEN MEDIA METRICS AND DJIA INDEX VOLATILITY (MOTIVATING EVIDENCE). This table reports Pearson correlations (ρ) between daily media metrics observed on day t and DJIA index-level volatility at lagged ($t-1$), contemporaneous (t), and leading ($t+1$) horizons, computed on the DJIA index. Panel A conditions on the level of media attention (Buzz) and reports correlations with Sentiment; Panel B conditions on sentiment tone and reports correlations with Buzz. Conditioning groups are defined by the bottom 30% (Low/Negative), middle 40% (Medium/Neutral), and top 30% (High/Positive) of the respective conditioning variable. Significance levels: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

PANEL A: Correlation of Sentiment$_t$ with Volatility, conditional on Buzz level

	Returns		Volatility		
Buzz level	t	$t+1$	$t-1$	t	$t+1$
News channel					
Low	0.124^{***}	0.000	-0.035	-0.036	-0.075^{**}
Medium	0.116^{***}	0.020	-0.099^{***}	-0.109^{***}	-0.137^{***}
High	0.106^{***}	0.034	-0.238^{***}	-0.294^{***}	-0.308^{***}
Social media channel					
Low	0.073^{**}	-0.012	-0.071^{**}	-0.040	-0.029
Medium	0.132^{***}	-0.005	-0.160^{***}	-0.166^{***}	-0.207^{***}
High	0.100^{***}	0.016	-0.229^{***}	-0.276^{***}	-0.280^{***}

PANEL B: Correlation of Buzz$_t$ with Volatility, conditional on Sentiment tone

	Returns		Volatility		
Sentiment tone	t	$t+1$	$t-1$	t	$t+1$
News channel					
Positive	-0.036	0.027	-0.136^{***}	-0.195^{***}	-0.192^{***}
Neutral	-0.010	0.003	-0.030	-0.030	-0.054^{*}
Negative	-0.053^{*}	-0.019	0.013	0.024	0.035
Social media channel					
Positive	-0.031	0.016	-0.120^{***}	-0.121^{***}	-0.090^{***}
Neutral	0.004	0.004	-0.048^{*}	-0.053^{*}	-0.026
Negative	-0.043	-0.007	-0.002	0.030	0.148^{***}

Section 6. The pattern has the same shape as for HARmPCAopt, with smaller gains: the median incremental fit at $\tau = 0.975$ is 0.5% against 1.4%, and the largest firm-level gain is 16.3% (CSCO) against 25.2%.

D.3 Predictor Importance: Static and Rolling

Figures D.2 and D.3 are the variable-importance diagnostics summarised in Section 5.3. Both are computed on the estimation sample and describe which predictors carry the in-sample fit; they are not measures of forecast value.

Section 5.3 documents *where* sentiment adds in-sample fit (the upper tail of the volatility distribution, for low-persistence stocks) and offers a candidate reason (less of the variation is carried by the past). A natural question is then whether these dynamics are stable over time or themselves time-varying. Figure D.2 confirms that in a static full-sample framework, the three HAR volatility lags dominate explanatory power, leaving little room for exogenous predictors.

The rolling-window results in Figure D.3 reveal a more nuanced picture: the hierarchy of predictors rotates over time, with bursts of sentiment and attention importance periodically

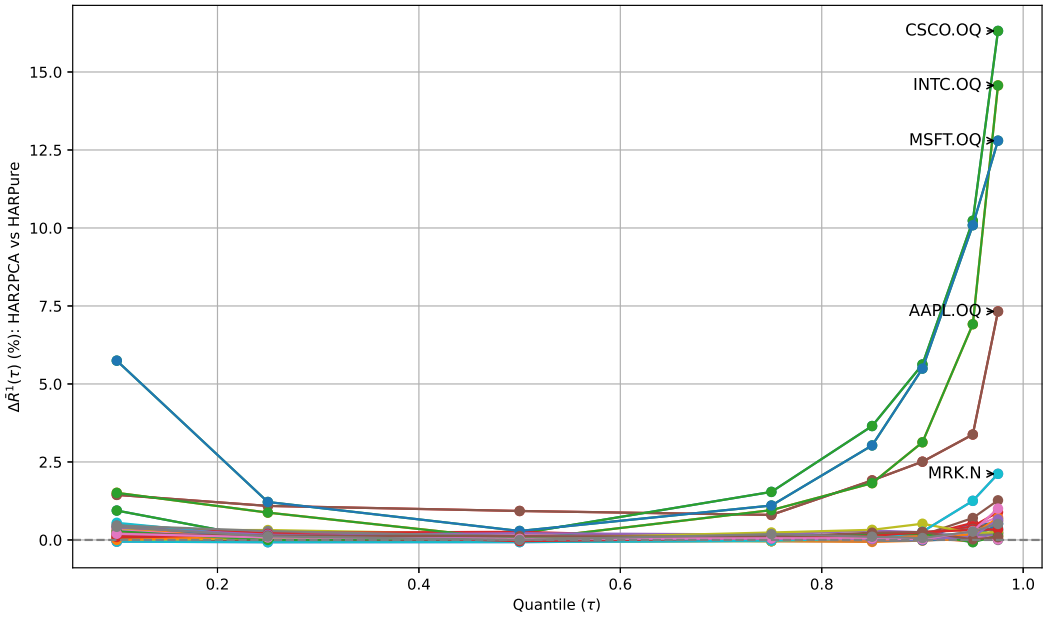


FIGURE D.1: QUANTILE-DEPENDENT INCREMENTAL EXPLANATORY POWER: HAR2PCA. As Figure 1, for the HAR2PCA specification, the specification shown in bold for the primary interaction regime in Table 5. Each line plots, for one DJIA constituent, the percentage change in the degrees-of-freedom-adjusted quantile pseudo-R^2 of HAR2PCA relative to HARPURE (Section 4.3) across $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.85, 0.90, 0.95, 0.975\}$. The five firms with the largest gains at $\tau = 0.975$ are labelled.

overtaking the autoregressive components (Panel C). Importance also decays rapidly as the estimation window lengthens from 3 to 6 and 12 months (Panel D), confirming that media narrative signals are episodic rather than persistent. Notably, sentiment importance collapses during the 2008 GFC itself, when raw volatility momentum was so dominant that it drowned out all other signals.

This set of results serves a diagnostic purpose, characterising how quickly the informational content of sentiment dissipates as the estimation horizon extends. Importantly, the 3-, 6-, and 12-month windows used here are not the estimation window of the out-of-sample evaluation; rather, they isolate the decay rate of the media predictors' in-sample importance under progressively longer lookback periods. The 252-day window adopted in the out-of-sample design is chosen to balance parameter estimation stability against the adaptability needed to track regime changes, a choice that the rapid decay of importance at longer horizons directly motivates. Together, these dynamics reinforce the necessity of the adaptive rolling-window design used in the out-of-sample evaluation.

E Additional Out-of-Sample Results

E.1 Evaluation Details

This section expands on four aspects of the out-of-sample design of Section 4.5: the choice of the estimation window, the retransformation bias in QLIKE, the choice of regime cut-offs and a threshold-free alternative to the regime tables, and the inference used in the regime tables of the main text.

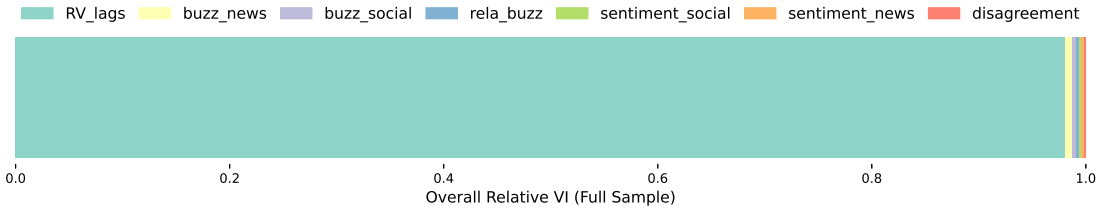


FIGURE D.2: IN-SAMPLE STATIC VARIABLE IMPORTANCE (FULL SAMPLE). The figure displays the in-sample relative importance of predictor groups for the HARmPCAopt model, computed from the full-sample estimates. The HAR volatility lags ($\log RV^{(d)}$, $\log RV^{(w)}$, $\log RV^{(m)}$) collectively dominate, leaving limited unconditional room for exogenous sentiment predictors consistent with the modest average gains reported in Table 4.



FIGURE D.3: IN-SAMPLE DYNAMIC VARIABLE IMPORTANCE (ROLLING WINDOW). This figure illustrates the time-varying in-sample predictor importance estimated within each 3-month rolling window; it measures which predictors carry the fit within a window, not their contribution to forecasts of the following day. Panel A shows the average importance share of each predictor group over time. Panel B ranks the top 10 individual predictors by average importance. Panel C reports quarterly aggregated importance shares by predictor group. Panel D contrasts total sentiment importance across 3-, 6-, and 12-month rolling windows; the rapid decay with window length confirms that media narrative signals are episodic rather than persistent drivers of predictive power. The shaded region marks the 2008 Global Financial Crisis, during which sentiment importance collapses as raw volatility momentum dominates. Predictor labels follow the naming conventions of Table A.3: `sentiment_news` aggregates news-channel tone features; `rela_buzz` denotes buzz-weighted sentiment; `disagreement` captures cross-channel divergence metrics; and `RV_D`, `RV_W`, `RV_M` correspond to $\log RV^{(d)}$, $\log RV^{(w)}$, and $\log RV^{(m)}$, respectively. The suffix `PC` denotes principal components extracted via the two-stage PCA strategy (Section 3.2).

Window length. Two features of the rolling scheme deserve comment in the light of Chassot and Audrino (2026), who document across 1,445 U.S. stocks that the forecasting performance of HAR models depends heavily on the fitting scheme. Their first and most important finding is that daily re-estimation is essential, with substantial deterioration when the model is refitted less often; our design re-estimates every specification every day. Their second finding is that longer training windows, of roughly two and a half to four years, produce somewhat lower unconditional forecast errors than shorter ones. Our window is shorter than that, for two reasons that are specific to our question. The persistence classification requires a fractional-integration estimate that tracks recent dynamics, and a multi-year window would average over exactly the episodes of changing persistence that we wish to isolate; and, as the rolling variable-importance analysis in Section 5.3 shows, the importance of media predictors decays quickly as the lookback lengthens, so a long window would tend to average their contribution away. The cost of this choice is that the unconditional performance of every specification, benchmark and augmented alike, is likely to be somewhat worse than it would be under a longer window. Because the same window is used for all specifications, this affects the level of forecast losses rather than their comparison, but we have not verified that the regime-conditional results are invariant to the window length, and we flag this as an open robustness question.

Retransformation. Because the models are estimated for log volatility, $\exp(\hat{y}_{t+1})$ understates the conditional mean of RV_{t+1} by the retransformation factor $\exp(\sigma^2/2)$, where σ^2 is the conditional variance of the log-scale forecast error (Granger & Newbold, 1976). We report QLIKE without this correction for comparability with the bulk of the HAR literature, which does the same, but note that the omitted factor differs across specifications because their residual variances differ, so that QLIKE comparisons carry a small specification-dependent bias. The log-space MSE and MAE are unaffected, and we therefore treat agreement between the three loss functions, rather than QLIKE alone, as the relevant criterion. The Monte Carlo study in Appendix F quantifies the size of the retransformation effect in a setting calibrated to ours; it is second order.

Cut-offs and a threshold-free alternative. The cut-offs of the regime classification (the top and bottom deciles of volatility, the top and bottom deciles of persistence) are conventional rather than optimised, and nothing in what follows depends on their exact values in a way that we have tuned. They are nevertheless a choice, and it is fair to ask whether the results reflect a joint volatility–persistence pattern or the particular partition. Two responses are available. First, the Monte Carlo study in Appendix F applies the same classification, with the cut-offs used here and with neighbouring alternatives, to simulated panels in which the truth is known; the signature that identifies state-dependent predictive content is stable across the cut-offs considered, and it is not generated when the predictive content of the exogenous signal is constant. Second, a threshold-free alternative is to model the firm-day loss differential directly as a function of persistence and volatility,

$$\Delta L_{i,t+1} = \alpha_i + \beta_1\, d_{i,t} + \beta_2 \log RV_{i,t} + \beta_3 \left(d_{i,t} \times \log RV_{i,t}\right) + u_{i,t+1}, \tag{E.1}$$

where $\Delta L_{i,t+1}$ is the loss of the sentiment-augmented model minus the loss of the benchmark, so that the interaction coefficient β_3 provides a single-parameter summary of whether the

gains from sentiment depend jointly on persistence and volatility. Because the interaction term is the product of two continuous variables, β_3 is more transparently interpreted through the implied loss surface than through its sign alone; we use Eq. (E.1) in Appendix F to characterise what the regime tables can and cannot reveal. A related approach is to let the data determine the threshold, as in the state-dependent evaluation tests of Odendahl et al. (2023), in which the loss differential follows a threshold model with an unknown threshold value; this is the natural formal test of the joint hypothesis and we return to it in the conclusion.

Statistical inference. Forecast comparisons are based on the loss differential between each sentiment-augmented specification and the HARPure benchmark. Three features of the design bear on how the resulting statistics should be read. First, the benchmark is nested in every augmented specification, which in a population comparison of models would make the standard Diebold–Mariano (DM) statistic degenerate under the null of equal accuracy and call for the adjustment of Clark and West (2007). Under a fixed-length rolling estimation window, however, the object being compared is the forecasting *method*, including its estimation error, rather than the population model; Giacomini and White (2006) show that in this setting the DM-type statistic retains a standard limiting distribution whether or not the models are nested, because parameter estimation error does not vanish as the evaluation sample grows. Our use of the DM statistic rests on this rolling-window justification. Second, all tests are two-sided: the null is equal predictive accuracy, and a rejection in a cell whose relative loss exceeds one indicates that the augmented model is significantly *worse* than the benchmark, not better. We say so explicitly in the tables because a reader could otherwise take a starred cell with a loss ratio above one for a favourable result. Third, the natural way to summarise the cross-section of stock-level DM statistics is Fisher's combined probability test (Becker, 1994), which combines the stock-level p-values under the assumption that they are independent. Forecast errors for DJIA constituents are strongly correlated across stocks, and the reference distribution of the combined statistic is therefore too tight: the Monte Carlo evidence in Appendix F indicates that, for 33 stocks and a cross-sectional correlation of loss differentials of 0.3, the nominal 5% test rejects a true null about 15% of the time, and that the proportion of stocks with a loss ratio below one, read as a test, is affected considerably more. A further defect is specific to the combination of two-sided p-values: the combined statistic rejects when stocks differ from the benchmark in *either* direction, so that a starred cell can combine stocks that are significantly better with stocks that are significantly worse, so that, applied to Table 5, it would put three stars on several cells in which fewer than half of the stocks outperform. A correctly sized alternative is to average the loss differentials across stocks on each day and apply a single DM test to the resulting series, which absorbs the cross-sectional dependence, or, for the regime-conditional hypothesis specifically, to apply the state-dependent tests of Odendahl et al. (2023). The stars in Tables E.1 and 5 are therefore from the pooled test, which is reported in full in Section 6.2 (Table 7) together with the Giacomini–White conditional test (Section E.4, Table E.2); the outperformance proportions in the same tables are descriptive summaries of the cross-section rather than tests.

E.2 Univariate Regime Diagnostics

Table E.1 conditions on the volatility regime and on the persistence regime separately. It is the diagnostic counterpart of Table 5 and is referred to in Section 6.1; as in that table, the volatility regime is the ex-post one and the persistence regime is real-time.

TABLE E.1: OUT-OF-SAMPLE PERFORMANCE: UNIVARIATE REGIME DIAGNOSTICS. This table reports average relative loss ratios (model loss divided by HARPURE loss) for MSE, MAE, and QLIKE, averaged across all DJIA constituents, conditioning separately on the volatility regime (Panel A) and the persistence regime (Panel B). Volatility regimes are defined by the top and bottom deciles of each firm's log-realised volatility distribution over the out-of-sample period, January 2004 to December 2016. Persistence regimes are defined by the top and bottom deciles of the cross-sectional distribution of the ARFIMA parameter d on each day (Section 4.5). Ratios below 1.0 favour the sentiment-augmented model; ratios above 1.0 favour the HARPURE benchmark. Ratios are computed stock by stock and averaged across the stocks that have at least 20 firm-days; the set of contributing stocks therefore differs across regimes and between this table and Table 5. Parentheses report the proportion of contributing stocks for which the extended model yields lower loss. Stars report the two-sided pooled Diebold–Mariano test of Section 4.5, which averages the loss differential across the stocks in the regime on each day and applies a single test with a Newey–West variance (five lags) to the daily series, so that cross-sectional dependence is absorbed; a starred cell with a ratio above 1.0 indicates significantly *worse* performance than the benchmark. Fisher's combination of the stock-level tests, the natural alternative, overstates significance under cross-sectional dependence (Appendix F) and is not used here. The proportions remain descriptive. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

		HARPURE	HAR2PCA	HARPCA	HARLASSO	HARMPCA	HARMPCAOPT
			Rel. Loss (Proportion of stocks outperforming)				
Loss	**Regime**	Rel. Loss					
Panel A: Volatility regime only							
MSE	High	1.0	1.002 (51.5%**)	1.002 (48.5%)	1.014 (33.3%)	1.004 (36.4%)	1.027 (18.2%*)
	Low	1.0	1.001 (45.5%*)	1.000 (54.5%)	1.019 (24.2%***)	1.007 (39.4%***)	1.027 (15.2%***)
MAE	High	1.0	1.000 (42.4%***)	1.000 (54.5%***)	1.007 (39.4%)	1.002 (48.5%**)	1.013 (15.2%)
	Low	1.0	0.997 (63.6%)	0.996 (60.6%)	1.004 (45.5%***)	0.999 (57.6%)	1.000 (48.5%)
QLIKE	High	1.0	1.000 (57.6%)	1.001 (54.5%)	1.012 (33.3%)	1.003 (48.5%)	1.031 (24.2%***)
	Low	1.0	1.000 (45.5%)	0.999 (57.6%)	1.015 (27.3%***)	1.005 (42.4%***)	1.019 (27.3%***)
Panel B: Persistence regime only							
MSE	High	1.0	1.081 (10.7%)	1.022 (14.3%***)	1.040 (3.6%***)	1.032 (3.6%***)	1.076 (0.0%***)
	Low	1.0	1.023 (21.4%***)	1.018 (28.6%***)	1.029 (21.4%***)	1.030 (10.7%***)	1.078 (7.1%***)
MAE	High	1.0	1.013 (7.1%***)	1.010 (10.7%***)	1.019 (3.6%***)	1.015 (7.1%***)	1.039 (0.0%***)
	Low	1.0	1.008 (25.0%***)	1.005 (28.6%***)	1.009 (28.6%***)	1.012 (25.0%***)	1.035 (10.7%***)
QLIKE	High	1.0	1.021 (21.4%**)	1.014 (21.4%***)	1.026 (14.3%***)	1.021 (21.4%***)	1.066 (7.1%***)
	Low	1.0	1.017 (25.0%*)	1.013 (28.6%*)	1.030 (17.9%**)	1.027 (17.9%**)	1.086 (10.7%***)

Table E.1 shows that conditioning on either volatility or persistence alone yields little systematic benefit from sentiment augmentation. Relative loss ratios are predominantly above 1.0 across both panels, indicating that naive sentiment augmentation offers little improvement (and can mildly worsen forecasts) relative to the HARPURE benchmark on average within these broad states. Two features of the construction matter for reading this table together

with Table 5, and one of them is easy to misread. First, a univariate regime pools firm-days across the other dimension: the low-persistence row of Panel B, for instance, contains the low-persistence firm-days at every volatility level, of which the high- and low-volatility firm-days in Table 5 are only the two extreme deciles. Second, the ratios are averages of stock-level ratios over the stocks that contribute at least one firm-day to the regime, and these sets differ: all constituents contribute to the volatility regimes, slightly fewer to the persistence regimes, and considerably fewer to each of the four joint regimes, in which a stock must be in an extreme decile of its own volatility on a day on which it is also in an extreme decile of the cross-sectional persistence distribution. A specification can therefore have an average ratio above one in a univariate regime and below one in both of its joint sub-regimes, as HARLASSO does for low persistence, because the joint sub-regimes cover a minority of the firm-days and a different set of stocks. Nothing in the two tables is arithmetically inconsistent, but the comparison across them is not an aggregation and should not be read as one.

A consistent pattern across both panels is that high-persistence regimes impose the largest penalties on sentiment models. When volatility is strongly persistent, the HAR benchmark performs well , and adding exogenous predictors introduces estimation error that outweighs any informational benefit. This boundary condition (sentiment augmentation costs accuracy where the autoregressive structure is strongest) is the out-of-sample counterpart of the cross-sectional pattern in Section 5.3.

E.3 Cumulative Forecast Performance in the Interaction Regime

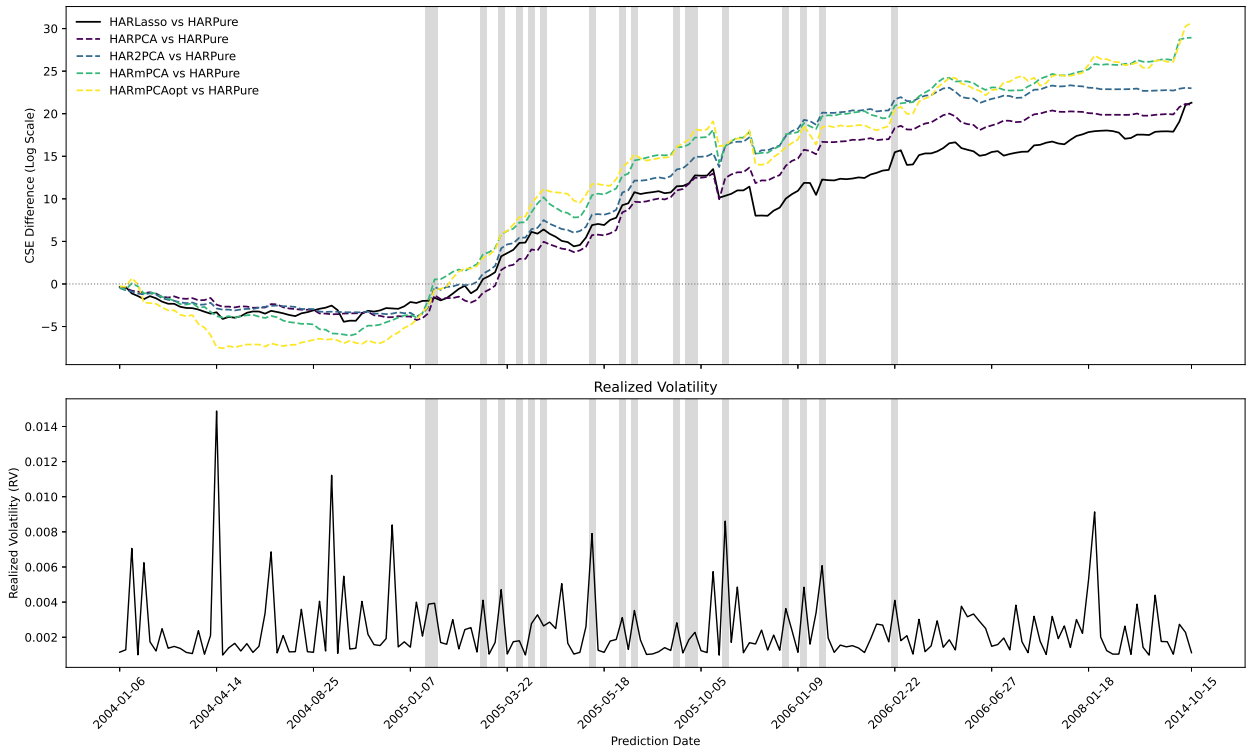
To assess whether forecasting gains accumulate consistently over time rather than concentrating in isolated episodes, we also compute the cumulative squared prediction error (CSPE) difference between the HARPure benchmark and each sentiment model:

$$\Delta\text{CSPE}_t = \sum_{j=1}^{t} \left(e^2_{\text{HARPure},j} - e^2_{\text{Model},j} \right), \tag{E.2}$$

where j indexes trading days within the evaluation period or regime. A positive and upward-trending ΔCSPE_t indicates that the sentiment model accumulates lower forecast errors than the benchmark consistently over time, which is stronger evidence of genuine predictive gains than a favourable average loss ratio alone.

A regime-specific loss reduction of approximately 1% is modest, and we do not argue otherwise. What can be examined is whether it is earned steadily across the regime or by a handful of days. Figure E.1 illustrates this for Apple Inc. (AAPL), plotting ΔCSPE_t from Eq. (E.2) restricted to the high-volatility, low-persistence interaction regime.

All five specifications end above zero for AAPL (HAR2PCA at +23.0), so the regime-level gain is not the product of one or two days. It is not earned evenly either: the curves are below zero through 2004, rise from early 2005, and most of AAPL's 178 regime days fall in 2004–2006, so that the horizontal axis, which indexes regime days rather than calendar time, compresses a decade into its right-hand third. Across the panel, the high-volatility, low-persistence firm-days are concentrated in 2008–2009 (779 of 1,487) and 2004–2006 (427); the pooled HAR2PCA MSE ratio is below one in 2005 (0.94), 2010 and 2011 (0.98), close



Notes: Grey-shaded areas highlight days of particular predictive power (HAR2PCA vs HARPure).

FIGURE E.1: CUMULATIVE FORECAST PERFORMANCE FOR APPLE INC. (AAPL) IN THE HIGH-VOLATILITY, LOW-PERSISTENCE REGIME. The top panel plots ΔCSPE_t (Eq. (E.2)) for AAPL, restricted to firm-days classified as the high-volatility and low-persistence interaction regime as defined in Section 4.5. An upward slope indicates that the sentiment-augmented model accumulates lower squared forecast errors than the HARPURE benchmark. The bottom panel displays daily realised volatility for context, identifying the specific episodes that drive the largest incremental forecasting gains.

to one in 2008 (1.00) and above one in 2009 (1.02). For AAPL, results are illustrative; the year-by-year pattern just described is computed on all 33 constituents.

E.4 Giacomini–White Conditional Test

Table E.2 reports the Giacomini–White (2006) test of conditional predictive ability with the real-time high-volatility, low-persistence indicator as the conditioning variable, formerly Panel C of Table 7. The null of equal conditional predictive ability is rejected for every specification under MSE and for all but HARMPCAOPT under QLIKE, but the rejection is driven entirely by the firm-days outside the regime, on which the augmented specifications are significantly worse than the benchmark (t-statistics between 2.6 and 20); inside the regime the mean loss differential is not significantly different from zero for any specification under either loss. The test therefore identifies conditional heterogeneity of the unfavourable kind: the augmented specifications are reliably worse outside the regime and indistinguishable from the benchmark inside it.

TABLE E.2: GIACOMINI–WHITE CONDITIONAL TEST WITH THE REAL-TIME REGIME INDICATOR. The table reports the Giacomini–White (2006) test of conditional predictive ability with instruments $h_t = (1, I_t)$, where I_t is the real-time high-volatility, low-persistence indicator of Table 6, Panel A: the Wald statistic W (asymptotically χ^2_2 under the null of equal conditional predictive ability) with its p-value, and the mean loss differential (augmented minus HARPURE) inside and outside the regime with t-statistics; all use a variance clustered by day with five Newey–West lags. A rejection of the Giacomini–White null accompanied by a positive mean differential indicates that the augmented specification is *worse*, not better. Entries marked n.c. are not computable for the reason given in the note to Table 7.

Specification	Loss	W	p-value	Mean ΔL in regime (t)	Mean ΔL outside (t)	Firm-days in regime
HAR2PCA	MSE	6.9	0.031	+0.0035 (+0.55)	+0.0077 (+2.58)	1,470
	QLIKE	42.7	0.000	-0.0054 (-0.78)	+0.0031 (+6.50)	
HARPCA	MSE	144.5	0.000	+0.0015 (+0.24)	+0.0044 (+12.02)	1,470
	QLIKE	55.5	0.000	-0.0071 (-0.97)	+0.0026 (+7.40)	
HARLASSO	MSE	385.7	0.000	+0.0231 (+2.63)	+0.0107 (+19.60)	1,470
	QLIKE	93.1	0.000	+0.0122 (+1.27)	+0.0061 (+9.64)	
HARMPCA	MSE	277.4	0.000	+0.0080 (+0.90)	+0.0071 (+16.57)	1,470
	QLIKE	72.5	0.000	-0.0070 (-0.78)	+0.0042 (+8.48)	
HARMPCAOPT	MSE	11.8	0.003	+0.0303 (+2.57)	+0.0372 (+2.22)	1,470
	QLIKE	n.c.	n.c.	+0.0157 (+1.17)	n.c.	

E.5 Period-Based Conditioning, Forecast Bias, Cut-Offs and the Loss-Differential Regression

Four further checks support the reading of the level-shift channel given in Section 6.2. Table E.3 applies the placebo comparison of Table 8 to sets of firm-days selected in other ways. Selecting a *period* identified ex post, the NBER recession, leaves the mean benchmark error at +0.04 (the benchmark under-forecasts slightly in the recession) and the placebo within 0.001 of one; selecting on the realised outcome without the persistence condition moves the mean error to ±0.5; selecting on the origin-day volatility, the real-time definition, leaves it at +0.02. Period-based conditioning is thus not immune to the channel, since any selection with a non-zero mean benchmark error rewards a shift in that direction, but the mean error it induces is an order of magnitude smaller than the one induced by selecting on the outcome, and so is the artefact. Table E.4 documents the level difference from the other side: on low-persistence firm-days all specifications over-forecast slightly, and the three factor specifications over-forecast by more than the benchmark (−0.022 against −0.018, both with t-statistics near −2), so that the upward shift relative to the benchmark is, on average, in the wrong direction.

Two further checks are reported for completeness. Table E.5 varies the cut-offs of the ex-post classification; the high-volatility, low-persistence ratios of the three factor specifications remain below one at most combinations, which is what the level-shift channel predicts, since it operates at any cut-off that selects on the outcome. Correcting the QLIKE forecast for retransformation bias, $F_{t+1} = \exp(\hat{y}_{t+1} + \hat{\sigma}_t^2/2)$ with $\hat{\sigma}_t^2$ the in-window residual variance of each specification, moves the ex-post high-volatility, low-persistence QLIKE ratios from 0.988 to 0.991 (HAR2PCA), 0.992 to 0.995 (HARPCA), 0.992 to 0.998 (HARMPCA), 1.004 to 1.016 (HARLASSO) and 1.030 to 1.050 (HARMPCAOPT): the correction penalises the specifications with larger residual variance, as the retransformation discussion in Section E.1 anticipated, and it narrows the gains further.

Table E.6 reports the threshold-free regression of Eq. (E.1), estimated on the firm-day panel with origin-day volatility, firm fixed effects and day-clustered Newey–West standard errors.

TABLE E.3: PERIOD-BASED VERSUS OUTCOME-BASED CONDITIONING. Cross-firm average relative losses (QLIKE / MSE) of the three factor specifications and of the placebo forecast of Table 8 (benchmark plus the HAR2PCA level difference on low-persistence firm-days) on sets of firm-days selected in different ways: the whole sample; the NBER recession, a period identified ex post; the recession's low-persistence firm-days; the top and bottom deciles of the *realised* target-day volatility (outcome-based, as in Table 5, here without the persistence condition); and the top and bottom deciles of the origin-day volatility (real-time). The second column is the mean HARPURE forecast error on each set; by Eq. (11), the placebo's expected relative MSE on a set S is $1 + (c^2 - 2c\,E[e_b \mid S])/E[e_b^2 \mid S]$, so that the artefact scales with the mean benchmark error that the selection induces. For the outcome-based high-volatility, low-persistence cell this formula gives 0.9949 (placebo in Table 8, MSE: 0.9942) and for the low-volatility cell 1.0091 (1.0101).

Selection	Firm-days	Mean e_b	HAR2PCA	HARPCA	HARMPCA	Placebo
All firm-days	107,877	-0.003	1.019 / 1.029	1.015 / 1.016	1.025 / 1.026	1.000 / 1.000
NBER recession (period)	13,101	+0.037	1.010 / 1.014	1.009 / 1.013	1.016 / 1.021	0.999 / 1.000
Recession × low persistence	1,588	+0.056	1.018 / 1.014	1.015 / 1.011	1.020 / 1.018	0.994 / 0.998
Top decile, realised $\log RV_{t+1}$ (outcome)	10,791	+0.477	1.000 / 1.002	1.001 / 1.002	1.003 / 1.004	0.999 / 0.999
Bottom decile, realised $\log RV_{t+1}$ (outcome)	10,791	-0.560	1.000 / 1.001	0.999 / 1.000	1.005 / 1.007	1.001 / 1.001
Top decile, origin-day $\log RV_t$ (real-time)	10,791	+0.015	1.026 / 1.020	1.020 / 1.019	1.031 / 1.025	0.999 / 1.000
Bottom decile, origin-day $\log RV_t$ (real-time)	10,791	+0.016	1.019 / 1.015	1.022 / 1.015	1.035 / 1.028	1.000 / 1.000

TABLE E.4: MEAN FORECAST ERROR BY REAL-TIME REGIME. Mean of $\log RV_{i,t+1} - \hat{y}_{i,t+1}$ (a negative value is an over-forecast) with the t-statistic of the pooled daily mean (five Newey–West lags) in parentheses, for the benchmark and the five augmented specifications, unconditionally, on low- and high-persistence firm-days, and in the two real-time low-persistence cells. The augmented factor specifications over-forecast by more than the benchmark on low-persistence firm-days, which is the level difference that the placebo of Table 8 exploits.

Firm-days	n	HARPURE	HAR2PCA	HARPCA	HARLASSO	HARMPCA	HARMPCAOPT
All	107,877	-0.0028 (-0.4)	-0.0028 (-0.4)	-0.0027 (-0.4)	-0.0026 (-0.4)	-0.0028 (-0.4)	-0.0007 (-0.1)
Low persistence	13,076	-0.0179 (-2.0)	-0.0220 (-2.4)	-0.0219 (-2.4)	-0.0168 (-1.8)	-0.0207 (-2.3)	-0.0139 (-1.5)
High persistence	13,076	+0.0107 (+1.5)	+0.0085 (+1.2)	+0.0093 (+1.3)	+0.0090 (+1.2)	+0.0109 (+1.5)	+0.0128 (+1.8)
High vol. & low pers.	1,470	-0.0079 (-2.5)	-0.0204 (-3.1)	-0.0200 (-3.1)	-0.0088 (-2.7)	-0.0178 (-3.2)	-0.0068 (-2.7)
Low vol. & low pers.	1,070	-0.0629 (-1.4)	-0.0598 (-1.2)	-0.0586 (-1.2)	-0.0495 (-0.9)	-0.0571 (-1.2)	-0.0434 (-0.5)

TABLE E.5: SENSITIVITY OF THE HIGH-VOLATILITY, LOW-PERSISTENCE CELL TO THE REGIME CUT-OFFS (EX-POST CLASSIFICATION). Cross-firm average relative loss in the high-volatility, low-persistence cell of Table 5 when the volatility regime is defined by the top q_v percentile of the firm's realised target-day log volatility and the persistence regime by the bottom q_p percentile of the daily cross-section of $d_{i,t}$. The paper uses $q_v = 90$, $q_p = 10$. Firm-days and the number of contributing stocks (at least 20 firm-days) are given for each pair. The corresponding grid under real-time classification is reported in the text.

				HAR2PCA		HARPCA		HARLASSO		HARMPCA		HARMPCAOPT	
q_v	q_p	Firm-days	Firms	MSE	QLIKE	MSE	QLIKE	MSE	QLIKE	MSE	QLIKE	MSE	QLIKE
85	10	2,253	13	0.994	0.993	0.994	0.996	1.008	1.016	1.000	1.002	1.039	1.052
85	15	2,713	17	0.991	0.995	0.996	1.002	0.993	1.002	1.000	1.001	1.028	1.037
85	20	3,612	20	1.001	1.001	1.000	1.001	1.027	1.032	1.001	1.002	1.042	1.050
90	10	1,487	13	**0.990**	**0.988**	**0.991**	**0.992**	**0.999**	**1.004**	**0.991**	**0.992**	**1.020**	**1.030**
90	15	1,793	15	0.990	0.995	0.993	1.002	0.994	1.004	0.987	0.988	1.009	1.016
90	20	2,385	18	0.991	0.995	0.992	0.999	1.004	1.011	0.995	0.996	1.024	1.033
95	10	727	8	0.992	0.992	0.989	0.993	0.995	0.991	0.992	1.000	1.011	1.019
95	15	874	11	1.003	1.004	1.000	1.003	1.000	1.000	0.999	1.000	1.029	1.032
95	20	1,144	14	0.995	1.000	0.994	1.003	1.029	1.039	0.996	0.998	1.033	1.042

It says the same thing as the real-time regime tables in continuous form: no coefficient, and in particular no interaction coefficient β_3, is significant for any specification or loss function, and the fitted differentials at the low-persistence, high-volatility corner are positive.

TABLE E.6: LOSS-DIFFERENTIAL REGRESSION, EQ. (E.1). Firm-day panel regression of the loss differential $\Delta L_{i,t+1}$ (augmented minus HARPURE) on the rolling persistence estimate $d_{i,t}$, the origin-day log volatility $\log RV_{i,t}$ and their product, with firm fixed effects; standard errors are clustered by day with five Newey–West lags (t-statistics in parentheses). Coefficients are multiplied by 10^3. The last four columns report the fitted differential ($\times 10^3$, relative to the firm mean) at the 10th and 90th percentiles of the within-firm distributions of d and $\log RV$: a negative entry in the (d_{10}, RV_{90}) column would indicate gains from the augmented specification when persistence is low and volatility high. $n = 107{,}844$ firm-days.

Specification	Loss	β_1 (d)	β_2 ($\log RV$)	β_3 ($d \times \log RV$)	(d_{10}, RV_{10})	(d_{10}, RV_{90})	(d_{90}, RV_{10})	(d_{90}, RV_{90})
HAR2PCA	MSE	-229.36 (-1.24)	13.05 (+1.59)	-22.68 (-1.13)	+9.5	+44.5	-32.7	-8.2
	QLIKE	-42.22 (-0.46)	2.42 (+0.52)	-4.61 (-0.46)	+1.7	+8.3	-6.0	-1.5
HARPCA	MSE	-66.61 (-0.50)	4.84 (+0.92)	-5.49 (-0.37)	+1.8	+14.0	-10.7	-1.0
	QLIKE	-5.91 (-0.06)	-0.06 (-0.01)	-0.05 (-0.00)	+0.7	+0.6	-0.5	-0.6
HARLASSO	MSE	-200.91 (-1.24)	12.02 (+1.80)	-18.96 (-1.04)	+7.8	+39.5	-29.4	-6.4
	QLIKE	-158.64 (-1.20)	8.03 (+1.20)	-15.81 (-1.11)	+7.6	+29.6	-21.6	-6.9
HARMPCA	MSE	-31.06 (-0.20)	3.15 (+0.50)	-1.34 (-0.08)	+0.0	+7.4	-6.0	+0.7
	QLIKE	15.05 (+0.12)	-1.12 (-0.18)	2.52 (+0.19)	-0.2	-3.4	+2.3	+0.3
HARMPCAOPT	MSE	-812.46 (-1.23)	51.09 (+1.24)	-92.53 (-1.17)	+27.3	+165.0	-119.8	-24.7
HARMPCAOPT	QLIKE	not estimable: one firm-day with a QLIKE loss of order 10^{16} dominates the regression						

E.6 Fluctuation Test and Robustness

Table E.7 asks whether the unconditional result hides periods in which the augmented specifications did better, which is the "pockets of predictability" question that the state-dependent framework of Odendahl et al. (2023) formalises and that the fluctuation test of Giacomini and Rossi (2010) answers without specifying the state. The rolling standardised loss differential never falls below zero by a significant margin for any specification under either loss function, in windows of 30% or of 10% of the sample; its most favourable value is +0.14 (HARPCA, QLIKE, ten-per-cent windows ending in January 2010) and its maximum exceeds the 5% critical value in the unfavourable direction for most specifications. The years 2005 and 2010–2011, in which the ex-post regime cell showed year-by-year gains (Section 6.1), do not stand out once all firm-days are included. There are no pockets.

Table E.8 collects the robustness checks. The real-time high-volatility, low-persistence cell shows no significant gain in either half of the sample, in the NBER recession or outside it; the point ratios are below one outside the recession and in 2010–2016 (the latter with 198 firm-days) and above one otherwise, with pooled statistics between -1.5 and $+0.8$, while the unconditional loss is significant in every subperiod. Replacing the ARFIMA d by the sum of the rolling HAR coefficients as the persistence measure, which agrees with d on 91% of the low-persistence flags, changes nothing: the real-time cell is above one, the ex-post cell below one under MSE, and all low-persistence firm-days together show significant losses. Sign-randomisation p-values for the pooled daily differentials, which assume nothing about the distribution of the losses, agree with the pooled Diebold–Mariano test: 0.01 to 0.04 under MSE in the ex-post cell for the three factor specifications, above 0.3 under QLIKE, nothing below 0.10 in the real-time cell except for HARMPCAOPT in the unfavourable direction, and zero unconditionally.

Table E.7: Fluctuation Test of Giacomini and Rossi (2010). For each specification and loss function, the loss differential against HARPure is averaged across stocks on each day; the fluctuation statistic is the standardised mean of the differential over a rolling window of μT days (T = 3,269), using the full-sample HAC long-run variance (five Newey–West lags). A negative value favours the augmented specification. The table reports the most favourable value of the statistic over all windows (its minimum), the date on which that window ends, and the largest absolute value. Critical values for the two-sided test at the 10, 5 and 1% levels are obtained by simulating the limiting process (20,000 paths): 2.81, 3.07 and 3.58 for $\mu = 0.3$; 3.25, 3.49 and 3.98 for $\mu = 0.1$. A minimum below minus the critical value would indicate a window of significant gains from the augmented specification; none occurs. The HARmPCAopt QLIKE series is not computable for the reason given in Table 7.

		$\mu = 0.3$ (window of 981 days)			$\mu = 0.1$ (window of 327 days)		
Specification	Loss	Minimum	Window ends	Max. $\lvert F\rvert$	Minimum	Window ends	Max. $\lvert F\rvert$
HAR2PCA	MSE	+0.63	2012-08-15	2.96	+0.19	2010-01-12	3.88
	QLIKE	+2.05	2011-12-16	4.18	+0.26	2010-01-12	4.06
HARPCA	MSE	+4.75	2012-08-15	9.52	+1.23	2010-01-21	6.95
	QLIKE	+2.22	2012-01-17	5.47	+0.14	2010-01-12	5.00
HARLasso	MSE	+8.59	2012-08-13	14.23	+3.67	2010-12-31	9.80
	QLIKE	+3.24	2011-08-03	7.30	+0.50	2010-10-19	6.56
HARmPCA	MSE	+7.20	2012-08-08	11.35	+3.25	2012-08-27	8.49
	QLIKE	+2.62	2012-07-25	6.20	+0.61	2012-08-27	5.89
HARmPCAopt	MSE	+0.61	2016-12-28	2.52	+0.29	2010-05-20	3.60
	QLIKE	n.c.	n.c.	n.c.	n.c.	n.c.	n.c.

Table E.8: Robustness of the Real-Time Result: Subperiods, Persistence Measure and Randomisation Inference. Panel A reports, for the three factor specifications, the cross-firm average relative loss in the real-time high-volatility, low-persistence cell (Table 6, Panel A) with the pooled Diebold–Mariano statistic in parentheses, and the unconditional pooled Diebold–Mariano statistic, for the full sample, the two halves of the evaluation period (which are also the two sources of the saved forecasts), and the NBER recession (December 2007 to June 2009) against the rest. Panel B replaces the ARFIMA d by the sum of the rolling HARPure coefficients as the persistence measure (cross-sectional deciles; the two low-persistence flags agree on 91% of firm-days). Panel C reports two-sided sign-randomisation p-values for the mean of the pooled daily loss differential (20,000 random sign assignments), a distribution-free counterpart of the pooled Diebold–Mariano test of Table 7.

Panel A: subperiods, real-time high-volatility, low-persistence cell; QLIKE ratio (pooled DM) and unconditional pooled DM

Sample	Firm-days	HAR2PCA	HARPCA	HARmPCA	Unconditional DM, HAR2PCA MSE / QLIKE
2004–2016	1,470	1.010 (-1.04)	1.003 (-1.26)	1.006 (-1.32)	+2.60 / +6.25
2004–2009	1,272	1.017 (-0.94)	1.009 (-1.18)	1.007 (-1.46)	+1.83 / +3.50
2010–2016	198	0.971 (-0.72)	0.963 (-0.57)	0.999 (+0.81)	+9.87 / +5.62
NBER recession	698	1.015 (+0.42)	1.009 (+0.33)	1.006 (-0.47)	+2.19 / +2.32
Outside recession	772	0.981 (-1.08)	0.974 (-1.29)	0.969 (-1.28)	+2.44 / +5.95

Panel B: persistence measured by the HAR coefficient sum; QLIKE ratio (pooled DM)

Cell	Firm-days	HAR2PCA	HARPCA	HARmPCA
High vol. & low pers. (real-time)	1,514	1.014 (-1.40)	1.011 (-1.59)	1.021 (-0.69)
High vol. & low pers. (ex-post)	1,516	0.997 (-0.89)	1.001 (-0.23)	0.995 (-0.17)
All low persistence	13,076	1.013 (+1.82)	1.015 (+2.23)	1.018 (+2.79)

Panel C: sign-randomisation p-values, MSE / QLIKE

Cell	Days	HAR2PCA	HARPCA	HARmPCA	HARLasso / HARmPCAopt
High vol. & low pers. (ex-post)	801	0.013 / 0.375	0.039 / 0.553	0.032 / 0.438	0.300/0.522 ; 0.398/0.603
High vol. & low pers. (real-time)	792	0.949 / 0.311	0.851 / 0.212	0.527 / 0.162	0.106/0.556 ; 0.028/0.408
Unconditional	3269	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000/0.000 ; n.c.

F Monte Carlo Assessment of the Regime-Conditional Evaluation Design

The out-of-sample evidence in Section 6.1 rests on a particular evaluation design: forecast losses from a rolling-window comparison are sorted ex post into regimes defined by a firm's own volatility and by its persistence relative to the cross-section, and the pattern of relative losses across regimes is read as evidence about where media signals carry information. A natural objection is that such a design might produce the pattern in Table 5 even if the predictive content of the media signals were the same in every state, simply because the benchmark forecasts worse in some states than in others. A second objection is that the pattern might depend on the particular cut-offs used to define the regimes, or on the fact that the volatility thresholds use full-sample information. A third is that the cross-sectional summaries in the tables, the Fisher-combined significance levels and the proportions of outperforming stocks, presume independence across stocks that does not hold. This appendix examines each objection by applying the evaluation design of Section 4.5, unchanged, to simulated panels in which the truth is known. Because the volatility classification of Table 5 uses the realised target-day volatility (Section 4.5), every exercise is run under both classifications: the real-time one, on the origin-day volatility, and the ex-post one, on the target-day volatility.

F.1 Design

We simulate panels of $N = 33$ firms and $T = 4{,}780$ trading days (the length of the DJIA sample), discard a burn-in of 300 days, and evaluate one-step-ahead forecasts over the final 3,270 days, which matches the length of the 2004–2016 evaluation period. Log realised volatility follows a HAR process whose coefficients switch between a high-persistence state H and a low-persistence state L:

$$y_{i,t+1} = c_{i,s} + a_s^{(d)}\, y_{i,t} + a_s^{(w)}\, y_{i,t}^{(w)} + a_s^{(m)}\, y_{i,t}^{(m)} + \gamma_s\, x_{i,t} + \sigma_s\, \varepsilon_{i,t+1}, \qquad s = s_{i,t} \in \{H, L\}, \quad \text{(F.1)}$$

where $y^{(w)}$ and $y^{(m)}$ are the five- and 22-day averages of Eq. (3), $x_{i,t}$ is an exogenous predictor observed at t, and the state $s_{i,t}$ follows a firm-specific two-state Markov chain whose probability of entering the low-persistence state is ten times higher during market-wide stress episodes, themselves generated by a common two-state chain. The innovations $\varepsilon_{i,t+1}$ contain a common component so that forecast errors are correlated across firms, as they are for DJIA constituents. In the high-persistence state the HAR coefficients sum to 0.95; in the low-persistence state they sum to 0.45, the intercept is higher and the innovation standard deviation is larger, so that low persistence and high volatility co-occur, imperfectly, as in the data. Table F.1 lists all parameter values.

The augmented forecasting model adds four exogenous terms to the HAR benchmark: the predictor $x_{i,t}$ and three pure-noise AR(1) series, so that the augmented model carries the estimation burden of a specification with four media components, as the two-stage specifications of Table 3 do, while only one of the four terms carries information. The three DGPs differ only in the relationship between x and the state:

> **DGP-0 (constant content).** $\gamma_H = \gamma_L = 0.05$ and x is independent of the state. The predictor carries the same, small, information in every state. This is the null against

which the design must be tested: any regime pattern that appears here is produced by the design, not by the predictor.

DGP-1 (state-dependent content). $\gamma_H = 0.05$ and $\gamma_L = 0.22$, with x independent of the state. The predictor is more informative in the low-persistence state. This is the conjecture of the paper.

DGP-2 (state proxy). $\gamma_H = \gamma_L = 0.05$, but the mean of x is one standard deviation higher in the low-persistence state. The predictor's content is constant, but its level tracks the state, so that a model containing it can partially absorb the intercept shift between states. This is a second alternative to the conjecture: a media signal that merely reflects whether a firm is in a turbulent phase.

DGP-3 (sharp state proxy). As DGP-2, but with HAR coefficients summing to 0.15 rather than 0.45 in the low-persistence state and spells about twice as long, so that the rolling estimate of d identifies the state more accurately (two thirds of flagged firm-days are in the true low-persistence state, against 45% under DGP-2). It asks whether the DJIA configuration of Section 6.2 appears once the classifier is sharper.

The parameter values are chosen so that the unconditional and high-persistence outcomes are of the same order as in the DJIA panel. Under DGP-0 the population share of variance explained by x in the high-persistence state is about 0.8%, while the cost of estimating four additional coefficients in a 252-day window is about 1.6% of the forecast error variance, so that the augmented model is expected to be slightly worse than the benchmark in most states, as it is in Table E.1. Under DGP-1 the population gain in the low-persistence state is about 5%, of which estimation error consumes part; the resulting high-volatility, low-persistence relative loss (Table F.2) is close to that of HAR2PCA in Table 5.

Table F.1: Monte Carlo: Data-Generating Process Parameters. Parameters of Eq. (F.1) and of the evaluation pipeline. The three DGPs share all parameters except the two in the last block.

Panel	$N = 33$ firms; $T = 4{,}780$ days after a 300-day burn-in; evaluation over the last 3,270 days; $R = 252$
HAR coefficients	high persistence $(a^{(d)}, a^{(w)}, a^{(m)}) = (0.35, 0.30, 0.30)$; low persistence $(0.15, 0.15, 0.15)$
Intercepts	$c_{i,s} = \mu_i\,(1 - a_s^{(d)} - a_s^{(w)} - a_s^{(m)}) + 0.5 \cdot \mathbb{I}(s = L)$, $\mu_i \sim N(0, 0.5^2)$
Innovations	$\sigma_H = 0.55$, $\sigma_L = 0.90$; cross-sectional correlation 0.30 through a common shock
Persistence states	$\Pr(H \to L) = 0.003$ (calm) or 0.030 (market stress); $\Pr(L \to H) = 0.015$
Market stress	common chain, $\Pr(\text{on}) = 0.005$, $\Pr(\text{off}) = 0.025$ (about 17% of days)
Predictor x	AR(1), $\phi = 0.5$, unit variance, cross-sectional correlation 0.20
Noise predictors	three independent AR(1) series, $\phi = 0.5$, uninformative
Estimation	rolling OLS, daily re-estimation; benchmark (HAR) and augmented (HAR + 4 terms)
Persistence estimate	Whittle estimate of d on the 252-day window, grid step 0.01
Regimes	volatility: top/bottom decile of own $y_{i,t}$ (real-time) or $y_{i,t+1}$ (ex-post); persistence: top/bottom decile of the day's cross-section of $\hat{d}_{i,t}$
Cell statistics	as in Section 4.5; a firm contributes to a cell if it has at least 20 firm-days in it
DGP-0	$\gamma_H = \gamma_L = 0.05$; no shift in x
DGP-1	$\gamma_H = 0.05$, $\gamma_L = 0.22$; no shift in x
DGP-2	$\gamma_H = \gamma_L = 0.05$; mean of x higher by 1.0 in state L
DGP-3	as DGP-2; low-persistence coefficients $(0.05, 0.05, 0.05)$; $\Pr(H \to L) = 0.002$ (calm) or 0.020 (stress), $\Pr(L \to H) = 0.008$

The evaluation pipeline replicates Section 4.5 step by step. Both models are re-estimated by OLS every day on a 252-day rolling window. The persistence estimate $\hat{d}_{i,t}$ is obtained on the same window by the Whittle (frequency-domain) estimator for the ARFIMA$(0, d, 0)$ model, a computationally cheap approximation to the time-domain estimator used in the paper that

preserves the classification logic; firms are then ranked cross-sectionally on each day and the top and bottom deciles are labelled high and low persistence. High- and low-volatility firm-days are the top and bottom deciles of each firm's own log-volatility distribution, computed on the full simulated sample as in the paper, applied to the origin-day volatility $y_{i,t}$ (real-time classification) and to the target-day volatility $y_{i,t+1}$ (ex-post classification), and, for the comparison in Panel D of Table F.3, on the trailing sample available at t. Within each cell we compute the stock-level ratio of average losses, the cross-firm average of these ratios, the proportion of firms with a ratio below one, and Fisher's combination of two-sided Diebold–Mariano p-values, exactly as in Tables E.1 and 5. We also estimate the loss-differential regression in Eq. (E.1) on the simulated firm-day panel with firm fixed effects and standard errors clustered by day. All results are averages over 100 replications; the persistence cut-off is the cross-sectional decile, as in Tables E.1 and 5.

F.2 Results

Table F.2: Monte Carlo: Regime-Conditional Relative Losses Under Three Data-Generating Processes. Each block reports, for one DGP, the same statistics as Table 5 computed on simulated panels of 33 firms: the cross-firm average of the stock-level relative loss (augmented model divided by the HAR benchmark), under the real-time classification (volatility regime from the origin-day volatility $y_{i,t}$; persistence regime from the cross-sectional deciles of the rolling $\hat{d}_{i,t}$), in parentheses the proportion of contributing firms with a ratio below one, and in the last column the frequency with which Fisher's combined test of the two-sided stock-level Diebold–Mariano statistics rejects at the 5% level (QLIKE loss). "Firm-days" and "Firms" are the average number of firm-days and of contributing firms in the regime. The same cells under the ex-post classification used in Table 5 are in Table F.5. Averages over 100 replications; the evaluation pipeline, regime definitions and cut-offs are those of Section 4.5. DGP parameters are listed in Table F.1.

Regime	Firm-days	Firms	MSE	MAE	QLIKE	Fisher rej.
DGP-0: constant predictive content						
High volatility & high persistence	1,063	18.7	1.009 (40%)	1.005 (41%)	1.013 (40%)	0.08
High volatility & low persistence	1,579	24.8	1.014 (35%)	1.007 (37%)	1.016 (39%)	0.14
Low volatility & high persistence	2,157	25.8	1.010 (38%)	1.006 (39%)	1.009 (43%)	0.05
Low volatility & low persistence	496	9.2	1.016 (40%)	1.008 (40%)	1.020 (40%)	0.05
Unconditional	107,910	33.0	1.012 (0%)	1.006 (1%)	1.013 (5%)	1.00
DGP-1: state-dependent predictive content						
High volatility & high persistence	1,087	19.0	0.996 (54%)	0.999 (52%)	0.996 (53%)	0.07
High volatility & low persistence	1,558	24.6	0.984 (61%)	0.994 (58%)	0.979 (62%)	0.16
Low volatility & high persistence	2,136	25.6	1.012 (39%)	1.007 (39%)	1.010 (43%)	0.09
Low volatility & low persistence	523	10.0	1.001 (51%)	1.001 (51%)	0.999 (52%)	0.10
Unconditional	107,910	33.0	0.998 (61%)	1.002 (30%)	0.992 (76%)	0.48
DGP-2: constant content, predictor proxies for the state						
High volatility & high persistence	1,031	18.3	1.020 (34%)	1.010 (36%)	1.015 (41%)	0.12
High volatility & low persistence	1,644	25.2	1.022 (30%)	1.013 (31%)	1.022 (37%)	0.28
Low volatility & high persistence	2,212	26.1	1.004 (46%)	1.003 (44%)	0.996 (53%)	0.11
Low volatility & low persistence	449	8.0	1.001 (50%)	0.999 (53%)	0.999 (51%)	0.05
Unconditional	107,910	33.0	1.007 (7%)	1.004 (9%)	0.997 (62%)	0.24
DGP-3: as DGP-2 with sharper persistence states						
High volatility & high persistence	1,076	18.1	1.019 (32%)	1.009 (36%)	1.013 (40%)	0.06
High volatility & low persistence	1,650	24.9	1.022 (29%)	1.011 (31%)	1.027 (33%)	0.23
Low volatility & high persistence	2,113	25.3	1.002 (48%)	1.003 (44%)	0.992 (54%)	0.07
Low volatility & low persistence	585	11.4	1.003 (48%)	0.999 (50%)	1.005 (47%)	0.06
Unconditional	107,910	33.0	1.009 (4%)	1.005 (5%)	1.001 (45%)	0.24

Table F.2 reports the regime-conditional relative losses under the three DGPs. Under DGP-0, in which the predictor carries the same information in every state, the augmented model is

worse than the benchmark in every cell, with relative losses between 1.009 and 1.020, and the high-volatility, low-persistence cell (1.016) is among the worst of the four rather than the best. The reason is arithmetic. A predictor with constant content reduces the forecast error variance by a fixed absolute amount, while the error variance of the benchmark is larger in the low-persistence, high-volatility state; the relative gain therefore shrinks in that state, whereas the cost of the additional estimated coefficients is roughly proportional to the error variance and does not. The mechanical channel that the objection has in mind, a benchmark that deteriorates in some states, moves the relative loss of a state-independent predictor towards or above one in those states; it does not move it below one. Across replications the high-volatility, low-persistence QLIKE ratio falls below one in 8% of cases and is the lowest of the four cells in 3% (Table F.3, Panel B).

Under DGP-1, in which the predictor is more informative in the low-persistence state, the signature of Table 5 appears: the high-volatility, low-persistence cell has relative losses of 0.984, 0.994 and 0.979 under MSE, MAE and QLIKE, with about 62% of contributing firms below one, while the low-volatility, high-persistence cell is above one under all three loss functions and the remaining two cells are close to one. The QLIKE ratio in the high-volatility, low-persistence cell is below one in 86% of replications and is the lowest of the four cells in 52%. The magnitudes are of the same order as those of HAR2PCA in Table 5 (0.990, 0.993 and 0.988, with 62% of firms below one). One feature of the simulated pattern differs from the empirical one and is informative: the high-volatility, high-persistence cell shows a small gain (0.996) rather than a loss, because the rolling persistence estimate is a noisy classifier (Panel A of Table F.3) and some true low-persistence firm-days are labelled high persistence; in the DJIA panel the corresponding cell shows losses for every specification, which suggests that the empirical classification separates the states at least as sharply as the simulated one. The low-volatility, low-persistence cell is close to one under DGP-1 (0.999), as it is in Table 5. This agreement should not be over-read, however: it concerns the ex-post table, and Section F.3 shows that DGP-1 does not reproduce the way the DJIA pattern behaves across the two classifications.

Under DGP-2, in which the predictor has constant content but its level tracks the state, the pattern is different again: the gains, such as they are, are located in the two low-volatility cells (0.996 and 0.999 under QLIKE) and both high-volatility cells show losses (1.015 and 1.022). The high-volatility, low-persistence ratio is below one in 6% of replications and is almost never (2%) the lowest of the four cells. A predictor that merely proxies for the state therefore does not generate the empirical pattern either; if anything, it generates its mirror image. We do not claim that DGP-2 exhausts the ways in which a state proxy could enter, but it is the natural one, and it is distinguishable from DGP-1 by the location of the gains.

Table F.3 collects the remaining diagnostics. Panel A shows that the real-time low-persistence flag is informative but far from exact: firm-days flagged as low persistence are in the true low-persistence state 45% of the time, against 18% for firm-days flagged as high persistence and 29% unconditionally. The 252-day window averages over spells of both states, so that the mean $\hat{d}$ differs between true states by only 0.04. The regime labels in the paper should be read in the same way, as identifying firm-days on which recent persistence is low *relative to the cross-section*, not as identifying a latent state without error; the empirical results are obtained despite this attenuation, not because of it. Panel C reports the interaction coefficient of Eq. (E.1). Its sign discriminates between the DGPs (positive in 78% of replications under

TABLE F.3: MONTE CARLO: DIAGNOSTICS OF THE REGIME-CONDITIONAL EVALUATION DESIGN. Panel A reports how well the real-time low-persistence flag (bottom cross-sectional quintile of the rolling Whittle estimate of d) identifies the true low-persistence state, and how strongly the high-volatility flag co-occurs with it. Panel B reports the frequency, across replications, of the signature in Table 5: a QLIKE ratio below one in the high-volatility, low-persistence cell; that cell being the lowest of the four; and both high-persistence cells at or above one. Panel C reports the interaction coefficient β_3 of the loss-differential regression in Eq. (E.1) (MSE loss, firm fixed effects, standard errors clustered by day): its average, the frequency with which it is positive, and the frequency of rejection at the 5% level. Panel D compares the high-volatility, low-persistence QLIKE ratio under full-sample (ex-post) and trailing (real-time) volatility thresholds, and reports the share of firm-days on which the two classifications agree. Panel E compares QLIKE computed from $\exp(\hat{y})$ with QLIKE computed from the retransformation-corrected forecast $\exp(\hat{y} + \hat{\sigma}^2/2)$ in the same cell. Averages over 100 replications.

	DGP-0	DGP-1	DGP-2	DGP-3
Panel A: classification				
Share of firm-days in the true low-persistence state	0.29	0.29	0.28	0.34
P(true low persistence | flagged low persistence)	0.46	0.45	0.46	0.67
P(true low persistence | flagged high persistence)	0.18	0.18	0.17	0.12
Mean $\hat{d}$ in true low / high persistence state	0.34 / 0.39	0.35 / 0.39	0.35 / 0.39	0.25 / 0.36
Forecast difference $\bar{f}_a - \bar{f}_b$: true low / high persistence	-0.0001 / +0.0000	-0.0004 / +0.0005	+0.0280 / -0.0115	+0.0199 / -0.0100
… on flagged low-persistence firm-days	+0.0001	+0.0001	-0.0074	-0.0072
… in true low persistence, first 60 days / later	–	–	–	+0.0334 / +0.0111
P(flagged | true low persistence): first 60 days / later	–	–	–	0.08 / 0.33
P(high-volatility day | true low / high persistence)	0.19 / 0.06	0.19 / 0.06	0.20 / 0.06	0.16 / 0.07
Panel B: frequency of the Table 5 signature (QLIKE)				
High-vol., low-pers. ratio below one	0.08	0.86	0.06	0.00
… and lowest of the four cells	0.03	0.52	0.02	0.00
Both high-persistence cells at or above one	0.70	0.32	0.31	0.21
… and under the ex-post (outcome-based) classification of Table F.5				
High-vol., low-pers. ratio below one	0.20	1.00	0.55	0.20
… with low-vol., low-pers. ratio above one	0.05	0.00	0.21	0.05
Panel C: loss-differential regression, Eq. (E.1)				
Mean $\hat{\beta}_3$	-0.0002	0.0058	-0.0097	-0.0076
Frequency $\hat{\beta}_3 > 0$	0.55	0.78	0.03	0.04
Rejection frequency, 5% level	0.03	0.12	0.50	0.49
Panel D: ex-post versus real-time volatility thresholds				
Agreement of the two high-volatility flags (share of firm-days)	0.988	0.988	0.988	0.988
High-vol., low-pers. QLIKE ratio, ex-post thresholds	1.016	0.979	1.022	1.027
High-vol., low-pers. QLIKE ratio, real-time thresholds	1.016	0.980	1.021	1.026
Panel E: QLIKE retransformation				
High-vol., low-pers. QLIKE ratio, $F = \exp(\hat{y})$	1.016	0.979	1.022	1.027
High-vol., low-pers. QLIKE ratio, $F = \exp(\hat{y} + \hat{\sigma}^2/2)$	1.016	0.984	1.018	1.022

DGP-1, in 55% under DGP-0 and in 3% under DGP-2), but with 33 firms and a noisy persistence estimate the t-test rejects at the 5% level in only 12% of replications under DGP-1. The regression is therefore a useful complement to the regime tables rather than a more powerful replacement for them, at least at this cross-sectional size. Panel D shows that replacing the full-sample volatility thresholds by trailing thresholds changes the classification on about 1% of firm-days and leaves the high-volatility, low-persistence ratio unchanged to three decimal places; the ex-post element of the volatility classification is immaterial in this setting. Panel E shows that correcting the QLIKE forecast for retransformation bias moves the ratio in the same cell by about 0.005 under DGP-1 and not at all under DGP-0, which is small relative to the gain of about 0.02 under DGP-1 and to the differences between cells; on the DJIA panel the correction is larger, 0.003 to 0.020 depending on the specification, because the specifications' residual variances differ more than in the simulation (Section 6.2).

Table F.4 varies the two cut-offs. Under DGP-1 the high-volatility, low-persistence QLIKE ratio lies between 0.977 and 0.985 for all nine combinations of a volatility cut-off at the 85th, 90th or 95th percentile with a persistence cut-off at the 10th, 15th or 20th percentile; under DGP-0 it lies between 1.015 and 1.017. Within this range, the cut-offs affect the size of the

TABLE F.4: MONTE CARLO: SENSITIVITY OF THE HIGH-VOLATILITY, LOW-PERSISTENCE RELATIVE LOSS TO THE REGIME CUT-OFFS. Each entry is the cross-firm average QLIKE ratio in the high-volatility, low-persistence cell, under the real-time classification (upper block) and the ex-post classification (lower block), when the volatility regime is defined by the top q_v percentile of the firm's own log-volatility distribution and the persistence regime by the bottom q_p percentile of the cross-sectional distribution of $\hat{d}$. The paper uses $q_v = 90$, $q_p = 10$. Averages over 100 replications.

	DGP-0 (constant content)			DGP-1 (state-dependent content)		
Persistence cut-off q_p:	10	15	20	10	15	20
Real-time classification						
Volatility cut-off $q_v = 85$	1.016	1.016	1.016	0.985	0.985	0.983
Volatility cut-off $q_v = 90$	1.016	1.016	1.015	0.979	0.980	0.981
Volatility cut-off $q_v = 95$	1.016	1.017	1.017	0.978	0.977	0.977
Ex-post classification						
Volatility cut-off $q_v = 85$	1.009	1.009	1.008	0.961	0.961	0.960
Volatility cut-off $q_v = 90$	1.007	1.007	1.006	0.951	0.951	0.952
Volatility cut-off $q_v = 95$	1.004	1.005	1.005	0.938	0.939	0.940

contributing set and the precision of the cell statistics, not their sign. This does not establish that the DJIA results are insensitive to the cut-offs, which requires recomputing Table 5 on the data, but it does establish that the design does not require a particular pair of cut-offs to display, or to suppress, the signature.

F.3 Outcome-Based Classification

TABLE F.5: MONTE CARLO: REGIME-CONDITIONAL RELATIVE LOSSES UNDER EX-POST (OUTCOME-BASED) VOLATILITY CLASSIFICATION. As Table F.2, but with the volatility regime defined on the realised target-day volatility $y_{i,t+1}$, as in Table 5. The last column reports the mean difference between the augmented and the benchmark forecast over firm-days flagged as low persistence, the quantity that drives the sign pattern under this classification (Section 6.2). Averages over 100 replications.

Regime	Firm-days	Firms	MSE	MAE	QLIKE	Fisher rej.	$\bar{f}_a - \bar{f}_b$ (low pers.)
DGP-0: constant predictive content							
High volatility & high persistence	1,085	19.2	1.003 (45%)	1.000 (50%)	1.005 (45%)	0.19	+0.0001
High volatility & low persistence	1,569	24.8	1.001 (46%)	0.996 (58%)	1.007 (41%)	0.27	
Low volatility & high persistence	2,163	25.8	1.006 (43%)	1.003 (44%)	1.007 (41%)	0.23	
Low volatility & low persistence	500	9.5	0.999 (52%)	0.996 (59%)	0.997 (56%)	0.14	
DGP-1: state-dependent predictive content							
High volatility & high persistence	1,108	19.3	0.982 (67%)	0.990 (65%)	0.976 (67%)	0.38	+0.0001
High volatility & low persistence	1,548	24.5	0.957 (82%)	0.968 (87%)	0.951 (78%)	0.88	
Low volatility & high persistence	2,143	25.7	1.000 (50%)	1.002 (47%)	1.004 (46%)	0.30	
Low volatility & low persistence	528	10.1	0.951 (83%)	0.971 (84%)	0.959 (84%)	0.83	
DGP-2: constant content, predictor proxies for the state							
High volatility & high persistence	1,054	18.7	0.980 (69%)	0.991 (65%)	0.970 (70%)	0.72	-0.0074
High volatility & low persistence	1,631	25.1	0.998 (52%)	0.996 (58%)	1.000 (50%)	0.40	
Low volatility & high persistence	2,218	26.1	1.012 (37%)	1.004 (43%)	1.011 (36%)	0.28	
Low volatility & low persistence	455	8.2	0.997 (51%)	0.994 (60%)	0.994 (54%)	0.29	
DGP-3: as DGP-2 with sharper persistence states							
High volatility & high persistence	1,096	18.3	0.980 (68%)	0.991 (64%)	0.971 (70%)	0.70	-0.0072
High volatility & low persistence	1,645	24.8	1.002 (46%)	0.998 (56%)	1.007 (43%)	0.26	
Low volatility & high persistence	2,120	25.5	1.012 (37%)	1.004 (43%)	1.010 (37%)	0.23	
Low volatility & low persistence	592	11.7	0.996 (54%)	0.994 (63%)	0.993 (59%)	0.26	

Table F.5 repeats the regime comparison with the volatility regime defined on the realised target-day volatility, as in Table 5. Three results follow. Under DGP-0 the augmented model remains at or above the benchmark in every cell (0.997 to 1.007 under QLIKE); the high-volatility, low-persistence ratio falls below one in 20% of replications but the joint signature

of the DJIA table, a ratio below one in that cell together with a ratio above one in the low-volatility, low-persistence cell, occurs in 5%. Sorting on the outcome does not, by itself, manufacture gains: when the augmented forecast differs from the benchmark only by zero-mean estimation noise, the noise is uncorrelated with the realised surprise and the sort has nothing to reward. Under DGP-1 the sort rewards the predictor's information in every cell, and the gains are larger than under real-time classification (0.951 in the high-volatility, low-persistence cell, 0.959 in the low-volatility, low-persistence cell), because a predictor that is informative about $y_{i,t+1}$ is, by construction, correlated with the realised surprise; the joint signature never occurs. Under DGP-2, in which the predictor is informative about the level of the outcome but too noisy to pay unconditionally, the sort produces a split along the volatility dimension, with gains in the high-volatility cells (0.970 in the high-persistence one) and losses in the low-volatility cells, and the joint signature occurs in 21% of replications; the last column shows the level difference between the two forecasts that the state proxy induces ($+0.028$ in the true low-persistence state, -0.012 in the true high-persistence state).

The general point is that outcome-based sorting rewards whatever component of the difference between two forecasts is correlated with the realised surprise, whether or not that component improves the forecast on average. A predictor with genuine but state-dependent content (DGP-1) is rewarded in every cell; a predictor that tracks the level of the outcome without paying unconditionally (DGP-2) is rewarded in the high-outcome cells and penalised in the low-outcome cells; and a forecast that is shifted by a small constant in a given state, the channel that the placebo of Table 8 isolates on the DJIA panel, is rewarded in the high-outcome cells of that state and penalised in its low-outcome cells, whether or not the shift is correct on average. In none of these cases does the pattern under outcome-based sorting say anything about gains available at the forecast origin, which is what the real-time classification measures. None of the four DGPs reproduces the particular DJIA configuration of a gain confined to the high-volatility, low-persistence cell, a loss in the low-volatility, low-persistence cell and nothing under real-time classification. DGP-3 shows why the state-proxy mechanism does not. With the sharper states the proxy still induces a level difference of $+0.020$ in the true low-persistence state, but Panel A of Table F.3 shows where in the spell it arises: $+0.033$ in the first sixty days of a low-persistence spell, when the rolling window is still dominated by high-persistence days and the estimated coefficient on x is applied to a value of x far above the window mean, against $+0.011$ later in the spell, when the window has caught up. The real-time flag, by contrast, is raised on only 8% of those early days and on 33% of the later ones, because the rolling $\hat{d}$ falls with a lag. The shift therefore occurs mostly before the flag is raised, and on flagged firm-days the difference nets to -0.007 once the one third of flagged days that are in the true high-persistence state are included. The DJIA configuration requires the forecast difference to be positive on the flagged days themselves, which the placebo of Table 8 imposes directly and which, in the data, reflects media predictors whose level remains elevated throughout a low-persistence episode rather than only at its onset. The joint signature occurs in 5% of DGP-3 replications and the high-volatility, low-persistence ratio falls below one under real-time classification in none.

F.4 Size of the Cross-Sectional Summaries

Table F.6 addresses the third objection by simulation under the null of equal predictive accuracy for every stock. When loss differentials are independent across stocks, Fisher's

TABLE F.6: MONTE CARLO: SIZE OF CROSS-SECTIONAL SUMMARIES OF STOCK-LEVEL FORECAST COMPARISONS UNDER CROSS-SECTIONAL DEPENDENCE. Loss differentials for $N = 33$ stocks are generated under the null of equal predictive accuracy with pairwise cross-sectional correlation ρ and T evaluation days. "Fisher" is the rejection frequency at the nominal 5% level of Fisher's combined probability test applied to two-sided stock-level Diebold–Mariano p-values, the test used in Tables E.1 and 5. "Sign" is the rejection frequency of a binomial test that the number of stocks with a negative mean loss differential equals $N/2$, which is the implicit test behind reading the proportion of outperforming stocks as evidence. "Pooled DM" first averages the loss differentials across stocks on each day and applies a single Diebold–Mariano test to the resulting series. 5,000 replications.

T	Test	$\rho = 0$	$\rho = 0.1$	$\rho = 0.2$	$\rho = 0.3$	$\rho = 0.5$
250	Fisher	0.053	0.075	0.108	0.144	0.195
	Sign	0.040	0.246	0.395	0.497	0.641
	Pooled DM	0.052	0.047	0.046	0.049	0.050
1,000	Fisher	0.048	0.067	0.115	0.147	0.190
	Sign	0.031	0.236	0.386	0.511	0.646
	Pooled DM	0.044	0.047	0.054	0.051	0.051
3,270	Fisher	0.051	0.070	0.111	0.147	0.194
	Sign	0.035	0.238	0.399	0.503	0.661
	Pooled DM	0.049	0.044	0.047	0.052	0.055

combined test has its nominal size. When they share a common component with pairwise correlation ρ, the test over-rejects: at $\rho = 0.3$, a plausible value for the loss differentials of large-capitalisation stocks in the same index, a nominal 5% test rejects about 15% of the time, and at $\rho = 0.5$ about 19%. The distortion is essentially independent of the evaluation length, because the dependence is cross-sectional. The proportion of stocks with a loss ratio below one is affected considerably more: read as a binomial test against one half, it rejects about half of the time at $\rho = 0.3$, because a common shock moves all stocks' loss differentials in the same direction at once, so that observing 60 or 70 per cent of stocks on one side of one is much less surprising than independence would suggest. Both summaries are therefore descriptive in Tables E.1 and 5, as stated in their notes. The last row of each block shows that a test which first averages the loss differentials across stocks on each day and then applies a single Diebold–Mariano test has the correct size at every ρ: the cross-sectional dependence is absorbed by the averaging, and only the time-series dependence of the pooled series remains to be handled, by a heteroskedasticity- and autocorrelation-robust variance in the usual way. This pooled test, and the state-dependent tests of Odendahl et al. (2023) applied to the pooled series, are the formal tests that the regime-conditional hypothesis calls for.

F.5 What the Study Does and Does Not Show

The study shows three things about the evaluation design. First, applied to a panel in which the exogenous predictor carries the same information in every state, the design does not produce relative losses below one in the high-volatility, low-persistence cell, even though the benchmark forecasts worse in that state; the pattern in Table 5 is not an artefact of sorting rolling-window losses into regimes. Second, the pattern is produced when the predictor's content rises in low-persistence states, at magnitudes comparable to the empirical ones. Third, a predictor that merely tracks the state produces a different pattern, with gains in the low-volatility cells. The design can therefore distinguish, at least between these alternatives, the conjecture of the paper from the two objections most naturally raised against it, provided the states are classified on information available at the forecast origin. Fourth, when the

volatility state is classified on the realised outcome, the comparison rewards any component of the forecast difference that is correlated with the realised surprise, whether or not it improves the forecast on average, so that gains under outcome-based sorting are not evidence of gains available in real time.

The study does not show that the DJIA results are generated by a process like DGP-1; Section 6.2 shows that they are not, since the DJIA gains exist only under outcome-based classification. Other processes that we have not simulated could produce a similar pattern; the most important is one in which a market-level predictor omitted from the benchmark, such as implied volatility or an earnings-announcement indicator, is correlated with both the media predictors and the persistence state, which is the concern raised in Section 4.1. Nor does the study say anything about the size or significance of the empirical gains, which are what they are in Tables E.1 and 5. It is a study of the properties of the evaluation design, and its conclusions are limited to that.